

TS

Typescript

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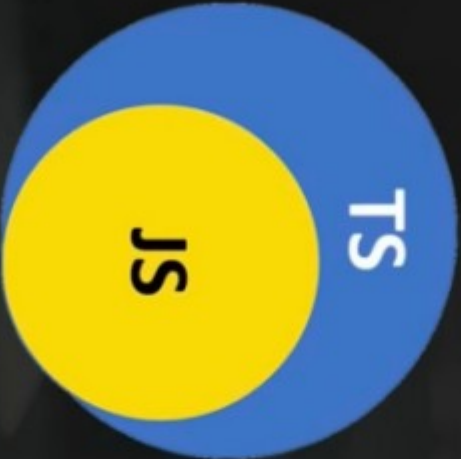
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A Venn diagram consisting of two concentric circles. The outer circle is blue and labeled 'TS' (TypeScript). The inner circle is yellow and labeled 'JS' (JavaScript). This visualizes that JavaScript is a subset of TypeScript.

What is TypeScript

- Better version of JavaScript
- With more features
- TypeScript is superset of JS
- All Features of JS
- And extra features and functionality

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TS

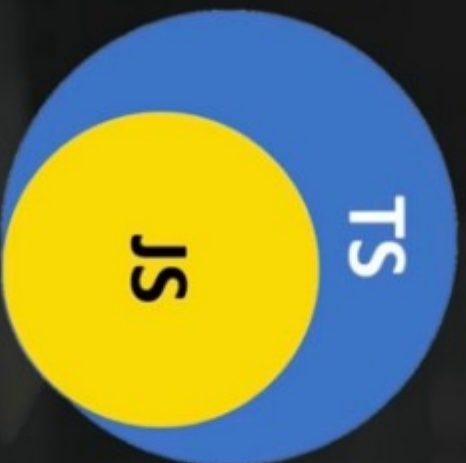
JS

Why TypeScript

- Static Typing or Define Data type.
- Improved Code Maintainability.
- Advance Features.
- Code Quality

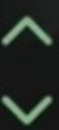
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Important TS features

- Define Data type.
- Interfaces
- Decorators
- Generics
- Namespaces
- Type Inference
- Advance Features.
- Code Quality
- Etc



via npm

TypeScript is available as a [package on the npm registry](#) available as "typescript".

You will need a copy of [Node.js](#) as an environment to run the package. Then you use a dependency manager like [npm](#), [yarn](#) or [pnpm](#) to download TypeScript into your project.

npm install typescript --save-dev

npm yarn pnpm

All of these dependency managers support lockfiles, ensuring that everyone on your team is using the same version of the language. You can then run the TypeScript compiler using one of the following commands:

npm tsc

npm yarn pnpm

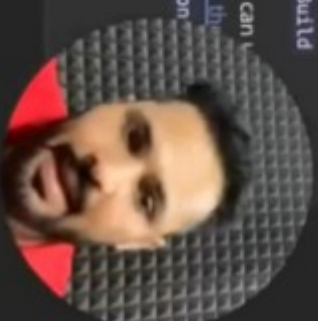
with Visual Studio

For most project types, you can get TypeScript as a package in NuGet for your MSBuild projects, for example an ASP.NET Core app.

When using NuGet, you can [install TypeScript through Visual Studio](#) using:

- The Manage NuGet Packages window (which you can get to by right-clicking on a project node)
- The NuGet Package Manager Console (found in Tools > NuGet Package Manager > Package Manager Console) and then running:
Install-Package Microsoft.TypeScript.MSBuild

For project types which don't support NuGet, you can [install the TypeScript Visual Studio extension](#). You can [install the extension](#) using Extensions > Manage Extension Studio.



The examples below are for more advanced use cases.

EXPLORE

OPEN EDITORS

TS app.ts

TYPESCRIPT-TUT

node_modules

TS app.ts

package-lock.json

package.json

← →

typescript-tut

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08 10 10

TS app.ts

X

ts app.ts > [6] userName

1 var userName:string="Anil sidhu"

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut % npm tsc app.ts

zsh +

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...

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X

Ln 1, Col 33

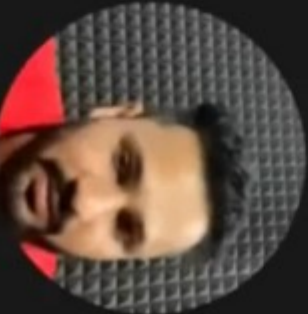
Spaces: 4

UTF-8

LF

() TypeScript

Go Live



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EXPLORE

OPEN EDITORS

TS app.ts

JS app.js

TYPESCRIPT-TUT

node_modules

JS app.js

TS app.ts

package-lock.json

package.json

← →

typescript-tut

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08 10 10

TS app.ts

JS app.js

TS app.ts > ...

1 var username:string="Anil sidhu"

2

3 console.log(username)

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut % npx tsc app.ts

anilsidhu@anils-MacBook-Air typescript-tut % npx tsc app.ts

anilsidhu@anils-MacBook-Air typescript-tut % npx tsc app.ts --watch

zsh + v

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OUTLINE

TIMELINE

Launchpad

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EXTENSIONS: MAR...

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typescript-tut

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08

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ts app.ts

user.html X

js app.js

```
1 <?user.html> <?html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Document</title>
7   <script src="/app.js"></script>
8 </head>
9 <body>
10   <h1>User file</h1>
11 </body>
12 </html>
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SPELL CHECKER

[12:35:49 AM] File change detected. Starting incremental compilation...

[12:35:49 AM] Found 0 errors. Watching for file changes.

□

node + v

□

□

...

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X

Ln 12, Col 8

Spaces: 4

UTF-8

LF

() HTML

Go Live

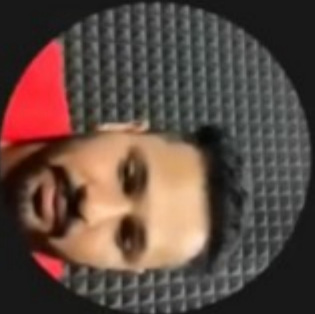
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Launchpad

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EXTENSIONS: MAR...

live

Live Server

Launch a development lo...

Ritwick Dey

Live Share

Real-time collaborative d...

Microsoft

Live Preview

Hosts a local server in yo...

Microsoft

Deprecated L...

Adds audio calling capab...

Microsoft

Live Sass Com...

Compile Sass or Scss to ...

Ritwick Dey

Live Sass Com...

Compile Sass or Scss to ...

Glenn Marks

Live Server (Fiv...

Launchpad

← →

typescript-tut

08 00 00

TS app.ts x

use terminal x

typescript-tut

```
1  var username:string= "All Star"
2  var age:number=30;
3  var college:string="IET alwar"
4
5
6  console.log(username)
7  console.log(age)
8  console.log(college)
9
10
```

PROBLEMS 1

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SPELL CHECKER 1

node + v

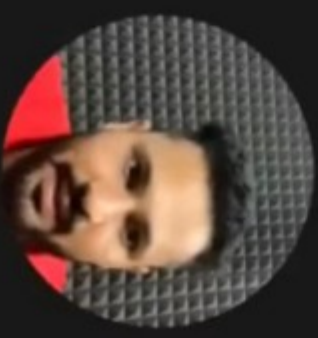
...

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X

[12:38:34 AM] File change detected. Starting incremental compilation...

[12:38:35 AM] Found 0 errors. Watching for file changes.



EXPLORER

OPEN EDITORS

TS app.ts 1

JS app.js

user.html

TYPESCRIPT-TUT

node_modules

JS app.js

TS app.ts 1

package-lock.json

package.json

user.html

TS app.ts 1

Cannot redeclare block-scoped variable 'name'. ts(2451)

TS app.ts > { lib.dom.d.ts(28611, 15): 'name' was also declared here.

1 var

2 var

3 var

4 var name="sam"

5

6 console.log(userName)

7 console.log(age)

8 console.log(college)

9

10

PROBLEMS 3

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SPELL CHECKER 2

node + v

...

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node_modules/typescript/lib/lib.dom.d.ts:28611:15 - error TS2451: Cannot redeclare block-scoped variable 'name'

28611 declare const name: void;

app.ts:4:15

var name="sam"

'name' was also declared here.

[12:39:53 AM] Found 2 errors. Watching for file changes.

Ln 4, Col 14

Spaces: 4

UTF-8

LF

TypeScript

Port: 5500

typescript-tut

08



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Globally Installing TypeScript

It can be handy to have TypeScript available across all projects, often to test one-off ideas. Long-term, codebases should prefer a project-wide installation over a global install so that they can benefit from reproducible builds across different machines.

via npm

You can use npm to install TypeScript globally, this means that you can use the `tsc` command anywhere in your terminal.

To do this, run `npm install -g typescript`. This will install the latest version (currently 5.8).

via Visual Studio Marketplace

You can install TypeScript as a Visual Studio extension, which will allow you to use TypeScript across many MSBuild projects in Visual Studio.

The latest version is available [in the Visual Studio Marketplace](#).

Working with TypeScript-compatible transpilers

There are other tools which convert TypeScript files to JavaScript files. You might use these tools for speed or consistency with existing build tooling.

Each of these projects handle the file conversion, but do not handle the type-checking aspects of the TypeScript compiler. So it's likely



Edit: Code

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https://slides.com/anilsidhu/code-182053/edit

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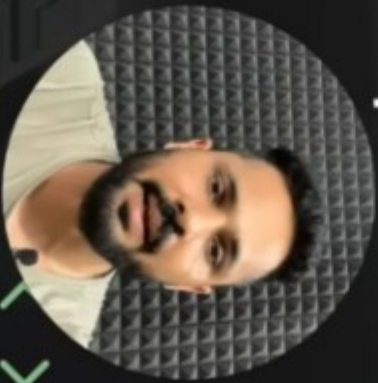
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Data Types in TypeScript



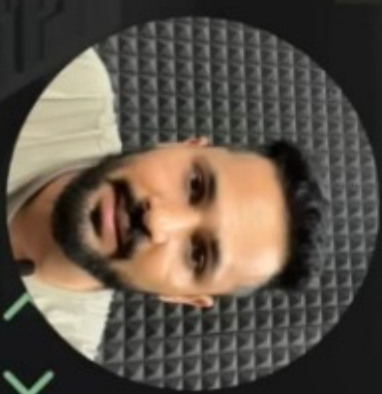
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What is Data Types



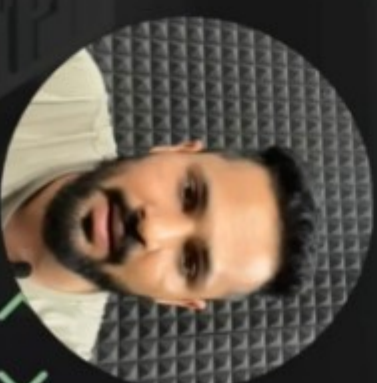
```
1 var userName:string="Anil sidhu
2 var age:number=30;
3 var male:boolean=true;
```





Categories

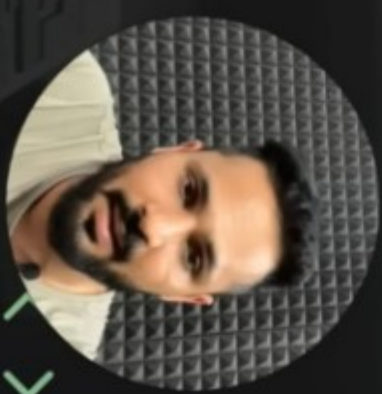
- Primitive Types
- Object Types
- Special Types
- Advanced Types
- Function Types

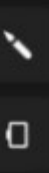




Primitive Types

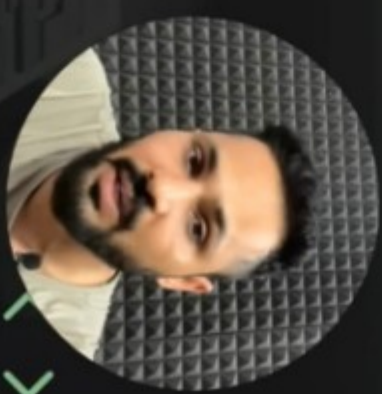
- number
- string
- boolean
- null
- undefined
- bigint
- symbol





Symbol

```
1 const sym1 = Symbol();  
2 const sym2 = Symbol('description');  
3 const sym3 = Symbol('description');  
4  
5 console.log(sym1 === sym2); // false  
6 console.log(sym2 === sym3); // false
```



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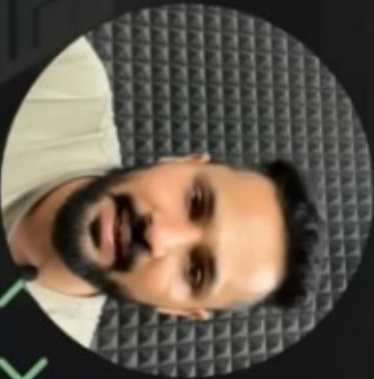
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Object Type

- Array
- Tuple
- Object

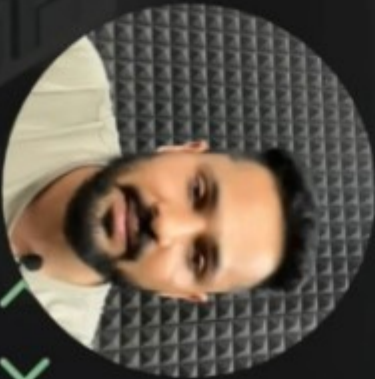






Special Types

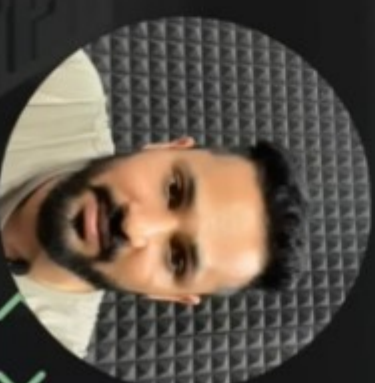
- any
- unknown
- void
- never





Advanced Types

- Union
- Intersection
- Type Alias
- Enum
- Literal Types



Edit: Code

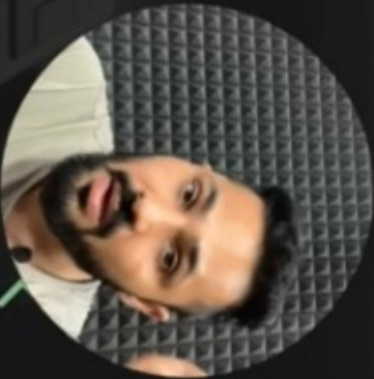
Typescript Data Types

https://slides.com/anilsidhu/code-182053/edit

Function Types

Defines the type of a function.

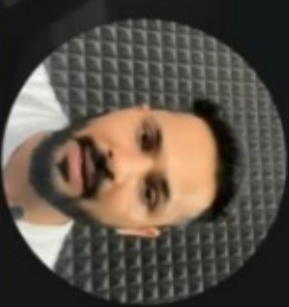






Number Data Type

- Apply number data type
- Redeclare issue
- How to Add number with data type
- How to use binary and hexadecimal numbers
- Convert string to number
- Type Inference with number
- Decimal with number data type



← →

typescript-tut

typescript-tut

...

TS numbers U X

TS numbers > ...

```
1 //var var_name : data_type = value
2 var num1:number=10;
3 var num2:number=30;
4
5 var total=num1+num2;
6 console.log(total);
7
```

JS numbers U X

JS numbers.js > ...

```
1 //var var_name : data_type = value
2 var num1 = 10;
3 var num2 = 30;
4 var total = num1 + num2;
5 console.log(total);
6
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

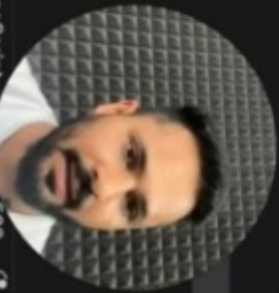
/dev/fd/13:18: command not found: compdef

anilsidhuganils-MacBook-Air typescript-tut % node number.js

40

anilsidhuganils-MacBook-Air typescript-tut %

Ln 6, Col 18 Spaces: 4 UTF-8 LF TypeScript 5500



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typescript-tut

TS numbers 3, U X

JS numbers U X

```
1 // var val_name : data_type = value
2 var num1:number=10;
3 var num2:number=30
4 var num3:number=30
5
6 var total:number=num1+num3;
7 console.log(total)
8
9 var oct:number=0b001;
10 var hexa:number=0x001;
11 var binary:number=00001;
12
13
```

var total:number
Octal literals are not allowed. Use the syntax '0o1'. ts(1121)
console.log(total)
View Problem (VS8) No quick fixes available
1

PROBLEMS 5

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER 2

number.ts:11:19 - error TS1121: Octal literals are not allowed. Use the syntax '0o1'.

11 var binary:number=00001;

[1:04:20 PM] Found 3 errors. Matching for file changes.

first-program

Launchpad

3

Ln 9, Col 17

Spaces: 4

UTF-8

LF

() TypeScript

5500

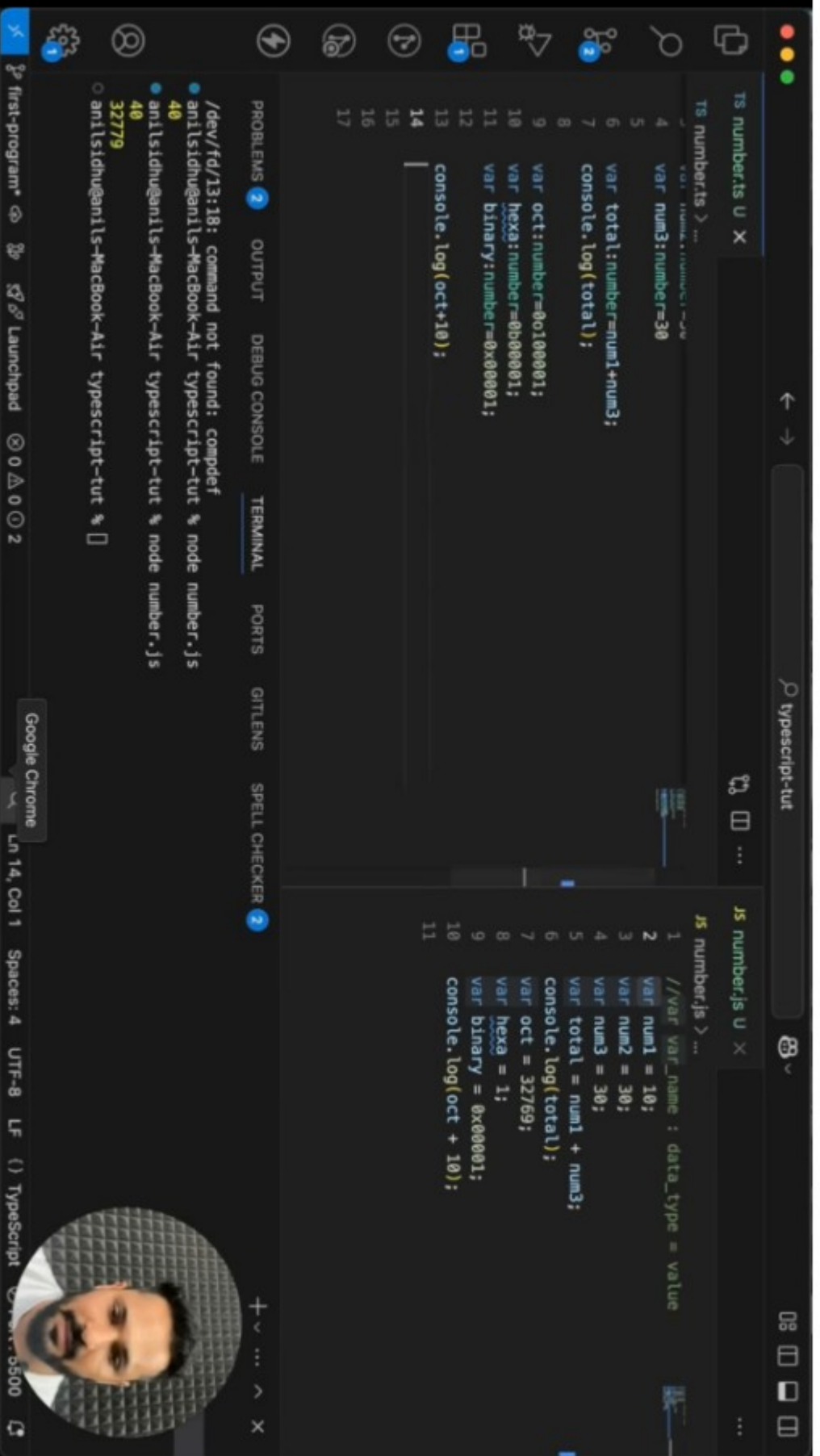
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...

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X

Subscribe



```

9  var oct:number=0o100001;
10 var hexa:number=0b00001;
11 var binary:number=0x00001;
12
13 console.log(oct+10);
14
15 var item:number=100;
16 var item2="50";
17 // var item2Converted=Number(item2)
18 var item2Converted+=item2
19 console.log(item+ +item2);
20
21
22
23
24

```

```

1 //var val_name : data_type = value
2 var num1 = 10;
3 var num2 = 30;
4 var num3 = 30;
5 var total = num1 + num3;
6 console.log(total);
7 var oct = 32769;
8 var hexa = 1;
9 var binary = 0x00001;
10 console.log(oct + 10);
11 var item = 100;
12 var item2 = "50";
13 // var item2Converted=Number(item2)
14 var item2Converted = +item2;
15 console.log(item + +item2);
16

```

- PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2
- anilsidhuganils-MacBook-Air typescript-tut & node number.js
150
40
32779
10050
 - anilsidhuganils-MacBook-Air typescript-tut & node number.js
40
32779
150
 - anilsidhuganils-MacBook-Air typescript-tut & node number.js
150



first-program launchpad 0 0 0 2



Ln 19, Col 21 Spaces: 4 UTF-8 LF () TypeScript 5500



Subscribe

TS numbers > ...

```
17 // 901 - AddContentOfItemAtEndOfList
18 var item2Converted+=item2
19 console.log(item+ 'item2Converted');
```

```
var data: number | string = 30
```

```
data="anil"
```

js number.js U X

```
js number.js > ...
```

```
2 var num1 = 10;
```

```
var total = num1 + num3;  
console.log(total);
```

PROBLEMS 2 OUTPUT

46
32779

10050

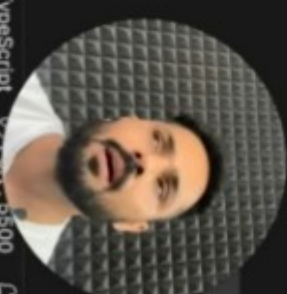
- anilsidhuganils-MacBook-Air typescript-tut % node number.js

46
32779

150

- anilsichuganils-MacBook-Air typescript-tut %
- first-program* ○  Launchpad ○ 0  0  0  2

Ln 28, Col 12 Spaces: 4 UTF-8 LF () TypeScript © 2018: 5500


 A red rectangular button with a white right-pointing arrow icon and the word "Subscribe" in white text.



String and Boolean

- Apply string data type
- Ways to define string
- Convert in string data type
- Apply boolean data type
- Possible boolean values
- Inference and declaration issues
- Run with html file



EXPLORER

OPEN EDITORS

TS string.ts

TYPESCRIPT-TUT

node_modules

app.js

app.ts

first.js

first.ts

number.js

number.ts

package-lock.json

package.json

string.ts

user.html

ts string.ts

ts string.ts

1

typescript-tut

typescript-tut

Ln 1, Col 1

Spaces: 4

UTF-8

LF

TypeScript

Port: 5500

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER

anilsidhu@anils-MacBook-Air: typescript-tut % tsc string.ts --watch

anilsidhu

X

← →typescript-tut

ts string.ts U X

ts string.ts > [0] info
1 var str:string="Hello How are you";
2 var str2:string='Hello How are you';
3 var str3:string="Hello How are you";
4
5 var age :number=30;
6 var userName:string="Anil sidhu";
7
8 var info:string=`my name is \${userName} and age is
9 \${age} years`
1

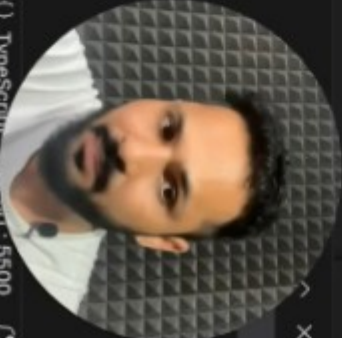
js string.js U X

js string.js > ...
1 var str = "Hello How are you";
2 var str2 = 'Hello How are you';
3 var str3 = "Hello How are you";
4 var age = 30;
5 var userName = "Anil sidhu";
6 var info = "my name is ".concat(userName, " and age
7

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[2:24:55 PM] File change detected. Starting incremental compilation...
[2:24:55 PM] Found 0 errors. Watching for file changes.
□

Ln 9, Col 14 Spaces: 4 UTF-8 LF () TypeScript Version: 5.5.0



ts string.ts U X

ts string.ts > ...
1 var str:string="Hello How are you";
2 var str2:string='Hello How are you';
3 var str3:string="Hello How are you";
4
5 var age :number=30;
6 var userName:string="Anil sidhu";
7
8 var info:string=`my name is \${userName} and age is
9 \${age} years`
10 console.log(info);
11

js string.js U X

js string.js > ...
1 var str = "Hello How are you";
2 var str2 = 'Hello How are you';
3 var str3 = "Hello How are you";
4 var age = 30;
5 var userName = "Anil sidhu";
6 var info = "my name is ".concat(userName, " and age
7 console.log(info);
8

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

- anilsidhuanils-MacBook-Air typescript-tut % node string.js
my name is Anil sidhu and age is 30 years
- anilsidhuanils-MacBook-Air typescript-tut %

Ln 11, Col 1 Spaces: 4 UTF-8 LF () TypeScript port: 5500



← → typescript-tut

ts string.ts U X

```
1 string.ts > ...
2 var str:string="Hello How are you";
3
4 var age :number=30;
5
6 var userName:string="Anil sidhu";
7
8 var info:string=`my name is ${userName} and age is `
9 console.log(info);
10
11 var num:number=100;
12 // var data:string=num.toString();
13 var data:string = "" + num
14
15
16
```

js string.js U X

```
1 string.js > ...
2 var str = "Hello How are you";
3 var str2 = 'Hello How are you';
4 var str3 = "Hello How are you";
5 var age = 30;
6 var userName = "Anil sidhu";
7 var info = "my name is ".concat(userName, " and age
8 console.log(info);
9 var num = 100;
10 // var data:string = num.toString();
11 var data = " " + num;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

- anilsidhuganils-MacBook-Air typescript-tut % node string.js
my name is Anil sidhu and age is
- anilsidhuganils-MacBook-Air typescript-tut % node string.js
my name is Anil sidhu and age is 30 years
- anilsidhuganils-MacBook-Air typescript-tut %

Ln 14, Col 1 Spaces: 4 UTF-8 LF () TypeScript Port: 5500



ts string.ts U X

ts string.ts > ...

10 var num:number=100;

11 // var data:string = num.toString();

12 // var data:string = " "+ num;

13 var boolData=true;

14 // var data:string = boolData.toString();

15 var data:string = "" +boolData;

16

17

18

19 var item:boolean=true;

20

js string.js U X

js string.js > ...

1 var str = "Hello How are you";

2 var str2 = 'Hello How are you';

3 var str3 = "Hello How are you";

4 var age = 30;

5 var userName = "Anil sidhu";

6 var info = "my name is ".concat(userName, " and age

7 console.log(info);

8 var num = 100;

9 // var data:string = num.toString();

10 // var data:string = " "+ num;

11 var boolData = true;

12 // var data:string = boolData.toString();

13 var data = "" + boolData;

14 var item = true;

15

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[2:28:59 PM] File change detected. Starting incremental compilation...

[2:28:59 PM] Found 0 errors. Watching for file changes.

first-program* Launchpad 0 0

Ln 19, Col 22 Spaces: 4 UTF-8 LF () TypeScript Server: 5500

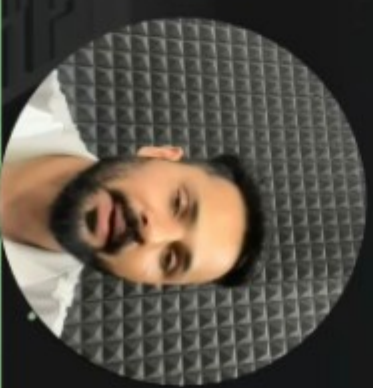


Subscribe



Null And Undefined

- What is Null Data Type
- How to use it
- What is Undefined Data Type
- How to use Undefined
- Possible Values
- Run with html file
- Interview Question



EXPLORED

OPEN EDITORS

GROUP 1

GROUP 2

JS null.js

TYPESCRIPT-TUT

node_modules

app.js

app.ts

first.js

first.ts

null.js

null.ts

number.js

numbers.ts

package-lock.json

package.json

string.js

string.ts

user.html

ts null.ts

ts null.ts

1

2

3

4

5

6

var data : null | string = null;

var userName: null | string = null;

var login = true;

if (login) {

userName = "Anil sidhu";

}

js null.js

js null.js

1

2

3

4

5

6

7

var data = null;

var userName = null;

var login = true;

if (login) {

userName = "Anil sidhu";

}

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

...

[3:05:06 PM] File change detected. Starting incremental compilation...

[3:05:06 PM] Found 0 errors. Watching for file changes.

first-program

Launchpad

0

0

0

Ln 6, Col 2

Spaces: 4

UTF-8

LF

TypeScript

Go Live

Subscribe

EXPLORER

OPEN EDITORS

GROUP 1

GROUP 2

JS null.js

TYPESCRIPT...

node_modules

undefine.

JS app.js

TS app.ts

JS first.js

TS first.ts

JS null.js

TS null.ts

JS number.js

TS number.ts

package-lock.json

package.json

JS string.js

TS string.ts

user.html

OUTLINE

TIMELINE

← →

typescript-tut

🔍

...

TS null.ts

U X

TS null.ts > ...

1 var data : null | string = null;

2 var username: null | string = null

3 var login = true

4 if (login) {

5 username = "Anil sidhu"

6 }

7

8 console.log(typeof data);

9

JS null.js

U X

JS null.js > ...

1 var data = null;

2 var username = null;

3 var login = true;

4 if (login) {

5 username = "Anil sidhu";

6 }

7

8 console.log(typeof data);

9

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER

/dev/fd/13:18: command not found: compdef

anilsidhu@anils-MacBook-Air typescript-tut % node null.js

object

anilsidhu@anils-MacBook-Air typescript-tut % node null.js

object

anilsidhu@anils-MacBook-Air typescript-tut % node null.js

string

anilsidhu@anils-MacBook-Air typescript-tut %

Ln 9, Col 1

Spaces: 4

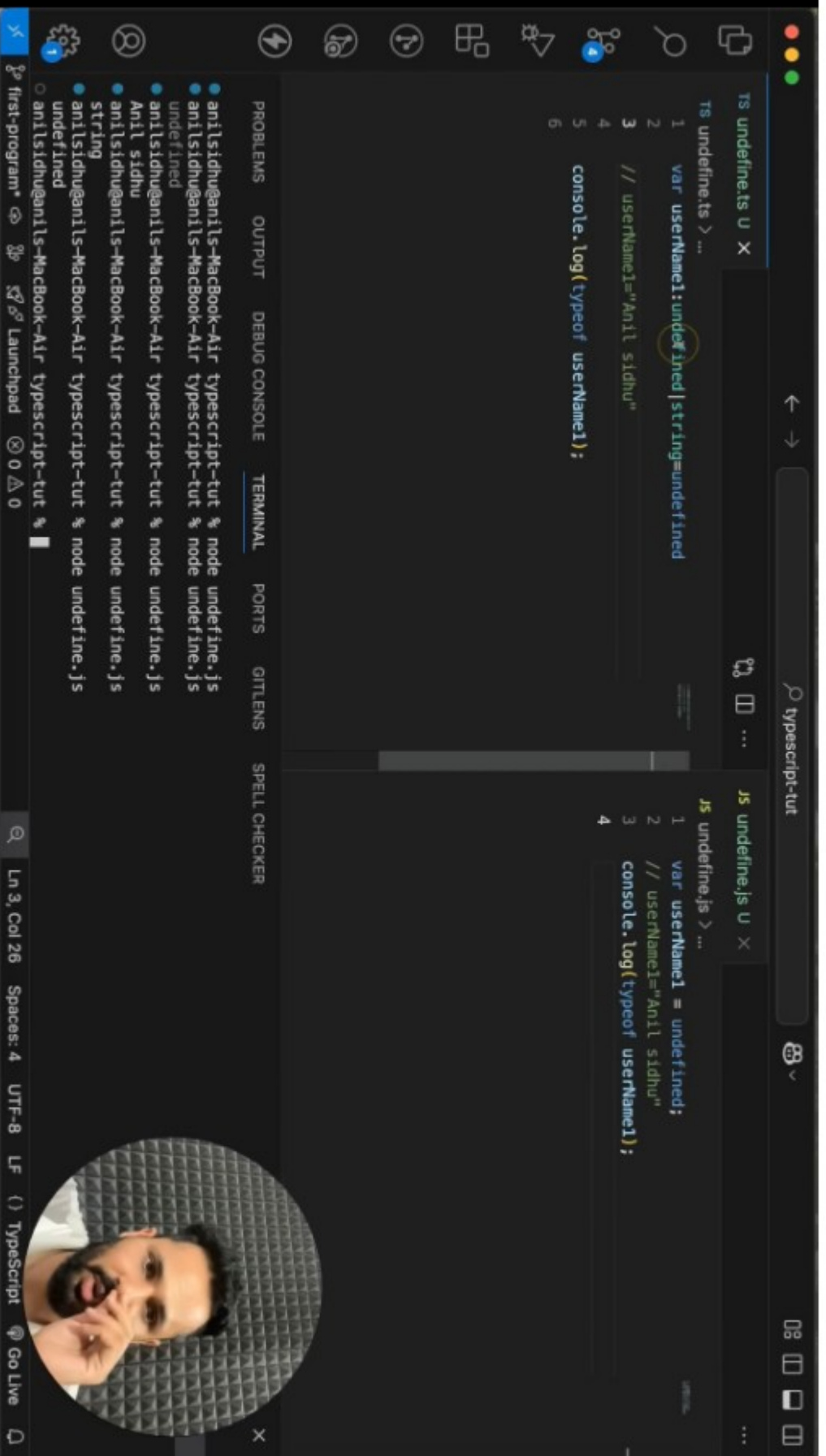
UTF-8

LF

()

TypeScript

Go Live



← →

typescript-tut

08

□

□

□

□

TS undefine.ts U X

TS undefine.ts > ...
1 var userName:undefined|string=undefined
2 var val=null
3
4 userName="Anil sidhu"
5
6 console.log(val==userName);
7

JS undefine.js U X

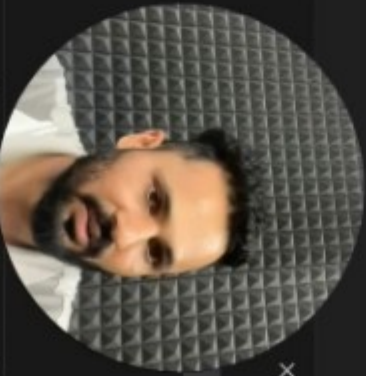
JS undefine.js > ...
1 var userName = undefined;
2 var val = null;
3 userName = "Anil sidhu";
4 console.log(val == userName);
5

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

undefined
anilsidhuganils-MacBook-Air typescript-tut % node undefine.js
Anil sidhu
anilsidhuganils-MacBook-Air typescript-tut % node undefine.js
string
anilsidhuganils-MacBook-Air typescript-tut % node undefine.js
undefined
anilsidhuganils-MacBook-Air typescript-tut % node undefine.js
false
anilsidhuganils-MacBook-Air typescript-tut %

first-program* Launchpad 0 0

Ln 6, Col 27 Spaces: 4 UTF-8 LF TypeScript Go Live

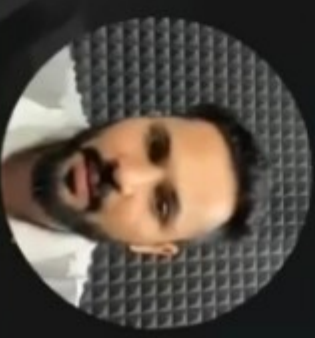


Subscribe



TypeScript Config File

- How to generate config file
- Use of config file
- Convert All TS file to js together
- Fix errors after config file
- Common use of config
- Interview Questions





The image is a screenshot of the Visual Studio Code (VS Code) editor interface. The main editor window displays a file named 'tsconfig.json' with the following content:

```
{
  "compilerOptions": {
    "disableSourceOfProjectReferenceRedirect": true,
    "disableSolutionSearching": true,
    "disableReferencedProjectLoad": true,

    /* Language and Environment */
    "target": "ES2024",
    // "lib": [],
    // "jsx": "preserve",
    // "libReplacement": true,
    // "experimentalDecorators": true,
    // "emitDecoratorMetadata": true,
    // "jsxFactory": "",
    // "jsxFragmentFactory": "",
    // "jsxImportSource": "",
    // "reactNamespace": "",
  }
}
```

The interface includes several panels: 'EXPLORER' on the left showing the file structure, 'OPEN EDITORS' at the top, and a 'Run and Debug' panel at the bottom. The 'Run and Debug' panel shows a terminal with the command 'tsc' being executed. The status bar at the bottom indicates the file is 'Ln 27, Col 1' and the encoding is 'UTF-8 LF'.

EXPLORED

OPEN EDITORS

tsconfig.json

ts null.ts

ts string.ts

ts number.ts

TYPESCRIPT-TUT

node_modules

output

app.js

app.ts

null.ts

number.js

number.ts

package-lock.json

package.json

string.js

string.ts

tsconfig.json

undefine.js

undefine.ts

TIMELINE

← →

typescript-tut

tsconfig.json

ts null.ts

ts string.ts

ts number.ts

tsconfig.json > {} compilerOptions

2 compilerOptions: {

54 /* Emit */

55 // "declaration": true,

56 // "declarationMap": true,

57 // "emitDeclarationOnly": true,

58 // "sourceMap": true,

59 // "inlineSourceMap": true,

60 // "noEmit": true,

61 // "outFile": "./",

62 "outDir": "./output",

63 // "removeComments": true,

64 // "importHelpers": true,

65 // "downlevelIteration": true,

66 // "sourceRoot": "",

67 // "mapRoot": "",

68 // "inlineSources": true,

69 // "emitBOM": true,

70 // "newline": "crlf",

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER

Ln 62, Col 30 Spaces: 2 UTF-8 LF {} JSON with Comments Go Live

/* Generate .d.ts files from TypeScript
/* Create source maps for d.ts files.
/* Only output d.ts files and not JavaS
/* Create source map files for emitted
/* Include source map files inside the
/* Disable emitting files from a compil
/* Specify a file that bundles all outp
/* Specify an output folder for all
/* Disable emitting comments.
/* Allow importing helper functions fr
/* Emit more compliant, but verbose an
/* Specify the root path for debuggers
/* Specify the location where debugger
/* Include source code in the sourcema
/* Emit a UTF-8 Byte Order Mark
/* Set the newline character



Subscribe

EXPLORED

OPEN EDITORS

tsconfig.json

ts tests

ts null.ts

ts string.ts

ts numbers.ts

TYPESCRIPT-TUT

js null.js

ts null.ts

js number.js

ts number.ts

package-lock.json

package.json

js string.js

ts string.ts

ts tests

tsconfig.json

js underfine.js

ts underfine.ts

user.html

OUTLINE

TIMELINE

← →

typescript-tut

tsconfig.json

ts tests

ts null.ts

ts string.ts

ts numbers.ts

tsconfig.json > {} compilerOptions

2 compilerOptions: {

50 // "allowJs": true,

51 // "checkJs": true,

52 // "maxNodeModuleJsDepth": 1,

53

54 /* Emit */

55 "declaration": true,

56 // "declarationMap": true,

57 // "emitDeclarationOnly": true,

58 // "sourceMap": true,

59 // "inlineSourceMap": true,

60 // "noEmit": true,

61 // "outFile": "./",

62 "outDir": "./output",

63 // "removeComments": true,

64 // "importHelpers": true,

65 // "downlevelIteration": true,

66 // "sourceRoot": "",

67 // "mapRoot": "",

outid

/* Enable error reporting in type-check

/* Specify the maximum folder depth use

/* Generate .d.ts files from TypeScript an

/* Create sourcemaps for d.ts files. */

/* Only output d.ts files and not JavaS

/* Create source map files for emitted

/* Include sourcemap files inside the e

/* Disable emitting files from a compil

/* Specify a file that bundles all outp

/* Specify an output folder for all

/* Disable emitting comments. */

/* Allow importing helper functions fro

/* Emit more compliant, but verbose

/* Specify the root path for

/* Specify the location whe

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

anilsidhuganils-MacBook-Air typescript-tut % tsc

anilsidhuganils-MacBook-Air typescript-tut % tsc

anilsidhuganils-MacBook-Air typescript-tut % tsc

anilsidhuganils-MacBook-Air typescript-tut % tsc

anilsidhuganils-MacBook-Air typescript-tut %

first-program

Launchpad

0 0 0 2

Ln 55, Col 27

Spaces: 2

UTF-8

LF

{}

JSON with Comments

Go Live

tsconfig.json U

ts string.d.ts out... U

ts tests U

ts null.ts M

ts string.ts M

ts numbers.ts M

node_modules

output

ts app.d.ts U

js app.js U

ts null.d.ts U

js null.js U

ts number.d.ts U

js number.js U

ts string.d.ts U

js string.js U

ts test.d.ts U

js test.js U

ts underline.d.ts U

tsconfig.json U

ts string.d.ts U X

ts tests U

ts null.ts M

ts string.ts M

ts number.ts M

output > ts string.d.ts > ...

1 declare var str: string;

2 declare var str2: string;

3 declare var str3: string;

4 declare var age: number;

5 declare var userName: string | null;

6 declare var info: string;

7 declare var num: number;

8 declare var boolData: boolean;

9 declare var str10: string;

10

PROBLEMS 2

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER 2

Ln 1, Col 1

Spaces: 4

UTF-8

LF

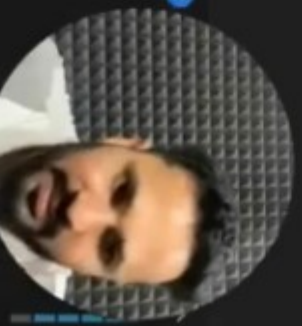
() TypeScript

Go Live

first-program

Launchpad

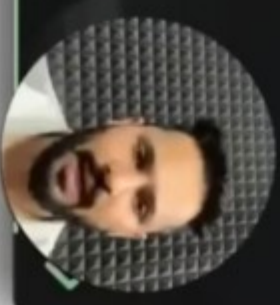
0 0 0 2





BigInt Data Type

- What is BigInt
- When we should use it
- Example
- Mixing issue
- Interview Questions





Why BigInt?

- In JavaScript and TypeScript, Number has a maximum safe integer
- `Number.MAX_SAFE_INTEGER = 253 - 1`
- 9007199254740991
- Beyond this limit, precision issues arise.

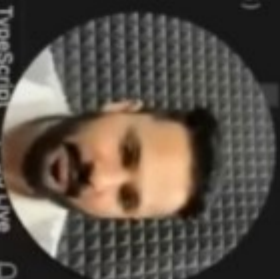


The screenshot shows the VS Code interface with the following details:

- Explorer Sidebar:** Shows a project structure with folders 'GROUP 1' and 'GROUP 2', and files 'TS bigint.ts', 'JS bigint.js', 'package-lock.json', 'package.json', 'string.js', 'OUTLINE', and 'TIMELINE'.
- Main Editor:** Displays the content of 'TS bigint.ts':


```
1 console.log("bigint example");
2
```
- Output Panel:** Shows a runtime error:


```
Error: Cannot find module \'/Users/amlslidhugan/Desktop/tutorials/typescript-tut/bigint.js\'
    at Function._resolveFilename (node:internal/modules/cjs/loader:1055:27)
    at Function._load (node:internal/modules/cjs/loader:1055:27)
    at TracingChannel.traceSync (node:diagnostics_channel:322:14)
    at wrapModuleLoad (node:internal/modules/cjs/loader:220:24)
    at Function.executeUserEntryPoint [as runMain] (node:internal/modules/run_main:170:5)
    at node:internal/main/run_main_module:36:49 {
  code: 'MODULE_NOT_FOUND',
  requireStack: []
}
```
- Status Bar:** Indicates the current file is 'Ln 2, Col 1'.



EXPLORE

OPEN EDITORS

GROUP 1

TS bigint.ts 2, U

GROUP 2

JS bigint.js output U

TYPESCRIPT-TUT

output

TS test.d.ts

JS test.js

TS undefine.d.ts

JS undefine.js

JS app.js

TS app.ts

TS bigint.ts 2, U

JS null.js

TS null.ts

JS number.js

TS number.ts

package-lock.json

package.json

string.js

OUTLINE

TIMELINE

← →

typescript-tut

TS bigint.ts 2, U X

1 var bignumber = 9007199254740991n

2 var x=1;

3 var y=2

4

5 console.log(bignumber+x);

6 console.log(bignumber+y);

7

JS bigint.js U X

output > JS bigint.js > ...

1 "use strict";

2 var bignumber = 9007199254740991n;

3 var x = 1;

4 var y = 2;

5 console.log(bignumber + x);

6 console.log(bignumber + y);

7

PROBLEMS 2

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut % node output/bigint.js

9007199254740992

anilsidhu@anils-MacBook-Air typescript-tut % node output/bigint.js

9007199254740992

anilsidhu@anils-MacBook-Air typescript-tut %

node

zsh

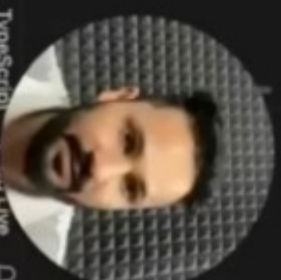
js bigint.js U X

```
output > js bigint.js > ...
```

```
1 use strict;
2 var bignumber = 9007199254740991n;
3 var x = 1;
4 var y = 2;
5 console.log(bignumber + x);
6 console.log(bignumber + y);
7
```

SPELL CHECKER

```
node
zsh
```



typescript-tut

ts bigint.ts U X

```
1  ts bigint.ts > {0} y
2  var bigInt=9007199254740991n
3  var x=1n;
4  var y=2n
5  console.log(bigInt+x);
6  console.log(bigInt+y);
7
```

js bigint.js U X

```
1  output > js bigint.js > ...
2  "use strict";
3  var bigInt = 9007199254740991n;
4  var x = 1n;
5  var y = 2n;
6  console.log(bigInt + x);
7  console.log(bigInt + y);
8
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut % node output/bigInt.js

9007199254740992

anilsidhu@anils-MacBook-Air typescript-tut % node output/bigInt.js

9007199254740992

anilsidhu@anils-MacBook-Air typescript-tut % node output/bigInt.js

9007199254740992

anilsidhu@anils-MacBook-Air typescript-tut %

Ln 3, Col 9

Spaces: 4

UTF-8

LF

() TypeScript

node

zsh



Edit: Code

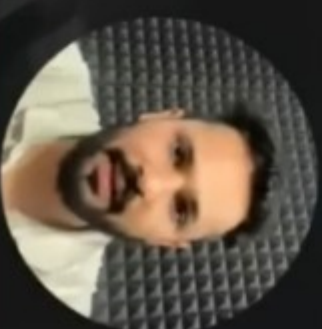


<https://slides.com/enlidh/code-182053/edit>



Symbol Data Type

- What is Symbol Data Type
- How to use it
- Example
- Where we can use it
- Interview Questions



EXPLORER

OPEN EDITORS

GROUP 1

GROUP 2

TYPESCRIPT-TUT

JS null.js

TS null.ts

JS numbers.js

TS numbers.ts

package-lock.json

package.json

string.js

TS string.ts

symbol.js

TS symbol.ts

test.ts

tsconfig.json

undefine.js

undefine.ts

user.html

ts symbol.ts

1 var sym= Symbol();

2 var sym2= Symbol();

3

4 var sym3=Symbol('abc');

5 var sym4=Symbol('abc');

6

7

8 console.log(sym==sym2);

9 console.log(sym);

10 console.log(sym2);

11

12

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js

false

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js

false

anils-MacBook-Air:typescript-tut anilsidhu\$

output > JS symbol.js

1 "use strict";

2 var sym = Symbol();

3 var sym2 = Symbol();

4 console.log(sym == sym2);

5 console.log(sym);

6 console.log(sym2);

7

node

node



Subscribe

EXPLORE

OPEN EDITORS

GROUP 1

GROUP 2

TYPESCRIPT-TUT

JS null.js

JS null.ts

JS number.js

TS number.ts

{} package-lock.json

{} package.json

JS string.js

TS string.ts

TS symbols

U

TS test.ts

tsconfig.json

JS undefine.js

TS undefine.ts

user.html

first-program

Launchpad

0 0 0

Ln 12, Col 23

Spaces: 4

UTF-8

LF

() TypeScript

Go Live

typescript-tut

terminal

ts symbols U X

ts symbols > ...

6

7

8

9

10

11

12

13

14

15

16

1

2

3

4

5

6

7

8

9

10

JS symbols U X

output > JS symbols > ...

1

2

3

4

5

6

7

8

9

10

node

node

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js

false

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js

false

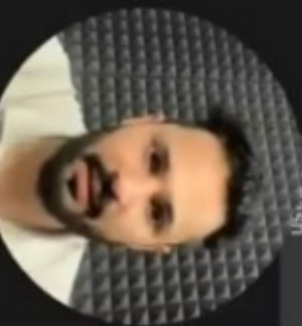
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js

Symbol()

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js

false

anils-MacBook-Air:typescript-tut anilsidhu\$




```

1  var sym= Symbol();
2  var sym2= Symbol();
3
4  var sym3=Symbol('abc');
5  var sym4= 'Symbol' only refers to a type, but is being used as a
6  value here. Do you need to change your target library?
7  Try changing the 'lib' compiler option to es2015 or
8  later. ts(2585)
9  console.l
10 console.l any
11
12 console.log(sym3==sym4)

```

output > JS symbol.js U X

```

1  "use strict";
2  var sym = Symbol();
3  var sym2 = Symbol();
4  var sym3 = Symbol('abc');
5  var sym4 = Symbol('abc');
6  console.log(sym3==sym2);
7  console.log(sym3);
8  console.log(sym4);
9  console.log(sym3 == sym4);

```

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 1

anils-MacBook-Air:typescript-tut anilsidnu\$ node symbol.ts

```

false
anils-MacBook-Air:typescript-tut anilsidnu$ node output/symbol.js
false
anils-MacBook-Air:typescript-tut anilsidnu$ node output/symbol.js
false
anils-MacBook-Air:typescript-tut anilsidnu$

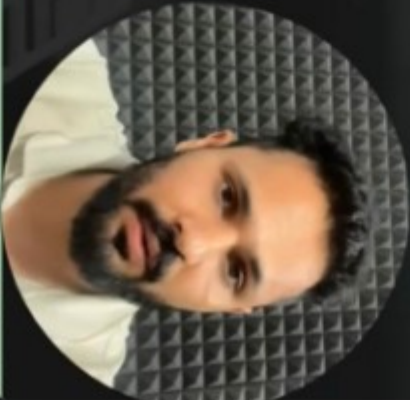
```


Edit: Code

https://slides.com/anilshih/code-182053/edit

TS

How to use TS
for Input fields



first-program Launchpad 0.0.0

Port: 5500

EXPLORED

OPEN EDITORS

inputs.html

inputs.ts

TYPESCRIPT-TUT

node_modules

output

app.js

app.ts

bigint.ts

inputs.html

inputs.ts

null.js

null.ts

number.js

number.ts

package-lock.json

package.json

string.js

string.ts

symbol.ts

test.ts

OUTLINE

TIMELINE

← → typescript-tut

inputs.html

inputs.ts

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Html Fields with TypeScript</title>
7 </head>
8 <body>
9   <input type="text" id="username" placeholder="enter user name">
10  <br>
11  <br>
12  <input type="text" id="email" placeholder="enter user email">
13  <br>
14  <br>
15  <input type="text" id="age" placeholder="enter user age">
16  <br>
17  <br>
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER

anilsidhuganils-MacBook-Air typescript-tut % tsc --watch

Ln 7, Col 8 Spaces: 4 UTF-8 LF () HTML Port: 5500

← → typescript-tut

inputs.html U X TS inputs.ts U

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Html Fields with TypeScript</title>
7   <script src="output/inputs.js"></script>
8 </head>
9 <body>
10  <input type="text" id="username" placeholder="enter user name">
11  <br>
12  <input type="text" id="email" placeholder="enter user email">
13  <br>
14  <input type="text" id="age" placeholder="enter user age">
15  <br>
16  <input type="text" id="age" placeholder="enter user age">
17  <br>
```

js inputs.js U X

output > js inputs.js

```
1 "use strict";
2
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[4:51:50 PM] Starting compilation in watch mode...

[4:51:50 PM] Found 0 errors. Watching for file changes.

Ln 7, Col 34 Spaces: 4 UTF-8 LF HTML Port: 5500



typescript-tut

inputs.html U

TS inputs.ts U X

function getInfo() {
 console.log("getInfo function called");
}

JS inputs.js U X

output > JS inputs.js > ...
1 "use strict";
2 function getInfo() {
3 console.log("getInfo fun
4 }
5

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

[4:52:46 PM] File change detected. Starting incremental compilation...

[4:52:46 PM] Found 0 errors. Watching for file changes.

Ln 2, Col 41

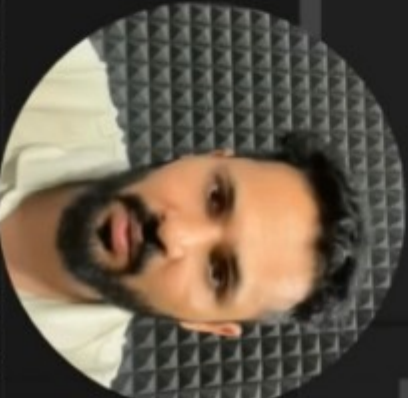
Spaces: 4

UTF-8

LF

() TypeScript

Port: 5500







← →typescript-tut

inputs.html U

TS inputs.ts U X

JS inputs.js U X

```
1 function getinfo(){
2   const nameInput= document.getElementById('username') as HTMLInput
3
4   const name:string=nameInput.value
5
6   const emailInput= document.getElementById('email') as HTMLInput;
7
8   const email:string=emailInput.value
9
10  const ageInput= document.getElementById('age') as HTMLInputElement;
11
12  const age:string=ageInput.value
13
14  console.log(name,email,age);
15
16
17 }
```

```
1 "use strict";
2 function getinfo() {
3   const nameInput = document.getEle
4   const name = nameInput.value;
5   const emailInput = document.getEl
6   const email = emailInput.value;
7   const ageInput = document.getElem
8   const age = nameInput.value;
9   console.log(name, email, age);
10 }
11
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

[4:57:04 PM] File change detected. Starting incremental compilation...

[4:57:04 PM] Found 0 errors. Watching for file changes.

□

first-program*

Launchpad

0 0 △ ▽

Ln 12, Col 30

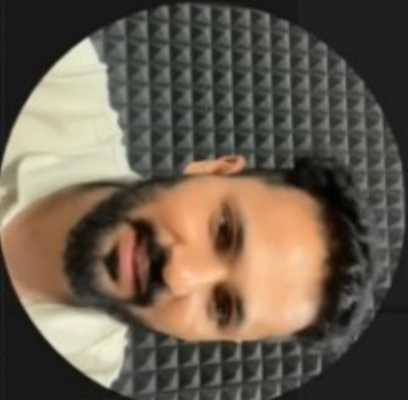
Spaces: 4

UTF-8

LF

() TypeScript

Port : 5500



← → typescript-tut

inputs.html U

TS inputs.ts U X

JS inputs.js U X

```
1 function getinfo() {
2   const nameInput = document.getElementById('username') as HTMLInput
3
4   const name:string=nameInput.value
5
6   const emailInput= document.getElementById('email') as HTMLInput
7
8   const email:string=emailInput.value
9
10  const ageInput= document.getElementById('age') as HTMLInputElement
11
12  const age:string=ageInput.value
13
14  console.log(name,email,age);
15
16
17 }
```

```
1 "use strict";
2 function getinfo() {
3   const nameInput = document.getEle
4   const name = nameInput.value;
5   const emailInput = document.getEl
6   const email = emailInput.value;
7   const ageInput = document.getElem
8   const age = ageInput.value;
9   console.log(name, email, age);
10 }
11
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

[4:57:07 PM] File change detected. Starting incremental compilation...

[4:57:07 PM] Found 0 errors. Watching for file changes.

□

first-program*

Launchpad

0 0 △ ▽

Ln 12, Col 30

Spaces: 4

UTF-8

LF

() TypeScript

Port : 5500

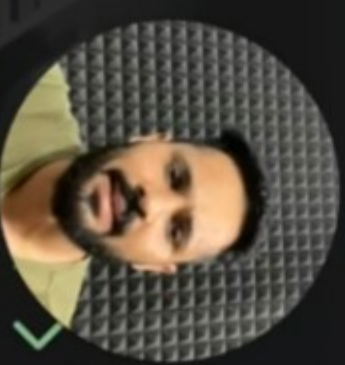






Array Data Type

- What is Array Data Type.
- How to use it.
- Example.
- Where we can use it.
- Interview Questions.

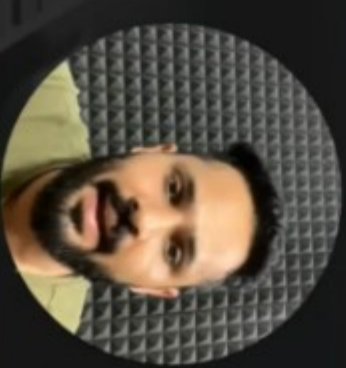




Array Data Type

- Collection of same Data type.
- Pre-defined data type.

```
1 let numbers: number[] = [1, 2, 3, 4, 5];  
2 let names: Array<string> = ["Alice", "Bob", "Charlie"];
```



← →typescript-tut

TS array.js U X

TS array.js > ...
1 var users:string[]=['anil','sam','peter'];
2 var marks:number[]=[50,60,92,36];
3 var students:Array<string>=['bruce','arya']
4 students.push('sidhu')
5 marks.push(100)
6 console.log(students);
7

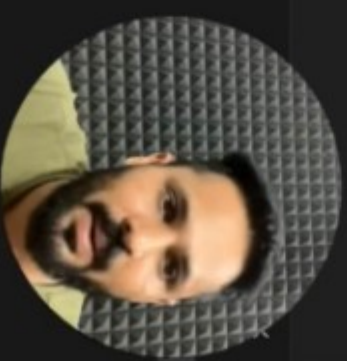
JS array.js U X

output > JS array.js > ...
1 "use strict";
2 var users = ['anil', 'sam', 'peter'];
3 var marks = [50, 60, 92, 36];
4 var students = ['bruce', 'arya'];
5 students.push('sidhu');
6 marks.push(100);
7 console.log(students);
8

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/array.js
 - anils-MacBook-Air:typescript-tut anilsidhu\$

Ln 5, Col 16 Spaces: 4 UTF-8 LF () TypeScript Port : 5500



← →typescript-tut

JS array.js U ×

output > JS array.js > ...
1 "use strict";
2 var users = ['anil', 'sam', 'peter'];
3 var marks = [50, 60, 92, 36];
4 var students = ['bruce', 'arya'];
5 students.push('sidhu');
6 marks.push(100);
7 var collegeName = 'IET alwar', 'IIT delhi';
8 console.log(students);
9

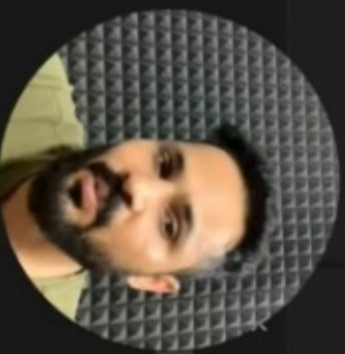
TS array.ts > ...
1 var users:string[]=['anil', 'sam', 'peter'];
2 var marks:number[]=[50,60,92,36];
3 var students:Array<string>=['bruce','arya'];
4 students.push('sidhu');
5 marks.push(100);
6
7 var collegeName:string='IET alwar';
8 console.log(students);
9
10

View Problem (⌘F8) No quick fixes available

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 4

- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/array.js
- anils-MacBook-Air:typescript-tut anilsidhu\$

Ln 8, Col 18 Spaces: 4 UTF-8 LF () TypeScript Port : 5500



← →typescript-tut

TS array.ts 1, 0

TS array.ts > ...

```
1 var users:string=['anil','sam','peter'];
2 var marks:number=[50,60,92,36];
3 var students:Array<string>=['bruce','arya']
4 students.push('sidhu')
5 marks.push(100)
6
7 var collegeName:ReadonlyArray<string>=['IET alwar','IIT delhi']
8 collegeName.push()
9 console.log(students);
10
```

JS array.js U X

output > JS array.js > ...

```
1 "use strict";
2 var users = ['anil','sam','peter'];
3 var marks = [50, 60, 92, 36];
4 var students = ['bruce','arya'];
5 students.push('sidhu');
6 marks.push(100);
7 var collegeName = ['IET alwar','IIT delhi'];
8 console.log(students);
9
```

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 4

- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/array.js
- anils-MacBook-Air:typescript-tut anilsidhu\$

Ln 8, Col 1 (18 selected) Spaces: 4 UTF-8 LF () TypeScript Port : 5500

Subscribe

Edit: Code

TypeScript Tutorial in Hindi 2

<https://slides.com/anilshih/code-182053/edit>



Tuple Data Type

- What is Tuple Data Type.
- Tuple VS Array
- Example.
- Where we can use it.
- Interview Questions.



Subscribe

- A **tuple** is a fixed-length,
- Ordered collection of **different types**.
- Each element in a tuple has a specific type.

```
1 let user: [number, string, boolean] = [1, "Anil Sidhu", true];
```



- A **tuple** is a fixed-length,
- Ordered collection of **different types**.
- Each element in a tuple has a specific type.

```
1 let user: [number, string, boolean] = [1, "Anil Sidhu", true];
```



← → typescript-tut

TS tuple.ts U X

```
1 u:string[]=['anil','sam','peter'];
2 idata:[string,number,boolean,number]='anil',30,true,100000];
3
4
5
```

JS tuple.js U X

```
output > JS tuple.js > [object] empData
1 "use strict";
2 var emp = ['anil','sam','peter'];
3 var empData = ['anil', 30, true, 100000];
4
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GTLENS SPELL CHECKER

[12:09:34 PM] File change detected. Starting incremental compilation...

[12:09:34 PM] Found 0 errors. Watching for file changes.

Ln 2, Col 42 Spaces: 4 UTF-8 LF TypeScript Go Live



```
1 use strict;
2 var emp = ['anil', 'sam', 'peter'];
3 var empata = ['anil', 30, true, 100000,];
4 empata.push('noida');
5
```

100



TS tuple, ts 1, U X

```
TS tuple.ts > [0] empData
```

```
1 var emp:string[]=['anil','sam','peter'];
2 var empdata:readonly[string,number,boolean,number,string]=['
3     🟡
4     empdata.push('noide');
5
6     console.log(empdata);
7
8
```

JS tuple.js U X

```
output > js tuple.js > ...
```

```
1 "use strict";
2 var emp = ['anil', 'sam', 'peter'];
3 var empData = ['anil', 30, true, 100000, ...];
4 empData.push('noids');
5 console.log(empData);
6
```

PROBLEMS 3

OUTPUT

DEBUG CONSOLE

PORTS

GIMLENS

SPELL CHECKER

12

```
anils-MacBook-Air:typescript-tut anils$dufs node output/tuple.js
```

```
[ 'all', 30, true, 100000, 'holds' ]
```

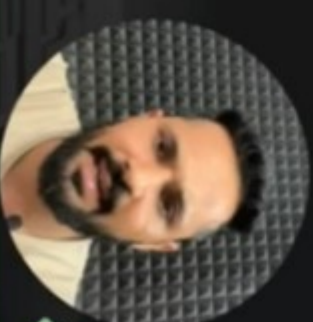
```
o anils-MacBook-Air:typescript-tut anils$ cd
```

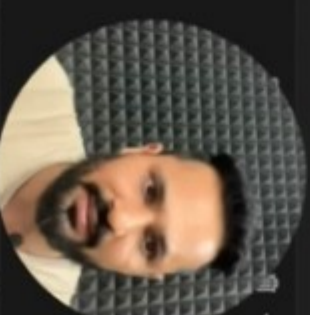
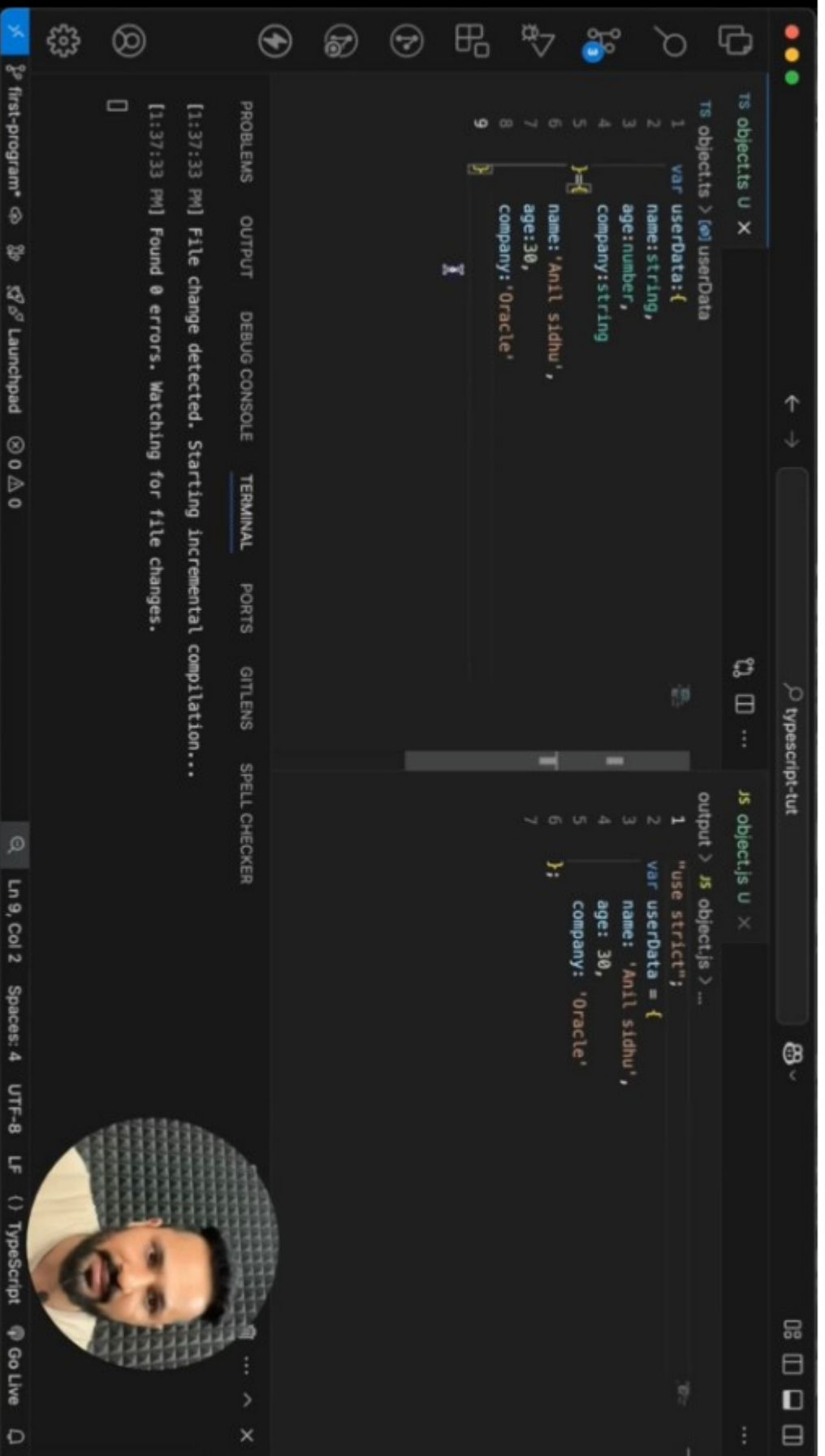




Object Data Type

- What is Object.
- Define data type in object
- Nested object
- Example.
- Interview Questions.





ts objects U X

ts objects > ...
11 // userData.age=25;
12
13 // console.log(userData);
14
15 var userData: {
16 name: string,
17 age: number,
18 company: string | undefined
19 }
20 name: 'Anil sidhu',
21 age: 30,
22 company: undefined
23 }
24 userData.company="HCL"
25

js objects U X

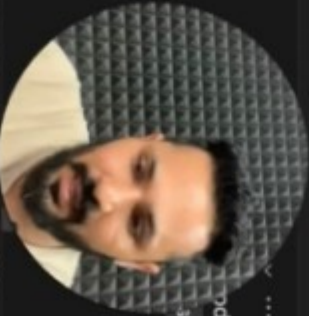
output > js objects > ...
1 "use strict";
2 // var userData: {
3 // name: string,
4 // age: number,
5 // company: string
6 // }=
7 // name: 'Anil sidhu',
8 // age: 30,
9 // company: 'Oracle'
10 // }
11 // userData.name="Sam";
12 // userData.age=25;
13 // console.log(userData);
14 var userData = {
15 name: 'Anil sidhu',
16 age: 30,
17 company: undefined

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

To update your account to use zsh, please run `chsh -s /bin/zsh`.
For more details, please visit <https://support.apple.com/kb/HT208050>.
• anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
 { name: 'Sam', age: 30, company: 'Oracle' }
• anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
 { name: 'Anil sidhu', age: 30, company: 'Oracle' }
• anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
 { name: 'Sam', age: 30, company: 'Oracle' }
• anils-MacBook-Air:typescript-tut anilsidhu\$

first-program Launchpad 0 0

Ln 24, Col 23 Spaces: 4 UTF-8 LF () TypeScript Go Live



ts objects U X

ts objects > ...

```
12 // console.log(userData);
13
14
15 var userData:{
16   [key:string]:string|number| undefined
17 }={
18   name:'Anil sidhu',
19   age:30,
20   company:undefined
21 }
22 userData.company="HCL";
23 userData.city="Moida"
24 console.log(userData);
```

js objects U X

output > js objects > ...

```
1 "use strict";
2 // var userData:{
3 //   name:string,
4 //   age:number,
5 //   company:string
6 // }={
7 //   name:'Anil sidhu',
8 //   age:30,
9 //   company:'Oracle'
10 // }
11 // userData.name="Sam";
12 // userData.age=25;
13 // console.log(userData);
14 var userData = {
15   name: 'Anil sidhu',
16   age: 30,
17   company: undefined
```

PROBLEMS 2

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER 2

- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
- { name: 'Sam', age: 30, company: 'Oracle' }
- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
- { name: 'Anil sidhu', age: 30, company: 'Oracle' }
- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
- { name: 'Sam', age: 30, company: 'Oracle' }
- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
- { name: 'Anil sidhu', age: 30, company: 'HCL' }
- anils-MacBook-Air:typescript-tut anilsidhu\$

first-program

Launchpad

0 0 0 2

Ln 22, Col 14

Spaces: 4

UTF-8

LF

()

TypeScript

Go Live

... ^ X

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← →typescript-tut

TS objects 1, U

TS objects > [0] userData

```
26 var userData:{
27   name:string,
28   age:number,
29   company:string,
30   address:
31     houseNo:string
32     sector:string
33     city:string
34   }
35   }={
36     name:'Anil sidhu',
37     age:30,
38     company:'Oracle',
39     address:{
40       houseNo:'house no 60',
41       sector:'70-a',
42       city:'gurgaon'
```

JS objects U

output > JS objects > ...

```
1 "use strict";
2 // var userData:{
3 //   name:string,
4 //   age:number,
5 //   company:string
6 // }={
7 //   name:'Anil sidhu',
8 //   age:30,
9 //   company:'Oracle'
10 // }
11 // userData.name="Sam";
12 // userData.age=25;
13 // console.log(userData);
14 // var userData:{
15 //   [key:string]:string|number| undefined
16 // }={
17 //   name:'Anil sidhu',
```

PROBLEMS 4

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER 4

- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
- { name: 'Sam', age: 30, company: 'Oracle' }
- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
- { name: 'Anil sidhu', age: 30, company: 'Oracle' }
- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
- { name: 'Sam', age: 30, company: 'Oracle' }
- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
- { name: 'Anil sidhu', age: 30, company: 'MCL' }
- anils-MacBook-Air:typescript-tut anilsidhu\$

first-program

Launchpad

0 0 0 4

Ln 33, Col 20

Spaces: 4

UTF-8

LF

()

TypeScript

Go Live

Subscribe

TS objects U X

ts objects > [0] userData
26 var userData:{
35 }={
36 name:'Anil sidhu',
37 age:30,
38 company:'Oracle',
39 address:{
40 houseNo:'house no 60',
41 sector:'70-a',
42 city:'gurgaon'
43 }
44 }

JS objects U X

output > JS objects > ...
1 "use strict";
2 // var userData:{
3 // name:string,
4 // age:number,
5 // company:string
6 // }=
7 // name:'Anil sidhu',
8 // age:30,
9 // company:'Oracle'
10 // }
11 // userData.name="Sam";
12 // userData.age=25;
13 // console.log(userData);
14 // var userData:{
15 // [key:string]:string|number|undefined
16 // }=
17 // name:'Anil sidhu',

PROBLEMS 4

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER 4

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Sam', age: 30, company: 'Oracle' }
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Anil sidhu', age: 30, company: 'Oracle' }
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Sam', age: 30, company: 'Oracle' }
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Anil sidhu', age: 30, company: 'MCL' }
anils-MacBook-Air:typescript-tut anilsidhu\$

first-program Launchpad 0 0 0 4

Google Chrome

Ln 33, Col 20 Spaces: 4 UTF-8 LF () TypeScript Go Live

ts objects U X

JS objects U X

output > JS objects > ...

1 "use strict";
2 // var userData:{
3 // name:string,
4 // age:number,
5 // company:string
6 // }=
7 // name:'Anil sidhu',
8 // age:30,
9 // company:'Oracle'
10 // }
11 // userData.name="Sam";
12 // userData.age=25;
13 // console.log(userData);
14 // var userData:{
15 // [key:string]:string|number|undefined
16 // }=
17 // name:'Anil sidhu',

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Sam', age: 30, company: 'Oracle' }
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Anil sidhu', age: 30, company: 'Oracle' }
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Sam', age: 30, company: 'Oracle' }
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Anil sidhu', age: 30, company: 'MCL' }
anils-MacBook-Air:typescript-tut anilsidhu\$

first-program Launchpad 0 0 0 4

Google Chrome

Ln 33, Col 20 Spaces: 4 UTF-8 LF () TypeScript Go Live

ts objects U X

JS objects U X

output > JS objects > ...

1 "use strict";
2 // var userData:{
3 // name:string,
4 // age:number,
5 // company:string
6 // }=
7 // name:'Anil sidhu',
8 // age:30,
9 // company:'Oracle'
10 // }
11 // userData.name="Sam";
12 // userData.age=25;
13 // console.log(userData);
14 // var userData:{
15 // [key:string]:string|number|undefined
16 // }=
17 // name:'Anil sidhu',

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Sam', age: 30, company: 'Oracle' }
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Anil sidhu', age: 30, company: 'Oracle' }
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Sam', age: 30, company: 'Oracle' }
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Anil sidhu', age: 30, company: 'MCL' }
anils-MacBook-Air:typescript-tut anilsidhu\$

first-program Launchpad 0 0 0 4

Google Chrome

Ln 33, Col 20 Spaces: 4 UTF-8 LF () TypeScript Go Live

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Edit: Code

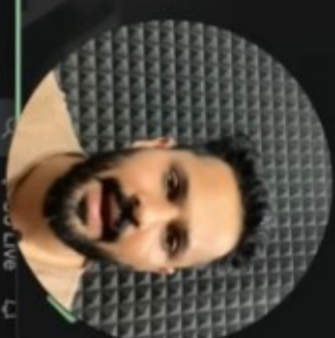
TypeScript Tutorial in Hindi 2

<https://slides.com/anilsidhu/code-182059/edit>



Any and Unknown

- What is any data type.
- When to use any.
- Example
- What is unknown data type
- Example of unknown.
- Interview Questions.

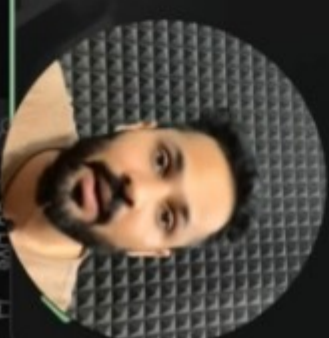


Subscribe



Any Data type

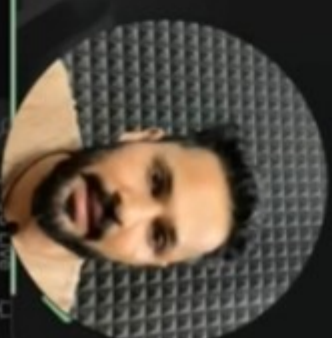
- Which allows a variable to hold values of any type.
- If you are not sure about data type of variable you can use any





Any Data Type Use

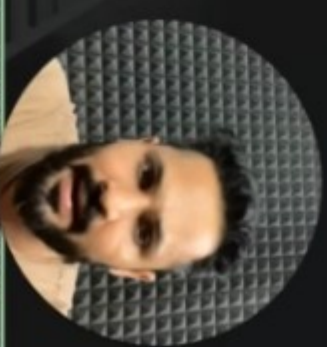
- When migrating JavaScript code to TypeScript.
- When dealing with dynamic values from APIs.
- When working with third-party libraries that lack type definitions.





Unknown Data type

- The unknown type in TypeScript is similar to any, but it's safer because it forces type checking before performing operations on the value.



TS any_unknown.ts U X

ts any_unknown.ts > ...

```
9
10 var value1:unknown = "anil";
11 value1=100;
12
13 value1={'anil'};
14 value1={};
15 value1="Anil sidhu"
16
17 if(typeof value1==='string'){
18
19 console.log(value1.toUpperCase());
20 }
```

JS any_unknown.js U X

output > JS any_unknown.js > ...

```
1 "use strict";
2 var value = "Anil";
3 value = 100;
4 value = {'anil'};
5 value = true;
6 value = {};
7 var value1 = "anil";
8 value1 = 100;
9 value1 = {'anil'};
10 value1 = {};
11 value1 = "Anil sidhu";
12 if (typeof value1 === 'string') {
13 console.log(value1.toUpperCase());
14 }
15 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[11:02:57 PM] File change detected. Starting incremental compilation...

[11:02:57 PM] Found 0 errors. Watching for file changes.

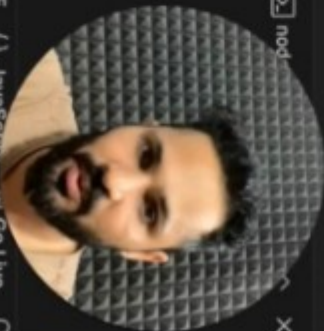
first-program

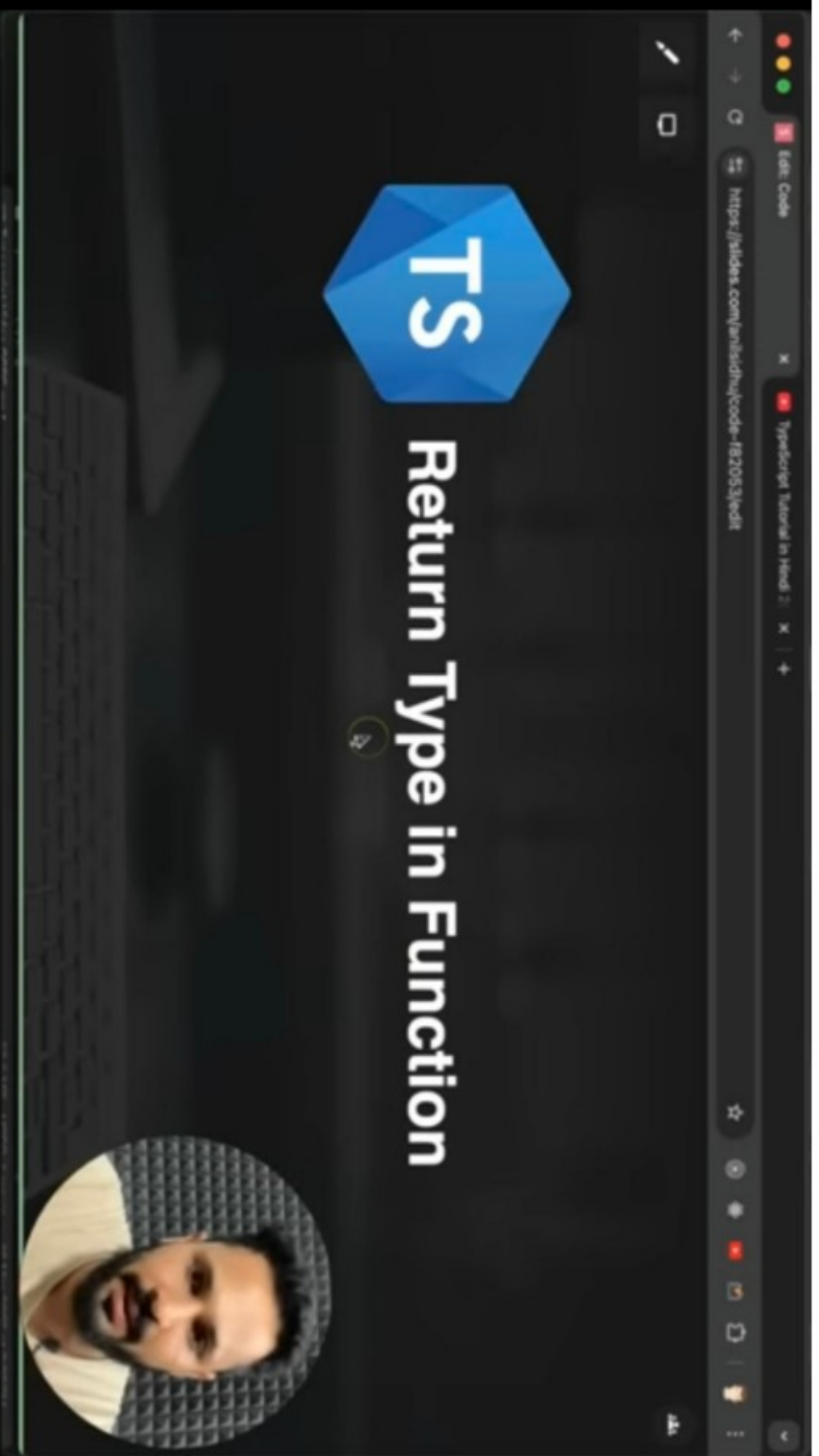
Launchpad

0 0

Ln 14, Col 2 Spaces: 4 UTF-8 LF {} JavaScript Go Live

mod





ts function.js U X

ts function.js > function fruits(): string

1 function fruits() {

2 return "Apple"

3 }

js function.js U X

output > js function.js > --

1 "use strict";

2 function fruits() {

3 return "Apple";

4 }

5

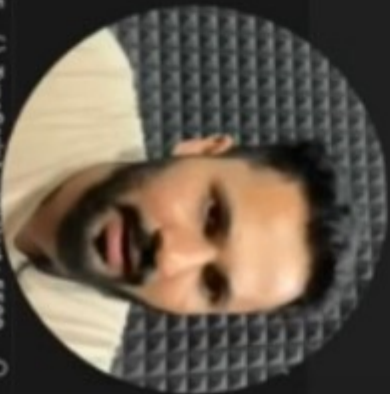
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS OUTPUTS SPELL CHECKER

[2:00:11 AM] File change detected. Starting incremental compilation...

[2:00:11 AM] Found 0 errors. Watching for file changes.

first-program launchpad 0.0.0

Ln 2, Col 19 Spaces: 4 UTF-8 LF () TypeScript Port: 5500



ts function.js U

ts function.js > simple

```
1 function fruits():number{
2   return 10
3 }
4
5 function simple():boolean{
6   return true
7 }
8
9
```

ts function.js X

output > ts function.js > --

```
1 "use strict";
2 function fruits() {
3   return 10;
4 }
5 function simple() {
6   return true;
7 }
8
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS OUTLENS SPELL CHECKER

[2:01:55 AM] File change detected. Starting incremental compilation...

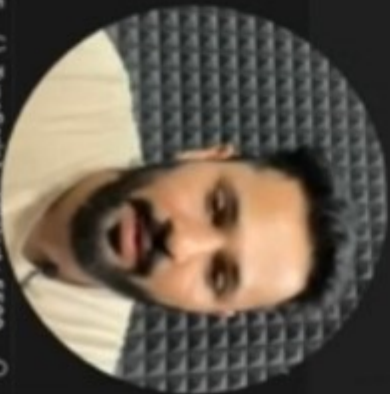
[2:01:55 AM] Found 0 errors. Watching for file changes.

first-program

launchpad

0 0 0

Ln 9, Col 1 Spaces: 4 UTF-8 LF () TypeScript Port: 5500



ts function.js U X

ts function.js > complex
5 function simple():boolean{
6 return true
7 }
8
9 function complex():number|string|boolean{
10 let data=10;
11 let name="anil sidhu"
12 let type="age";
13
14 if(type=="age"){
15 return data;
16 }else{
17 return name
18 }
19 }

js function.js U X

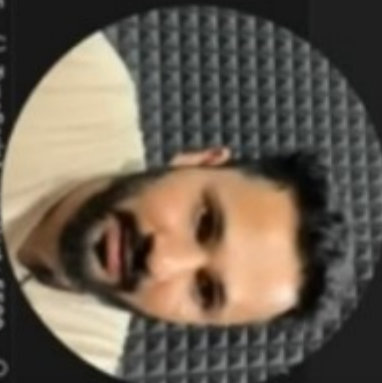
output > js function.js > ...
1 "use strict";
2 function fruits() {
3 return 10;
4 }
5 function simple() {
6 return true;
7 }
8 function complex() {
9 let data = 10;
10 let name = "anil sidhu";
11 let type = "age";
12 if (type == "age") {
13 return data;
14 }
15 else {
16 return name;
17 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QTLENIS SPELL CHECKER

[2:03:51 AM] File change detected. Starting incremental compilation...
[2:03:51 AM] Found 0 errors. Watching for file changes.
□

first-program launched 0 0 0

Ln 19, Col 2 Spaces: 4 UTF-8 LF () TypeScript Port: 5500



← → ↺

Edit: Code

× TypeScript Tutorial in Hindi 1 × +

https://slides.com/janishidhu/code-182053/edit

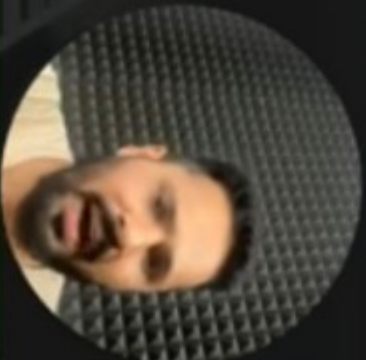
☆ 🔍 📄 🗑️ 📌 📁 📂 📅 📆 📇 📈 📉 📊 📋 📌 📍 📎 📏 📐 📑 📒 📓 📔 📕 📖 📗 📘 📙 📚 📛 📜 📝 📞 📟 📠 📡 📢 📣 📤 📥 📦 📧 📨 📩 📪 📫 📬 📭 📮 📯 📰 📱 📲 📳 📴 📵 📶 📷 📸 📹 📺 📻 📼 📽 📾 📿 📠 📡 📢 📣 📤 📥 📦 📧 📨 📩 📪 📫 📬 📭 📮 📯 📰 📱 📲 📳 📴 📵 📶 📷 📸 📹 📺 📻 📼 📽 📾 📿

⌂ 🔍 📄 🗑️ 📌 📁 📂 📅 📆 📇 📈 📉 📊 📋 📌 📍 📎 📏 📐 📑 📒 📓 📔 📕 📖 📗 📘 📙 📚 📛 📜 📝 📞 📟 📠 📡 📢 📣 📤 📥 📦 📧 📨 📩 📪 📫 📬 📭 📮 📯 📰 📱 📲 📳 📴 📵 📶 📷 📸 📹 📺 📻 📼 📽 📾 📿



Never in TypeScript

🔗



EXPLORER

...

OPEN EDITORS

TYPESCRIPT-TUT

inputs.html

TS inputs.ts

TS never.ts

JS null.js

TS null.ts

JS number.js

TS number.ts

TS objects

package-lock.json

package.json

JS string.js

TS string.ts

TS symbols

TS test.ts

tsconfig.json

TS tuple.ts

JS underfine.js

TS underfine.ts

user.html

OUTLINE

TIMELINE

first-program

launchpad

0 0 0

typescript-tut

typescript-tut

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

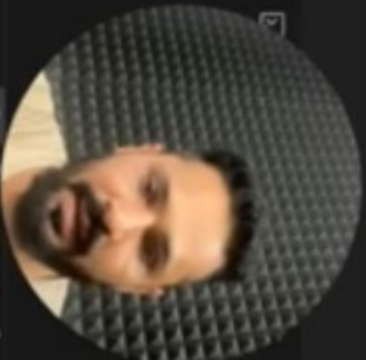
SPELL CHECKER

/dev/fd/13:18: command not found: compilef

anilsidhu@nills-MacBook-Air: typescript-tut % touch never.ts

anilsidhu@nills-MacBook-Air: typescript-tut % tsc --watch

Port: 5500



EXPLORER

OPEN EDITORS

GROUP 1

GROUP 2

TS never.ts

JS never.js output

U

TYPESCRIPT-TUT

JS app.js

TS app.ts

TS arrays.ts

TS bigint.ts

TS function.ts

TS inputs.html

TS inputs.ts

TS never.ts

TS null.js

TS null.ts

JS number.js

TS number.ts

TS objects

TS package-lock.json

TS package.json

OUTLINE

TIMELINE

TS never.ts

1

2

3

4

5

6

7

8

9

10

11

function loopFunction():never{

while (true) {

console.log("loop")

}

}

function simple():never{

console.log("simple")

}

JS never.js

output > JS never.js > ...

1

2

3

4

5

6

7

8

9

10

"use strict";

function loopFunction() {

while (true) {

console.log("loop");

}

}

function simple() {

console.log("simple");

}

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

...

never.ts:9:21 - error TS2534: A function returning 'never' cannot have a reachable function simple():never {

[12:28:25 PM] Found 1 error. Watching for file changes.

first-program

launchpad

1

Ln 9, Col 26 (5 selected)

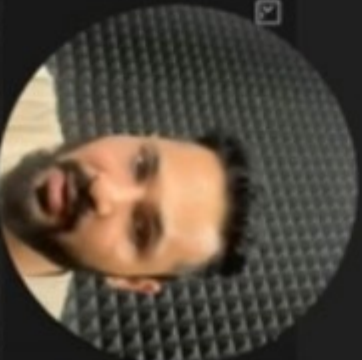
Spaces: 4

UTF-8

LF

TypeScript

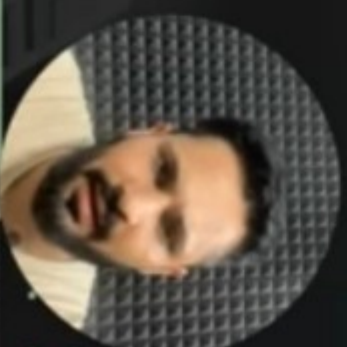
Port: 5500





Union Types

- What is Union Types.
- Use union type with variable data type.
- Use union type with function return type.
- Use union type with function params type.
- Type check with Union type



← → typescript-tut

08 11 2023 14:00

TS union.ts 1, U

```
1  var studentData:string| number | []="Anil";
2
3  studentData=9999;
4
5  studentData=[]
6
7  function fruitsData():string {
8      var item=1
9      if(item>1){
10         return ['apple','banana']
11     }else{
12         return "A"
13     }
14     console.log(customElements)
15 }
```

JS union.js U X

```
1  "use strict";
2  var studentData = "Anil";
3  studentData = 9999;
4  studentData = [];
5  function fruitsData() {
6      var item = 1;
7      if (item > 1) {
8          return ['apple', 'banana'];
9      }
10     return "Anil";
11 }
```

output > JS union.js > ...

```
1  "use strict";
2  var studentData = "Anil";
3  studentData = 9999;
4  studentData = [];
5  function fruitsData() {
6      var item = 1;
7      if (item > 1) {
8          return ['apple', 'banana'];
9      }
10     return "Anil";
11 }
```

console.log(fruitsData());

PROBLEMS 1

- log ElementInternals
- log MediaElementAudioSourceNode
- addEventListeners
- CustomElementRegistry
- removeEventListener

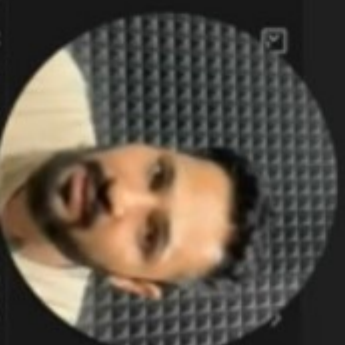
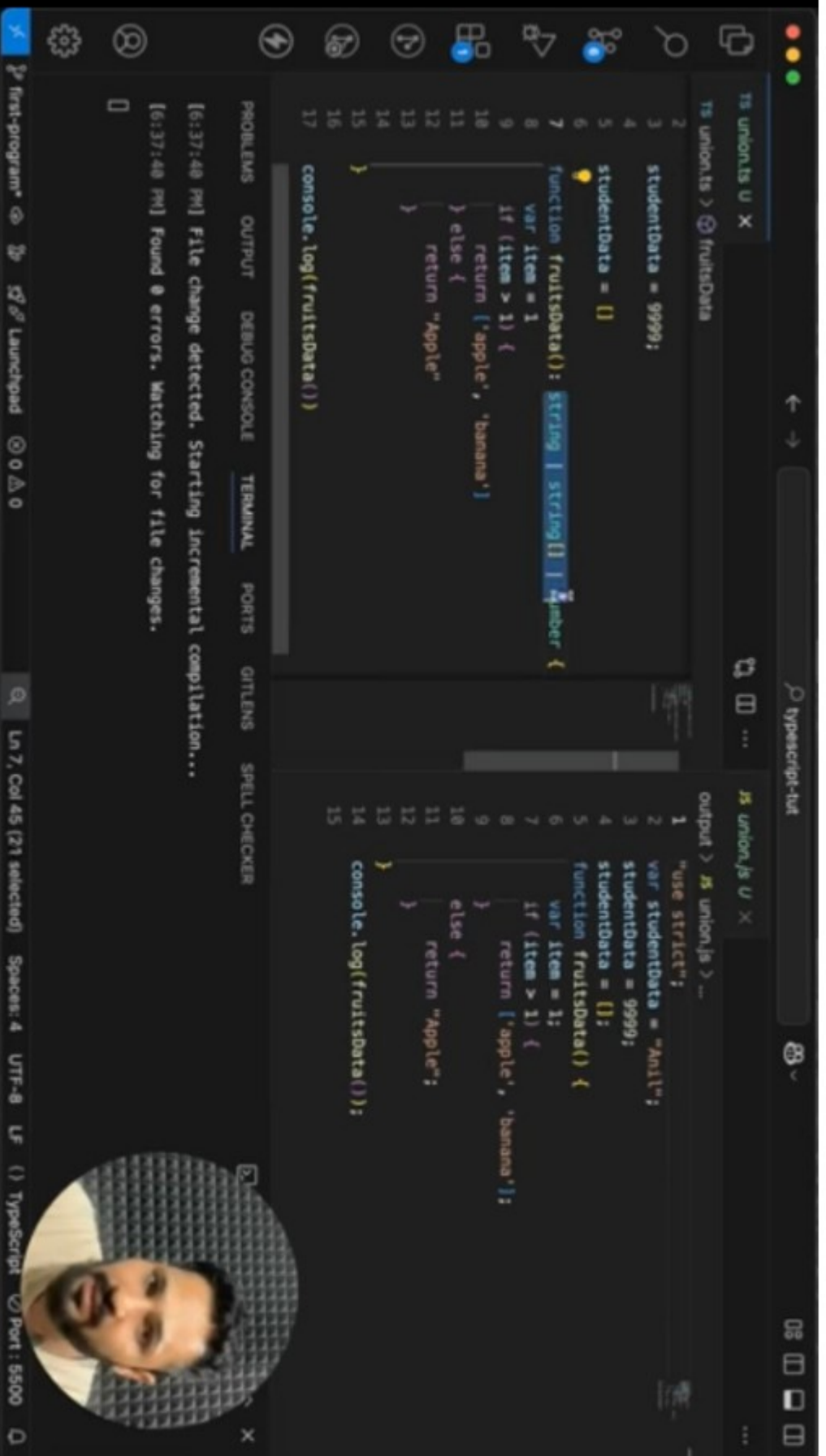
6:37:07 PM] File change detected. Starting incremental compilation...

union.ts:10:9 - error TS2322: Type 'string[]' is not assignable to type 'string'.

return ['apple','banana']

6:37:07 PM] Found 1 error. Watching for file changes.

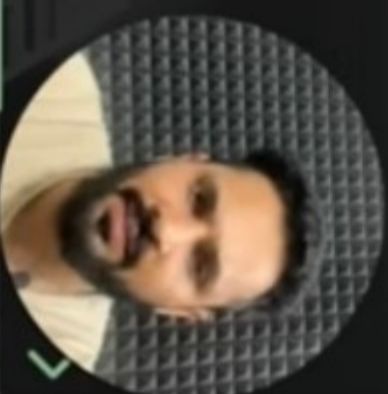
Ln 11, Col 9 Spaces: 4 UTF-8 LF () TypeScript Port: 5500

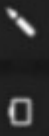




Interface

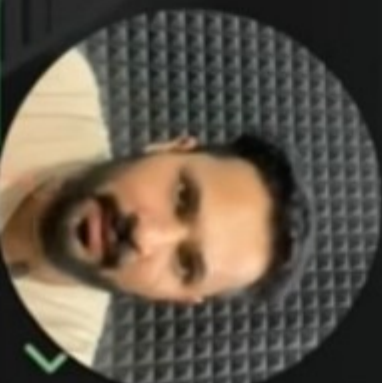
- What is Interface.
- Define Interface
- How to use Interface
- Extend Interface





Interface

- An **interface** is a way to define the structure of an object, function etc.



typescript-tut

ts interfaces U

```
1 interface Info {
2   name:string,
3   age:number,
4   college:string
5 }
6
7 var studentObj:Info={
8   name:'anil',
9   age:30,
10  college:'IET alwar'
11 }
12
```

js interfaces U

```
1 "use strict";
2 var studentObj={
3   name:'anil',
4   age:30,
5   college:'IET alwar'
6 };
7
```

PROBLEMS 2

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

OUTLENS

SPELL CHECKER 2

[3:29:52 AM] File change detected. Starting incremental compilation...

[3:29:52 AM] Found 0 errors. Watching for file changes.

Ln 12, Col 1

Spaces: 4

UTF-8

LF

TypeScript

Port: 8500

typescript-tut

TS interfaces U X

```
1 interfaces > [es] management
2 var studentObj:Info = {
3   name: 'anil',
4   age: 30,
5   college: 'IET alwar'
6 }
7
8 var teacherObj:Info = {
9   name: 'Sam',
10  age: 40,
11  college: 'IET alwar'
12 }
13
14 var management:Info = {
15   name: 'Sam',
16   age: 40,
17   college: 'IET alwar'
18 }
19
20
21
22
```

JS interfaces U X

```
1 output > JS interfaces > ...
2 "use strict";
3 var studentObj = {
4   name: 'anil',
5   age: 30,
6   college: 'IET alwar'
7 };
8
9 var teacherObj = {
10  name: 'Sam',
11  age: 40,
12  college: 'IET alwar'
13 };
14
15 var management = {
16   name: 'Sam',
17   age: 40,
18   college: 'IET alwar'
19 };
20
21
22
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

EXTENSIONS

SPELL CHECKER

[3:30:52 AM] File change detected. Starting incremental compilation...

[3:30:52 AM] Found 0 errors. Watching for file changes.

first-program

VS Code

0.0.0.0

Ln 22, Col 2

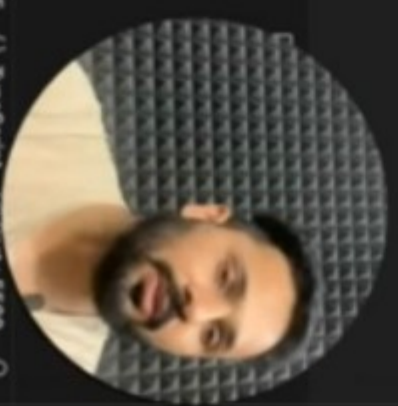
Spaces: 4

UTF-8

LF

TypeScript

Port: 8500



ts interfaces U X

ts interfaces > + TeacherType > subject

```
6 interface TeacherType extends Info {
7     subject:string
8 }
9
10 var studentObj:Info = {
11     name:'anil',
12     age:30,
13     college:'IET alwar'
14 }
15
16 var teacherObj:TeacherType={
17     name:'Sam',
18     age:40,
19     college:'IET alwar',
20     subject:'Math'
21 }
22
```

output > ts interfaces > ...

```
1 "use strict";
2 var studentObj = {
3     name: 'anil',
4     age: 30,
5     college: 'IET alwar'
6 };
7 var teacherObj = {
8     name: 'Sam',
9     age: 40,
10    college: 'IET alwar',
11    subject: 'Math'
12 };
13 var management = {
14    name: 'Sam',
15    age: 40,
16    college: 'IET alwar'
17 };

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS OUTLENS SPELL CHECKER

[3:32:08 AM] File change detected. Starting incremental compilation...

[3:32:08 AM] Found 0 errors. Watching for file changes.

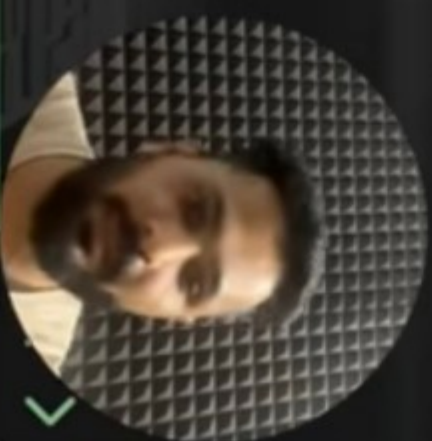
first-program launched 0 0 0 0 6 Ln 8, Col 12 (7 selected) Spaces: 4 UTF-8 LF TypeScript Port: 8500





Intersection Types in TypeScript

- What is Intersection Types.
- How to use it with type
- How to use it with interface
- Interview Questions



← → 🔍 Edit: Code

https://slides.com/jenishdhy/code-182053/edit

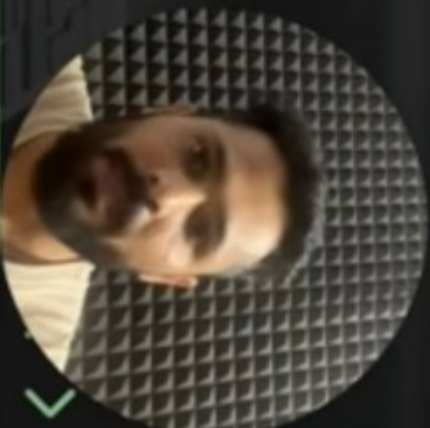
× TypeScript Tutorial in Hindi 1 × +

☆ 🔍 📄 📁 📌 📎 📏 📐 📑 📒 📓 📔 📕 📖 📗 📘 📙 📚 📛 📜 📝 📞 📟 📠 📡 📢 📣 📤 📥 📦 📧 📨 📩 📪 📫 📬 📭 📮 📯 📰 📱 📲 📳 📴 📵 📶 📷 📸 📹 📺 📻 📼 📽 📾 📿 📠 📡 📢 📣 📤 📥 📦 📧 📨 📩 📪 📫 📬 📭 📮 📯 📰 📱 📲 📳 📴 📵 📶 📷 📸 📹 📺 📻 📼 📿

⌂



- Intersection types allow you to Combine multiple types into one



✓

typescript-tut

TS Intersection: U

```
1 type personA={name:string}
2 type personB={age:number}
3 type personC=personA & personB
4
5
6
7 var PersonDataA:personA={name:'Anil sidhu'}
8 var PersonDataB:personB={age:30}
9
10
11 var PersonDataC : personC=
12 {name:'Sidhu',age:30}
```

JS Intersection: JS U

```
1 "use strict";
2 var PersonDataA = { name: 'Anil sidhu' };
3 var PersonDataB = { age: 30 };
4 var PersonDataC = { name: 'Sidhu', age: 30 };
5
```

output > JS Intersection: JS U

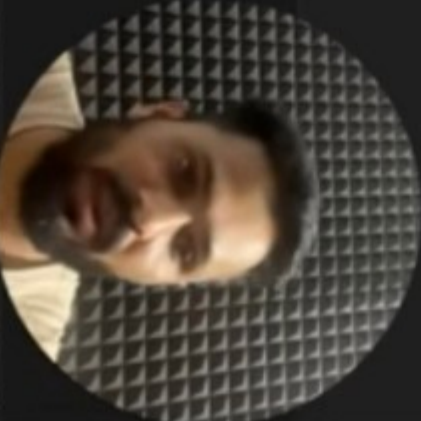
```
1 "use strict";
2 var PersonDataA = { name: 'Anil sidhu' };
3 var PersonDataB = { age: 30 };
4 var PersonDataC = { name: 'Sidhu', age: 30 };
5
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS OUTLENS SPELL CHECKER

[2:30:44 AM] File change detected. Starting incremental compilation...

[2:30:44 AM] Found 0 errors. Watching for file changes.

Ln 4, Col 1 Spaces: 4 UTF-8 LF TypeScript Port: 5500



typescript-tut

ts Intersection.ts U X

```
1 // type personTA={name:string}
2 // type personTB={age:number}
3 // type personTC=personTA & personTB
4
5 interface personTA {name:string}
6 interface personTB{age:number}
7 type personTC = personTA & personTB
8
9
10 var PersonDataA:personTA={name:'Anil sidhu'}
11 var PersonDataB:personTB={age:30}
12
13
14 var PersonDataC : personTC=
15   {name:'Sidhu',age:30}
```

js Intersection.js U X

```
1 "use strict";
2 // type personTA={name:string}
3 // type personTB={age:number}
4 // type personTC=personTA & personTB
5 var PersonDataA = { name: 'Anil sidhu' };
6 var PersonDataB = { age: 30 };
7 var PersonDataC = { name: 'Sidhu', age: 30 };
8
```

output > js Intersection.js > ...

```
1 "use strict";
2 // type personTA={name:string}
3 // type personTB={age:number}
4 // type personTC=personTA & personTB
5 var PersonDataA = { name: 'Anil sidhu' };
6 var PersonDataB = { age: 30 };
7 var PersonDataC = { name: 'Sidhu', age: 30 };
8
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS OUTLENS SPELL CHECKER

[2:33:48 AM] File change detected. Starting incremental compilation...

[2:33:48 AM] Found 0 errors. Watching for file changes.

first-program

launchpad

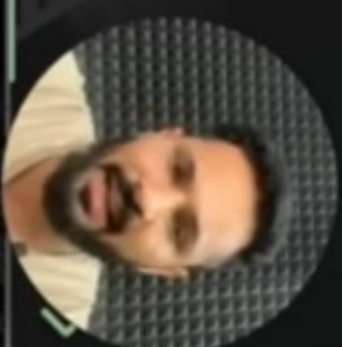
0 0 0

Ln 15, Col 22 Spaces: 4 UTF-8 LF () TypeScript Port: 5500



Type in TypeScript

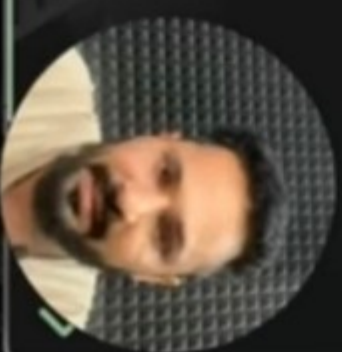
- What is Type.
- Define type
- How to use type
- Difference between interface and type





Type in TypeScript

- Define Custom data type for Object, function etc



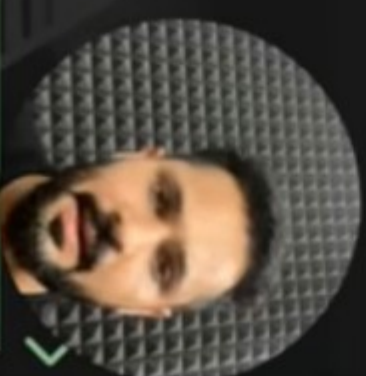
The image shows a screenshot of the Visual Studio Code editor interface. The main editor window displays a TypeScript file named 'typescript-tut.ts'. The code includes several type annotations and variable declarations. A yellow circle highlights the 'email' property in the 'studentData' object. The Explorer sidebar on the left shows the file structure, including 'typescript-tut.ts' and 'package-lock.json'. The Output panel at the bottom shows the command 'antlsidmugnlts-flacBook-Air typescript-tut %'. The status bar at the bottom indicates the file is at line 12, column 1, with 4 spaces, UTF-8 encoding, and LF line endings.



Enum Data Type



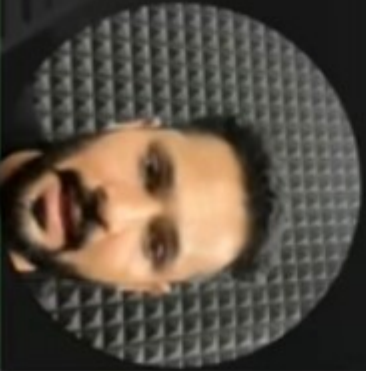
- What is Enum Type
- How to define enums
- How to use enums
- Interview Questions





Enum Data Type

- A data type that allows you to define a set of named constants



EXPLORER

OPEN EDITORS

TS enum.ts

TS enum.ts

TS any_unknown.ts

JS app.js

TS app.ts

TS array.ts

TS bigint.ts

TS enum.ts

TS function.ts

TS functionParams.ts

TS inputs.html

TS inputs.ts

TS interface.ts

TS intersection.ts

TS new.ts

JS null.js

JS number.js

TS number.ts

TS object.ts

OUTLINE

TIMELINE

ts enum.ts U X

ts enum.ts > ...

1 enum whoType{

2 student="student",

3 teacher="teacher",

4 management="management",

5 labStaff="labStaff",

6 }

7 }

8

9 var who:whoType= whoType.teacher;

10

11 who =whoType.teacher;

12

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER

Ln 11, Col 21

Spaces: 4

UTF-8

LF

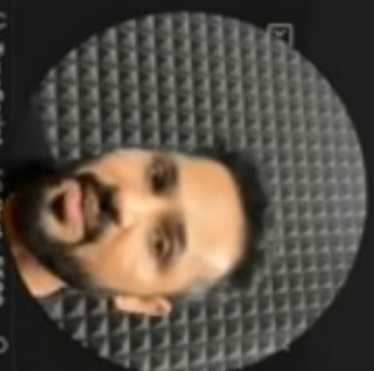
() TypeScript

Port: 5500

typescript-tut

08

...



EXPLORER

OPEN EDITORS

TS enum.ts U

TYPESCRIPT-TUT

any_unknown.ts

app.ts

app.ts

arrays.ts

bigint.ts

enum.ts

function.ts

functionParams.ts

inputs.html

inputs.ts

interface.ts

intersection.ts

never.ts

null.ts

number.ts

number.ts

objects.ts

OUTLINE

TIMELINE

← →

typescript-tut

08 10 11 ...

ts enum.ts U X

ts enum.ts > ...

1 enum whoType{

2 teacher:"teacher",

3 management:"management",

4 labStaff:"labStaff",

5 }

6 }

7 }

8 }

9 var who:whoType= whoType.teacher;

10 }

11 who =whoType.teacher

12 }

13 console.log(whoType.teacher);

14 }

15 }

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER

o anilsidhu@nils-MacBook-Air: typescript-tut %

1



first-program launched 0 0 0 Ln 13, Col 28 Spaces: 4 UTF-8 LF (1) TypeScript Port: 5500

EXPLORER

...

OPEN EDITORS

TS enum.ts U

TYPESCRIPT-TUT

output

js union.ts

ts any_unknown.ts

js app.ts

ts app.ts

ts arrays.ts

ts bigint.ts

TS enum.ts U

ts function.ts

ts functionParams.ts

inputs.html

ts inputs.ts

ts interface.ts

ts intersection.ts

ts never.ts

js null.ts

ts null.ts

js number.ts

OUTLINE

TIMELINE

← →

typescript-tut

08 10 10

...

ts enum.ts U X

ts enum.ts > ...

14

15

16

17

18

19

20

21

22

23

24

25

26

enum Fruit {

a="Apple",

b="Banana",

c="Cherry"

}

var myFruit:Fruit=Fruit.a

myFruit=Fruit.c

console.log(myFruit);

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GTITLENS

SPELL CHECKER

anils-MacBook-Air:typescript-tut anilsidhu\$ cd output/

anils-MacBook-Air:output anilsidhu\$ node enum.js

teacher

anils-MacBook-Air:output anilsidhu\$ node enum.js

management

anils-MacBook-Air:output anilsidhu\$ node enum.js

1

X

EXPLORER

OPEN EDITORS

TS enum.ts U

TYPESCRIPT-TUT

output

js union.js

ts any_unknown.ts

js app.js

ts app.ts

ts arrays.ts

ts bigint.ts

TS enum.ts U

ts function.ts

ts functionParams.ts

inputs.html

ts inputs.ts

ts interface.ts

ts intersection.ts

ts never.ts

js null.js

ts null.ts

js number.js

OUTLINE

TIMELINE

← →

typescript-tut

08 10 10

TS enum.ts U X

TS enum.ts > Roles > let manager

27 enum Roles {

28 admin,

29 manager,

30 qa,

31 developer,

32 user

33 }

34

35 var userRole: Roles = Roles.qa;

36

37 console.log(userRole);

38

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER

anils-MacBook-Air:output anilsidhu\$ node enum.js

anils-MacBook-Air:output anilsidhu\$ node enum.js

anils-MacBook-Air:output anilsidhu\$ node enum.js

anils-MacBook-Air:output an

Google Chrome

Ln 29, Col 8 Spaces: 4 UTF-8 LF (1) TypeScript

X

Edit: Code

(1) TypeScript Tutorial in Hindi

+

https://slides.com/enishidny/code-182053/edit

☆

🔍

🔖

📄

🔗

🏠

New Chrome available

⋮

✎

🖨

DOM Handling and TypeCasting



- How to access DOM element.
- Use exclamatory sign
- Typecasting
- Interview Questions



121

EXPLORER

OPEN EDITORS

dom.html U

TS dom.ts U

JS dom.js output U

TYPESCRIPT-TUT

output

any_unknown.ts

app.js

app.ts

arrays.ts

bigint.ts

dom.html U

dom.ts U

enums.ts

functions.ts

functionParams.ts

inputs.html

inputs.ts

interface.ts

intersection.ts

OUTLINE

TIMELINE

dom.html U

TS dom.ts U X

JS dom.js U

TS dom.ts > ...

1 var headingEl=document.querySelector('h1');

2 console.log(headingEl?.textContent);

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CH

[1:46:21 PM] File change detected. Starting incremental compilation...

[1:46:21 PM] Found 0 errors. Watching for file changes.

Ln 2, Col 23

Spaces: 4

UTF-8

LF

()

TypeScript

Port: 5500

EXPLORER

OPEN EDITORS

dom.html U

TS dom.ts U

JS dom.js output U

TYPESCRIPT-TUT

output

any_unknown.ts

app.js

app.ts

arrays.ts

bigint.ts

dom.html U

dom.ts U

enums.ts

function.ts

functionParams.ts

inputs.html

inputs.ts

interface.ts

intersection.ts

OUTLINE

TIMELINE

← →

typescript-tut

dom.html U

TS dom.ts U X

JS dom.js U

TS dom.ts > [0] headingEl

1 var headingEl=document.querySelector('h1')
2 console.log(headingEl.classList)

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CH

(1:47:26 PM) File change detected. Starting incremental compilation...

(1:47:26 PM) Found 0 errors. Watching for file changes.

□

first-program

Launchpad

0 0

Ln 1, Col 44 (1 selected)

Spaces: 4

UTF-8

LF

()

TypeScript

Port: 5500



EXPLORER

OPEN EDITORS

dom.html U

ts dom.ts U

js dom.js output U

TYPESCRIPT-TUT

output

any_unknown.ts

app.js

app.ts

arrays.ts

bigint.ts

dom.html U

dom.ts U

enums.ts

function.ts

functionParams.ts

inputs.html

inputs.ts

interface.ts

intersection.ts

OUTLINE

TIMELINE

← →

typescript-tut

dom.html U X

ts dom.ts U X

js dom.js U

dom.html > html > body > h1.headingStyle

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>DOM handling</title>
5 </head>
6 <body>
7   <h1 class="headingStyle">Heading h1</h1>
8   <a class="anchorStyle" href="/home">go to home</a>
9   <script src="/output/dom.js"></script>
10 </body>
11 </html>
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHEK

[1:50:23 PM] File change detected. Starting incremental compilation...

[1:50:23 PM] Found 0 errors. Watching for file changes.

□

Ln 7, Col 28 (12 selected)

Spaces: 4

UTF-8

LF

HTML

Port: 5500

EXPLORER

OPEN EDITORS

dom.html

TS dom.ts

JS dom.js

output

any_unknown.ts

app.js

app.ts

arrays.ts

bigint.ts

dom.html

dom.ts

enums.ts

function.ts

functionParams.ts

inputs.html

inputs.ts

interface.ts

intersection.ts

OUTLINE

TIMELINE

dom.html

TS dom.ts

JS dom.js

ts dom.ts

```
1 var headingEl=document.querySelector('h1')!;
2 var anchorEl=document.querySelector('a')!;
3 var anchorElClass=document.querySelector('.anchorStyle')! as HTMLAnchorElement;
4 var headingElClass=document.querySelector('.headingStyle')! as HTMLHeadingElement;
5 console.log(headingEl.classList)
6 console.log(anchorEl.href)
7 console.log(anchorElClass.href)
8 console.log(headingElClass.classList)
9
10
11
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CH

[1:51:15 PM] File change detected. Starting incremental compilation...

[1:51:15 PM] Found 0 errors. Watching for file changes.

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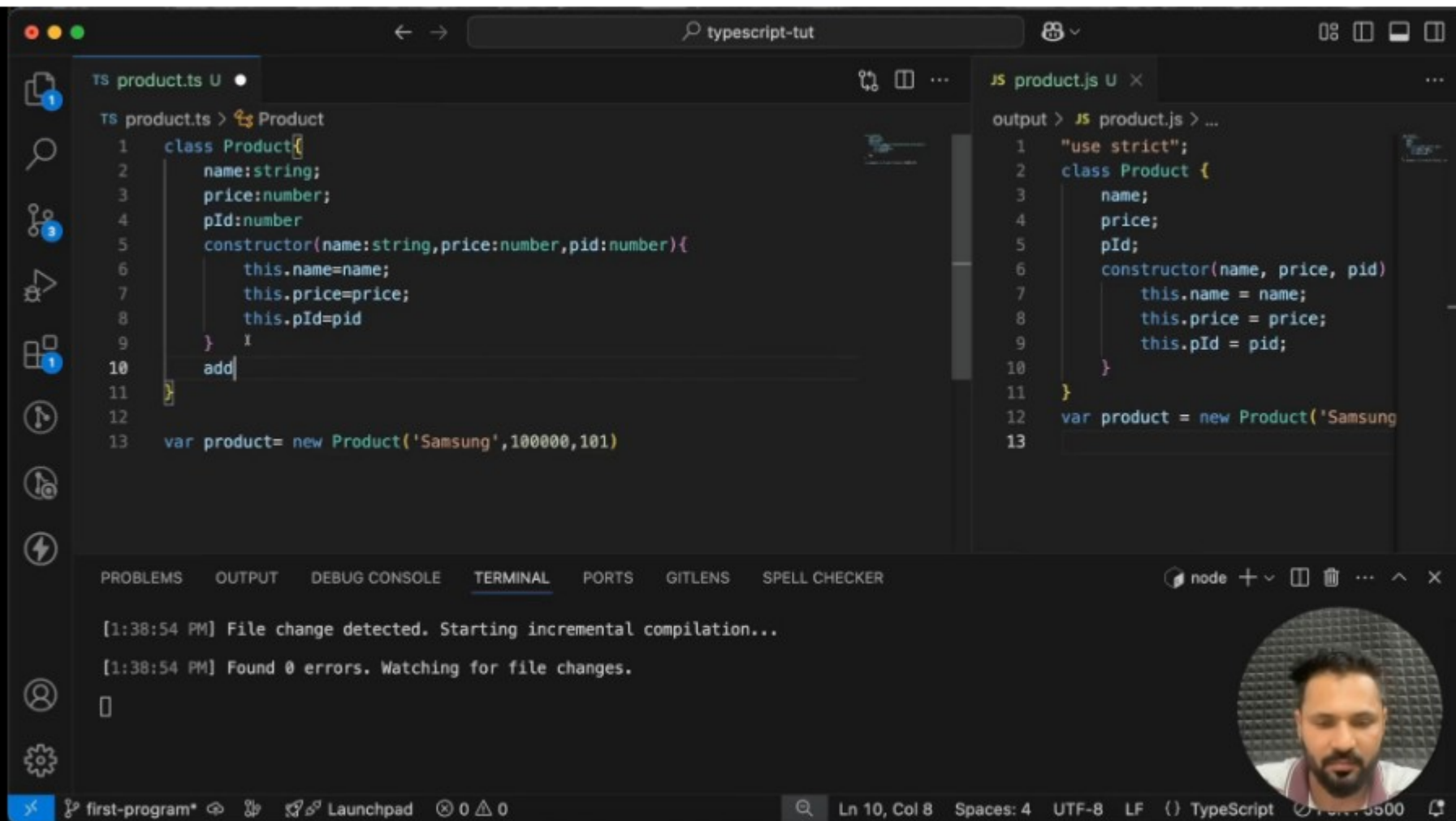
New Chrome available

TS

Classes in TypeScript

- What is Class
- Example of class
- Define Data type to class property and functions
- Interview Questions

Subscribe





← → typescript-tut

TS product.ts U X

TS product.ts > ...
1 class Product{
15 buyProduct():string{
17 return `product \${this.name} is ordered in \${this.price}`
18 }else{
19 return `no product in cart`
20 }
21 }
22 }
23 }
24 }
25 var product= new Product('Samsung',100000,101);
26 console.log(product.buyProduct());

JS product.js U X

output > JS product.js > ...
1 "use strict";
2 class Product {
3 name;
4 price;
5 pid;
6 inCart = false;
7 isOrdered = false;
8 constructor(name, price, pid)
9 {
10 this.name = name;
11 this.price = price;
12 this.pId = pid;
13 }
14 addToCart() {
15 this.inCart = true;
16 }
17 buyProduct() {
18 if (this.inCart) {

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

/dev/fd/13:18: command not found: compdef
● anilsidhu@anils-MacBook-Air typescript-tut % cd output
● anilsidhu@anils-MacBook-Air output % node product.js
no product in cart
○ anilsidhu@anils-MacBook-Air output %

Ln 26, Col 25 Spaces: 4 UTF-8 LF () TypeScript 5500



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typescript-tut

TS product.ts U X

TS product.ts > ...
1 class Product{
22 }
23 }
24
25 var product= new Product('Samsung',100000,101);
26 product.addToCart();
27 console.log(product.buyProduct())
28
29
30 var product= new Product('I phobne',200000,301);
31 product.addToCart();
32 console.log(product.buyProduct())

JS product.js U X

output > JS product.js > ...
1 "use strict";
2 class Product {
3 name;
4 price;
5 pid;
6 inCart = false;
7 isOrdered = false;
8 constructor(name, price, pid)
9 {
10 this.name = name;
11 this.price = price;
12 this.pId = pid;
13 }
14 addToCart() {
15 this.inCart = true;
16 }
17 buyProduct() {
18 if (this.inCart) {

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

anilsidhu@anils-MacBook-Air typescript-tut % cd output
anilsidhu@anils-MacBook-Air output % node product.js
product Samsung is ordered in 100000
product I phobne is ordered in 200000
anilsidhu@anils-MacBook-Air output %

Ln 29, Col 1 Spaces: 4 UTF-8 LF () TypeScript 3500

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Access Modifiers

- What is Access Modifiers.
- Public.
- Private.
- Protected.
- Interview Questions.



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TS

Inheritance in typeScript

- What is Inheritance?
- Example.
- Interview Questions.

< >

Notification bell icon

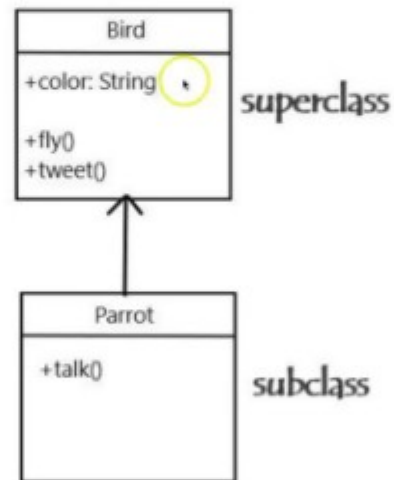
Volume slider

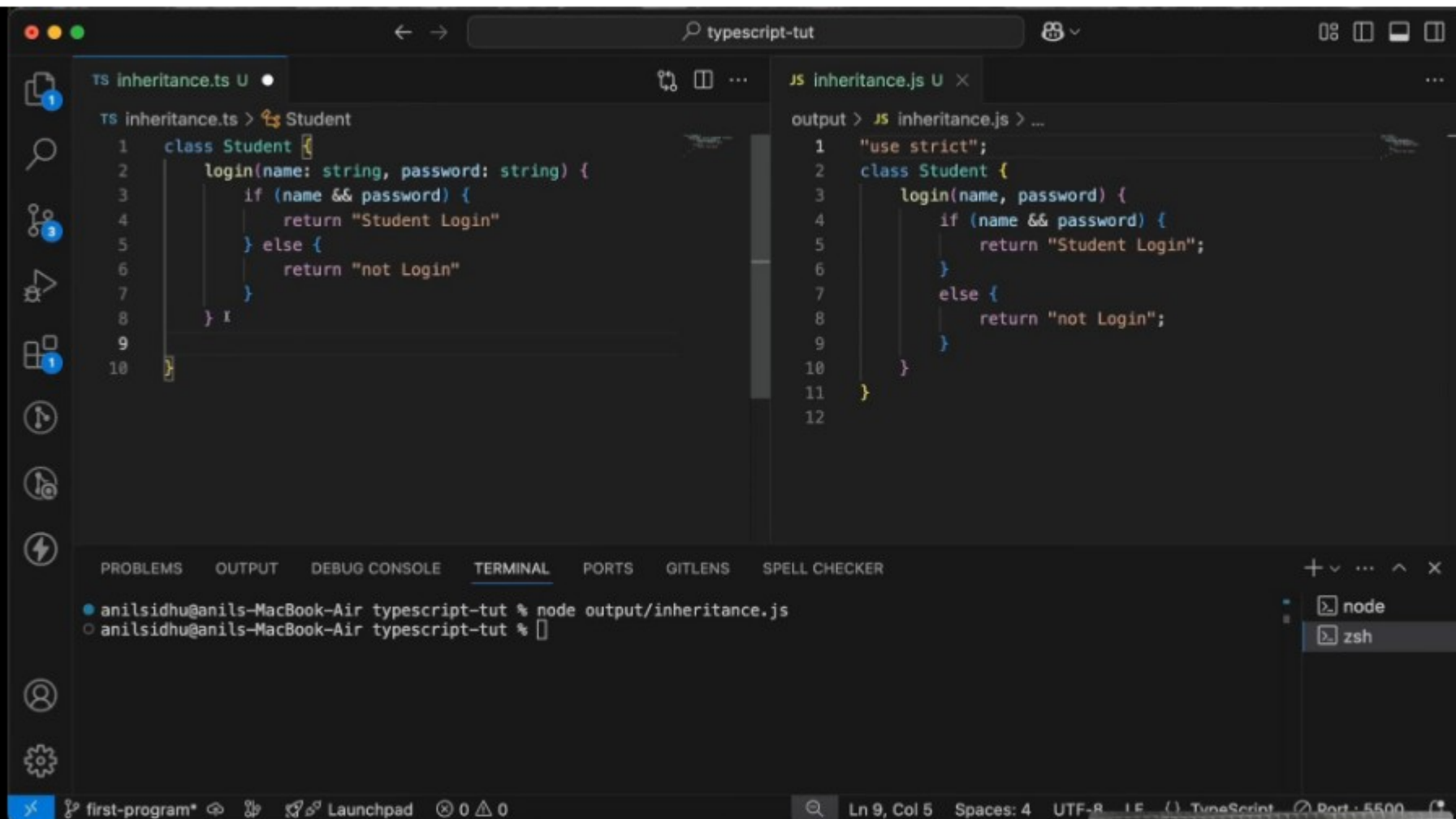
Speaker icon

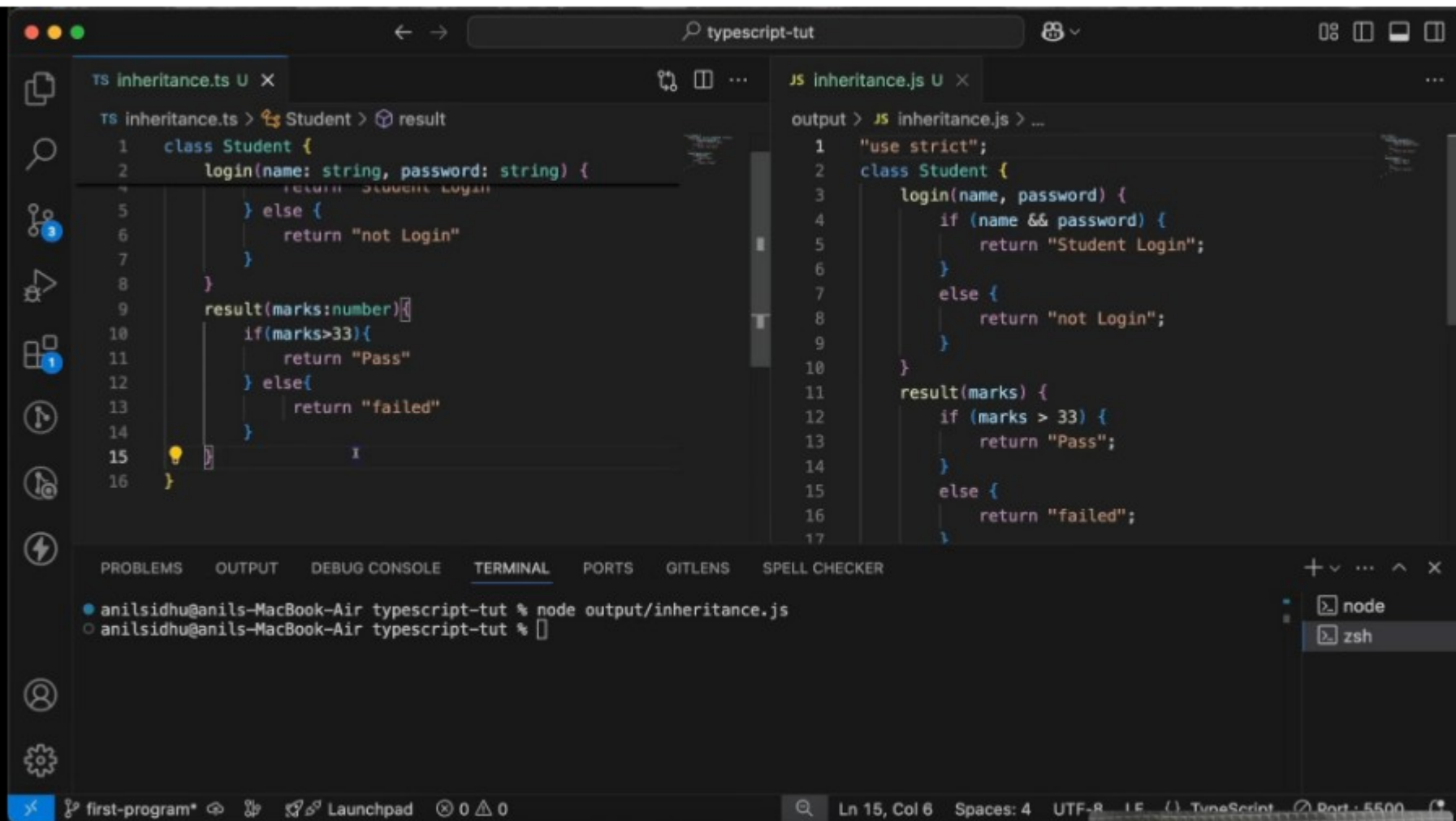
Settings icon

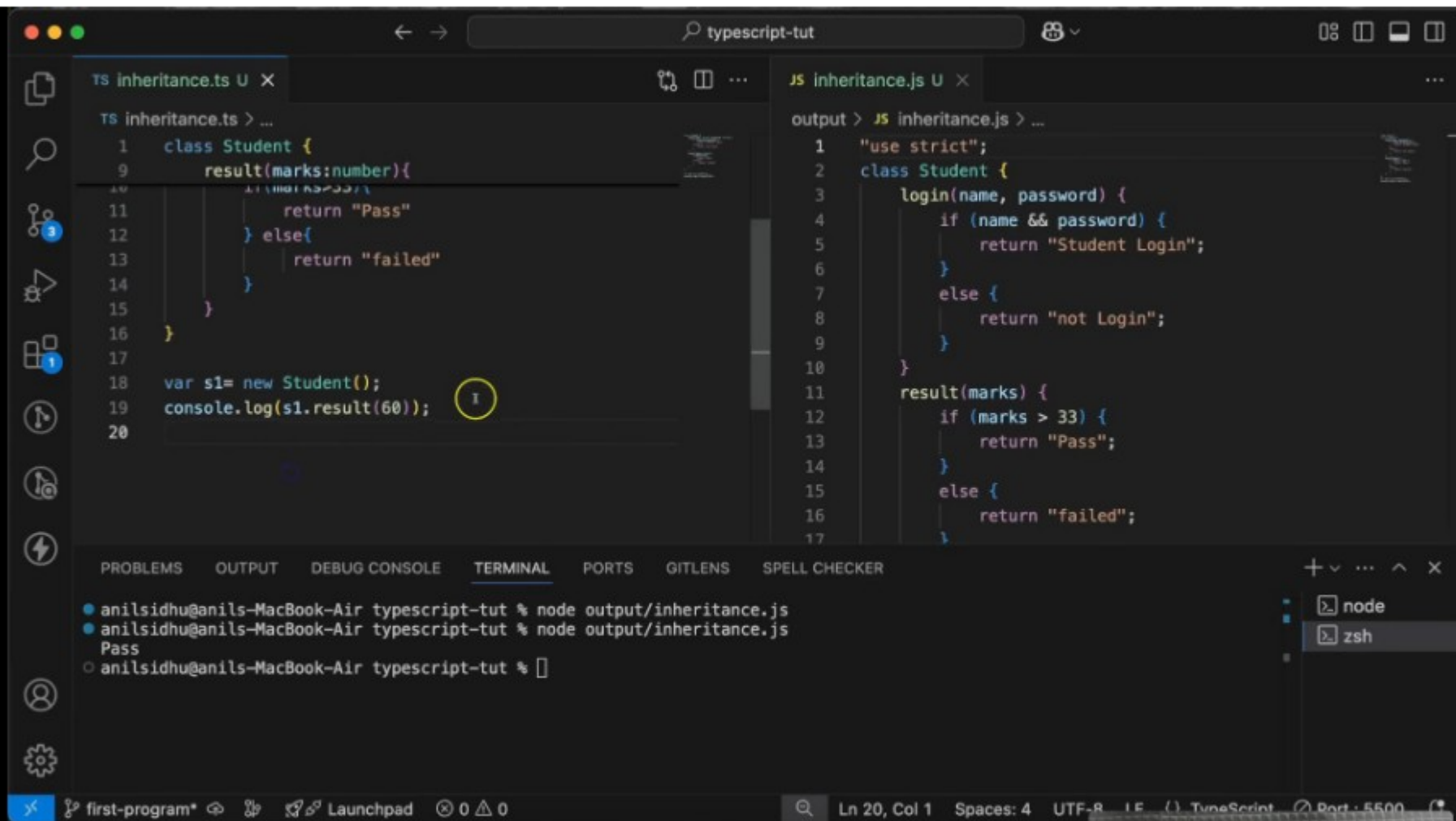


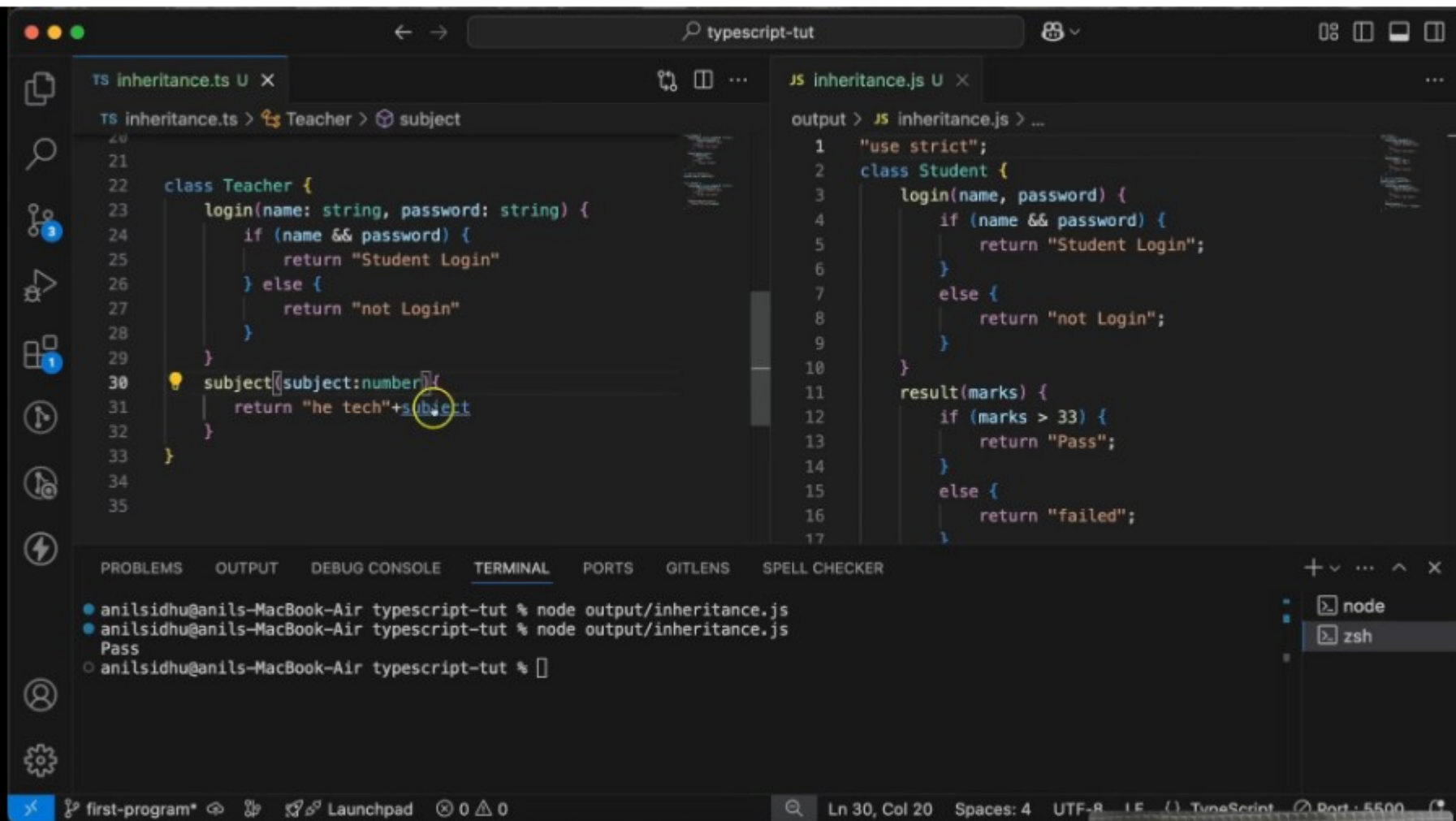
Inheritance











typescript-tut

TS inheritance.ts U X

TS inheritance.ts > Teacher

```
21
22 var s1= new Student();
23 console.log(s1.result(60));
24
25
26 class Teacher extends Auth {
27
28     subject(subject:number){
29         return "he tech"+subject
30     }
31 }
32
33 var t1 = new Teacher();
34 console.log(t1.login("sam","12345"));
35
36
```

JS inheritance.js U X

output > JS inheritance.js > ...

```
1 "use strict";
2 class Auth {
3     login(name, password) {
4         if (name && password) {
5             return "Student Login";
6         }
7         else {
8             return "not Login";
9         }
10    }
11 }
12 class Student extends Auth {
13     result(marks) {
14         if (marks > 33) {
15             return "Pass";
16         }
17         else {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut % node output/inheritance.js

anilsidhu@anils-MacBook-Air typescript-tut % node output/inheritance.js

Pass

anilsidhu@anils-MacBook-Air typescript-tut % node output/inheritance.js

Pass

Student Login

anilsidhu@anils-MacBook-Air typescript-tut % node output/inheritance.js

Pass

Student Login

anilsidhu@anils-MacBook-Air typescript-tut %

node

zsh

first-program* Launchpad 0 0

Ln 26, Col 28 Spaces: 4 UTF-8 LF TypeScript Port: 5500

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Module in typeScript



- What is modules in TS?
- Example.
- Interview Questions.

ScreenPal - 3.16.1

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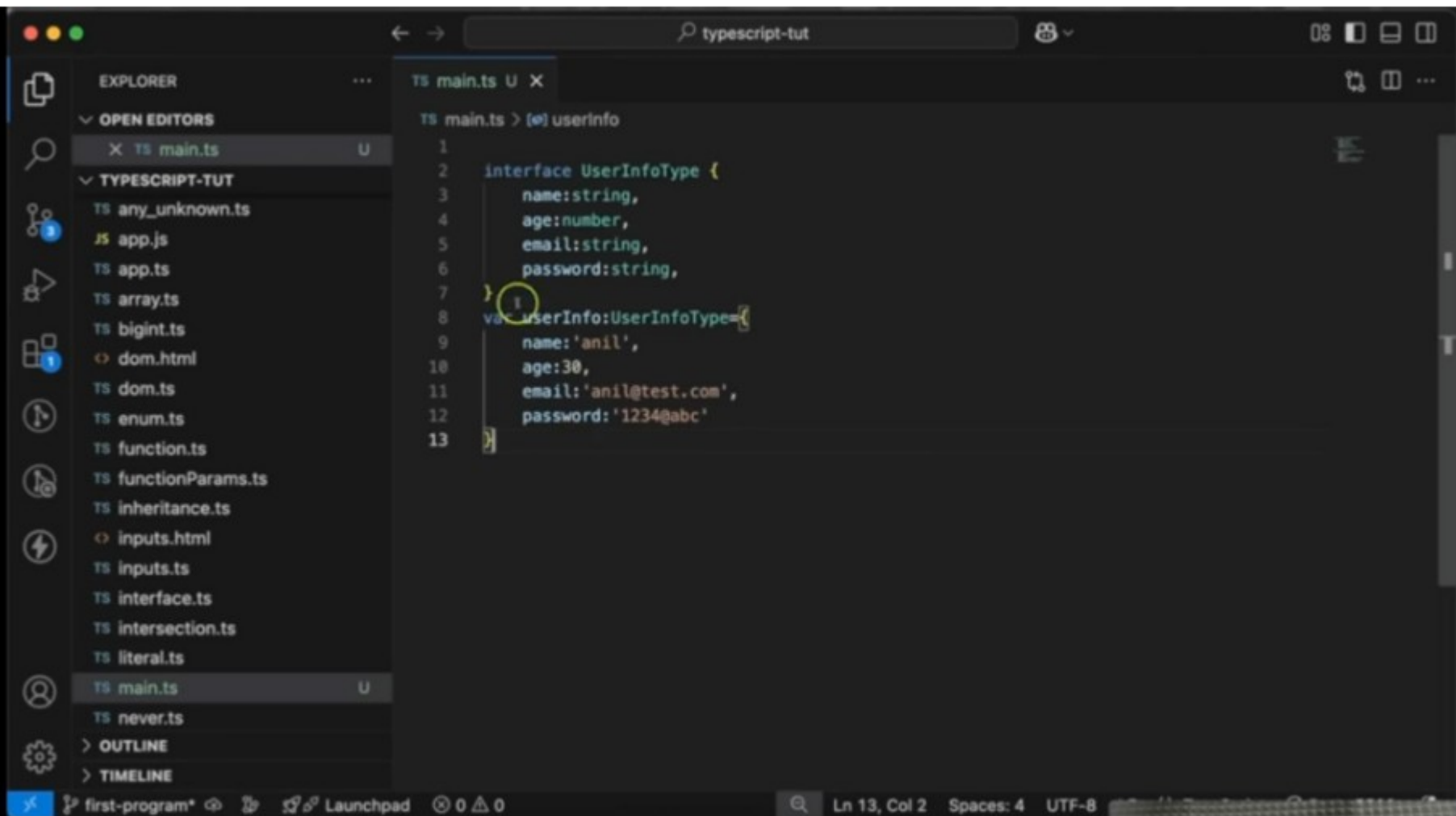
What is Module?

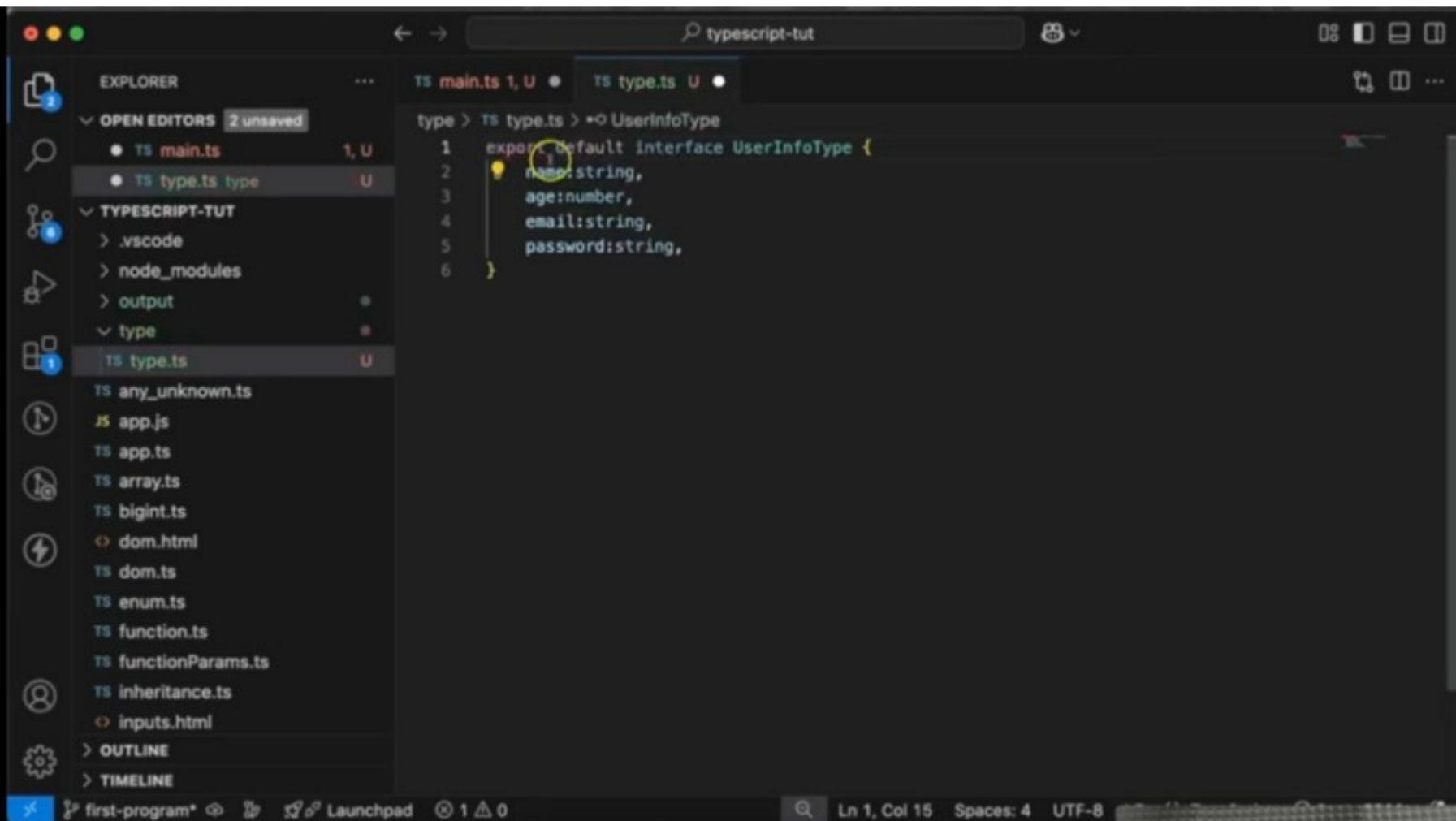


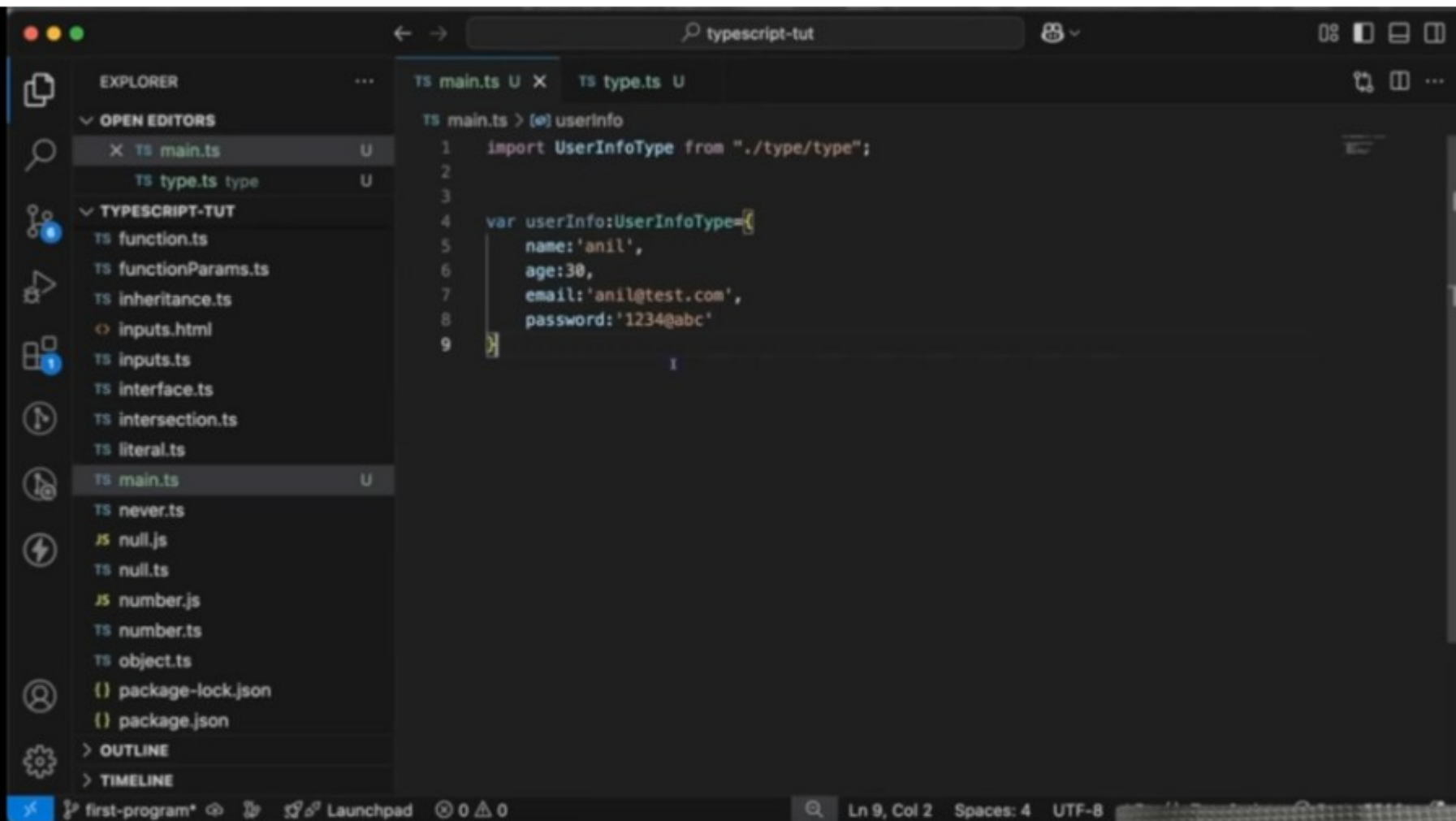
Self-contained unit of code that encapsulates related functionalities, such as classes, functions, and variables

ScreenPal - 3.16.1

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Getter and Setter in Typescript

- What is Getter and Example.
- What is Setter and Example.
- Interview Questions.



typescript-tut

TS get_set.ts U X

TS get_set.ts > EmpInfo > (get) name

```
1 class EmpInfo{
2   _name:string="Anil";
3
4   get name():string{
5     return "MR. " + this._name;
6   }
7 }
8
9 var emp1= new EmpInfo();
10 console.log(emp1.name);
11
```

JS get_set.js U X

output > JS get_set.js > ...

```
1 "use strict";
2 class EmpInfo {
3   _name = "Anil";
4   get name() {
5     return "MR. " + this._name;
6   }
7 }
8 var emp1 = new EmpInfo();
9 console.log(emp1.name);
10
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air output % node get_set.js

anilsidhu@anils-MacBook-Air output % node get_set.js

Anil

anilsidhu@anils-MacBook-Air output % node get_set.js

Anil

anilsidhu@anils-MacBook-Air output % node get_set.js

Anil

anilsidhu@anils-MacBook-Air output % node get_set.js

MR. Anil

anilsidhu@anils-MacBook-Air output %

node



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Launchpad

0 0

Ln 5, Col 21 (5 selected)

Spaces: 4

UTF-8

LF

() TypeScript

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typescript-tut

TS get_set.ts U X

TS get_set.ts > ...
1 class EmpInfo{
2 get name():string{
3
4
5
6
7
8
9 set email(val:string){
10 this._email="emp"+val
11 }
12 }
13
14 var emp1= new EmpInfo();
15
16 emp1.email="peter@test.com";
17 console.log(emp1._email);
18

JS get_set.js U X

output > JS get_set.js > ...
1 "use strict";
2 class EmpInfo {
3 _name = "Anil";
4 _email = "anil@test.com";
5 get name() {
6 return "MR. " + this._name;
7 }
8 set email(val) {
9 this._email = "emp" + val;
10 }
11 }
12 var emp1 = new EmpInfo();
13 emp1.email = "peter@test.com";
14 console.log(emp1._email);
15

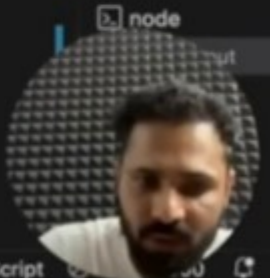
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air output % node get_set.js
anilsidhu@anils-MacBook-Air output % node get_set.js
Anil
anilsidhu@anils-MacBook-Air output % node get_set.js
Anil
anilsidhu@anils-MacBook-Air output % node get_set.js
Anil
anilsidhu@anils-MacBook-Air output % node get_set.js
MR. Anil
anilsidhu@anils-MacBook-Air output % node get_set.js

node

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Logo: A blue 3D hexagon with the white text "TS" inside.

Section Title: Interface with Class

Navigation: A green arrow pointing left is located at the bottom right of the slide.

Footer: A red "Subscribe" button with a YouTube icon is in the bottom right corner.

← → typescript-tut

TS collegeData.ts U ×

TS collegeData.ts > CollegeDataType

```
1 interface CollegeDataType{
2     name:string;
3     // displayTeachersName():void;
4 }
5
6 class CollegeData implements CollegeDataType{
7     name:string;
8     constructor(cName:string){
9         this.name=cName
10    }
11
12 }
13
14 var college1= new CollegeData('IIT Delhi')
```

JS collegeData.js U ×

output > JS collegeData.js > ...

```
1 "use strict";
2 class CollegeData {
3     name;
4     constructor(cName) {
5         this.name = cName;
6     }
7 }
8 var college1 = new CollegeData('IIT Delhi');
9
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

/dev/fd/13:18: command not found: compdef

● anilsidhu@anils-MacBook-Air typescript-tut % cd output

● anilsidhu@anils-MacBook-Air output % node collegeData.js

○ anilsidhu@anils-MacBook-Air output %

node

zsh output

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Ln 3, Col 15 Spaces: 4 UTF-8 LF () TypeScript Port : 5500

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Static Keyword



- Why static keyword?
- Advantage of static keyword?
- How to use static property
- How to use static functions
- Example.
- Interview Questions.



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Static Keywords



- use to define static property and methods.
- Memory Efficiency
- Utility Methods
- Global Constants



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typescript-tut

TS static.ts U X

JS static.js U X

TS static.ts > ...

output > JS static.js > ...

```
1 class Company{
2   static name:string="Google"
3 }
4
5
6 var c1= new Company();
7 // console.log(c1.name);
8 console.log(Company.name);
9
10
```

```
1 "use strict";
2 class Company {
3   static name = "Google";
4 }
5 var c1 = new Company();
6 // console.log(c1.name);
7 console.log(Company.name);
8
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

```
/dev/fd/13:18: command not found: compdef
• anilsidhu@anils-MacBook-Air typescript-tut % cd output
• anilsidhu@anils-MacBook-Air output % node static.js
Google
• anilsidhu@anils-MacBook-Air output % node static.js
Google
○ anilsidhu@anils-MacBook-Air output %
```

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TypeScript Tutorial in Hindi 2

is math a class in javascript

https://www.google.com/search?q=is+math+a+class+in+javascript+or+typescript&oq=is+&gs_lcrp=EgZjaHJvbW...

 is math a class in javascript or typescript

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 En  Listen

Math is not a class in either JavaScript or TypeScript. Instead, it is a built-in object in JavaScript that provides properties and methods for mathematical constants and functions. TypeScript inherits this behavior from JavaScript, treating Math as an object as well.

Since Math is an object and not a class, you do not create instances of it using the new keyword. All its properties and methods are static, meaning you can access them directly using Math.property or Math.method().

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 Stack Overflow

2 answers · 5 years ago

Is Math an object or a class? - javascript



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Typeguard in typescript

- What is TypeGuard.
- Types of TypeGuard.
- Examples.
- Interview Questions.



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Typeguard in typescript



- A **Type Guard** in TypeScript is a technique used to narrow down the type of a variable within a conditional block



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Typeguard in typescript



- `typeof`
- `instanceof`
- Custom Type



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Why Use Type Guards?



- Provides better **type safety**
- Apply conditions with data type
- Helps TypeScript **infer types automatically**



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typescript-tut

TS typeGuards.ts U X

JS typeGuards.js U X

TS typeGuards.ts > ...

output > JS typeGuards.js > ...

```
1 let userData20:number|string|boolean="Anil sidhu"
2
3 userData20=true
4 if(typeof userData20 == "boolean"){
5     console.log('this is a bool data type');
6 }
7
8 else if(typeof userData20 == "string"){
9     console.log('this is a string data type');
10 }
11 }else{
12     console.log("this is a number");
13 }
14 }
```

```
1 "use strict";
2 let userData20 = "Anil sidhu";
3 userData20 = true;
4 if (typeof userData20 == "boolean") {
5     console.log('this is a bool data type');
6 }
7 else if (typeof userData20 == "string") {
8     console.log('this is a string data type');
9 }
10 else {
11     console.log("this is a number");
12 }
13
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

/dev/fd/13:18: command not found: compdef

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anilsidhu@anils-MacBook-Air output %

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Ln 12, Col 34 Spaces: 4 UTF-8 LF () TypeScript



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typescript-tut

TS typeGuards.ts U X

JS typeGuards.js U X

TS typeGuards.ts > checkDataType

output > JS typeGuards.js > ...

```
15 }
16
17 function checkDataType(data:string | number){
18
19     if(typeof data == 'number'){
20         console.log("this is a number");
21     }else{
22         console.log("this is a string");
23     }
24 }
25
26
27
28 checkDataType(29)
```

```
1 "use strict";
2 let userData20 = "Anil sidhu";
3 // userData20=true
4 if (typeof userData20 == "boolean") {
5     console.log('this is a bool data type');
6 }
7 else if (typeof userData20 == "string") {
8     userData20;
9     console.log('this is a string data type');
10 }
11 else {
12     console.log("this is a number");
13 }
14 function checkDataType(data) {
15     if (typeof data == 'number') {
16         console.log("this is a number");
17     }
18     else {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a string data type

anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a string data type

anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a string data type

anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a string data type

anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a number

first-program* Launchpad 0 0 0

Ln 28, Col 15 Spaces: 4 UTF-8 LF () TypeScript



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typescript-tut

TS typeGuards.ts U X

JS typeGuards.js U X

TS typeGuards.ts > [0] o1

output > JS typeGuards.js > ...

```
37 var p1 = new Product21()
38 class Order21 {
39 }
40
41
42 var o1 = new Order21();
43
44
45 function checkDetails(data :Order21 |Product21 ){
46     if(data instanceof Order21){
47         console.log('this is a order');
48     }else{
49
50         console.log('this is a product');
51     }
52 }
53
```

```
1 "use strict";
2 let userData20 = "Anil sidhu";
3 // userData20=true
4 // if(typeof userData20 == "boolean"){
5 //     console.log('this is a bool data type');
6 // }
7 // else if(typeof userData20 == "string"){
8 //     console.log('this is a string data type');
9 // }else{
10 //     console.log("this is a number");
11 // }
12 // }
13 // function checkDataType(data:string | number){
14 //     if(typeof data == 'number'){
15 //         console.log("this is a number");
16 //     }else{
17 //         console.log("this is a string");
18 //     }
19 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

Product21 {}

anilsidhu@anils-MacBook-Air output % node typeGuards.js

this is a string

this is a product

anilsidhu@anils-MacBook-Air output % node typeGuards.js

this is a product

anilsidhu@anils-MacBook-Air output % node typeGuards.js

this is a order

anilsidhu@anils-MacBook-Air output %

first-program* Launchpad 0 0

Ln 42, Col 7 Spaces: 4 UTF-8 LF () TypeScript



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typescript-tut

TS typeGuards.ts U

JS typeGuards.js U

TS typeGuards.ts > ...

output > JS typeGuards.js > ...

```
45 function checkDetails(data :Order21 |Product21 ){
46     // console.log('this is a order');
47
48
49     }else{
50
51         console.log('this is a product');
52     }
53 }
54
55 checkDetails(o1);
56
57
58 |
```

```
1 "use strict";
2 let userData20 = "Anil sidhu";
3 // userData20=true
4 // if(typeof userData20 == "boolean"){
5 //     console.log('this is a bool data type');
6 // }
7 // else if(typeof userData20 == "string"){
8 //     userData20
9 //     console.log('this is a string data type');
10 // }else{
11 //     console.log("this is a number");
12 // }
13 // function checkDataType(data:string | number){
14 //     if(typeof data == 'number'){
15 //         console.log("this is a number");
16 //     }else{
17 //         console.log("this is a string");
18 //     }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

Product21 {}

- anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a string
this is a product
- anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a product
- anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a order
- anilsidhu@anils-MacBook-Air output %

first-program* Launchpad 0 0

Ln 58, Col 1 Spaces: 4 UTF-8 LF () TypeScript



Subscribe

typescript-tut

TS typeGuards.ts U X

TS typeGuards.ts > ...

```
50
57
58 interface userData{
59     name:string,
60     city:string
61 }
62
63 interface userInfo{
64     id:number,
65     email:string
66 }
67 var userData22:userData | userInfo
68
```

JS typeGuards.js U X

output > JS typeGuards.js > ...

```
1 "use strict";
2 let userData20 = "Anil sidhu";
3 // userData20=true
4 // if(typeof userData20 == "boolean"){
5 //     console.log('this is a bool data type');
6 // }
7 // else if(typeof userData20 == "string"){
8 //     console.log('this is a string data type');
9 // }
10 // }else{
11 //     console.log("this is a number");
12 // }
13 // function checkDataType(data:string | number){
14 //     if(typeof data == 'number'){
15 //         console.log("this is a number");
16 //     }else{
17 //         console.log("this is a string");
18 //     }
19 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

Product21 {}

- anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a string
this is a product
- anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a product
- anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a order

first-program* Launchpad 0 0

Ln 68, Col 1 Spaces: 4 UTF-8 LF () TypeScript



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typescript-tut

TS typeGuards.ts U X

TS typeGuards.ts > [0] userData21 > email

```
74 var userData21:userData | userInfo={
75   email:'anil@test.com'
76 }
77
78 function checkUserInfo(data:userData|userInfo){
79   if((data as userData).name!==undefined){
80     console.log('this is a user data');
81   }else{
82     console.log('this is a user info');
83   }
84 }
85
86 checkUserInfo(userData21)
```

JS typeGuards.js U X

output > JS typeGuards.js > ...

```
1 "use strict";
2 let userData20 = "Anil sidhu";
3 // userData20=true
4 // if(typeof userData20 == "boolean"){
5 //   console.log('this is a bool data type');
6 // }
7 // else if(typeof userData20 == "string"){
8 //   console.log('this is a string data type');
9 // }else{
10 //   console.log("this is a number");
11 // }
12 // }
13 // function checkDataType(data:string | number){
14 //   if(typeof data == 'number'){
15 //     console.log("this is a number");
16 //   }else{
17 //     console.log("this is a string");
18 //   }
19 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

Product21 {}

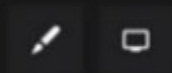
- anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a string
this is a product
- anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a product
- anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a order
- anilsidhu@anils-MacBook-Air output %

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Ln 76, Col 12 Spaces: 4 UTF-8 LF () TypeScript



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Generic in TypeScript



- What is Generic in TypeScript
- Example
- Interview Questions



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Generic in TypeScript



- Generics in TypeScript allow you to create reusable components that work with a variety of data types while maintaining type safety



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← → typescript-tut

TS generic.ts 1, U X

TS generic.ts > ...
1 function fruits(name:string|number):string|number{
2 | return name 1
3 }
4
5 fruits("apple")
6 fruits(100)
7 fruits(true)
8
9

JS generic.js U X

output > JS generic.js > ...
1 "use strict";
2 function fruits(name) {
3 | return name;
4 }
5 fruits("apple");
6 fruits(100);
7 fruits(true);
8

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

generic.ts:7:8 - error TS2345: Argument of type 'boolean' is not assignable to parameter of type 'string | number'.
7 fruits(true)
[12:55:01 AM] Found 1 error. Watching for file changes.

first-program* Launchpad 1 0 Ln 6, Col 12 Spaces: 4 UTF-8 LF () TypeScript Port: 5500



Subscribe

← → typescript-tut

TS generic.ts U ×

TS generic.ts > ...
1 function fruits10<T>(name:T):T{
2 return name
3 }
4
5 let onlyFruit=fruits10("apple")
6 let onlyNum=fruits10(100)
7 let onlyBool =fruits10(true)
8
9

JS generic.js U ×

output > JS generic.js > ...
1 "use strict";
2 function fruits10(name) {
3 return name;
4 }
5 let onlyFruit = fruits10("apple");
6 let onlyNum = fruits10(100);
7 let onlyBool = fruits10(true);
8

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[12:56:38 AM] File change detected. Starting incremental compilation...
[12:56:38 AM] Found 0 errors. Watching for file changes.
[]

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Ln 8, Col 1 Spaces: 4 UTF-8 LF () TypeScript Port: 5500



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typescript-tut

TS generic.ts U X

JS generic.js U X

TS generic.ts > ...

output > JS generic.js > ...

```
// let onlyFruit=fruitsAny('apple')
// let onlyNum=fruitsAny(100)
// let onlyBool =fruitsAny(true)

let onlyFruit=fruits10('apple')
let onlyNum=fruits10(100)
let onlyBool =fruits10(true)

function Users17<T>(data:T):T {
  return data
}

let userCollection = Users17(['anil','sam'])
```

```
"use strict";
function fruits10(name) {
  return name;
}
function fruitsAny(name) {
  return name;
}
// let onlyFruit=fruitsAny("apple")
// let onlyNum=fruitsAny(100)
// let onlyBool =fruitsAny(true)
let onlyFruit = fruits10("apple");
let onlyNum = fruits10(100);
let onlyBool = fruits10(true);
function Users17(data) {
  return data;
}
let userCollection = Users17(['anil', 'sam']);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[12:59:31 AM] File change detected. Starting incremental compilation...

[12:59:31 AM] Found 0 errors. Watching for file changes.

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Ln 22, Col 1 Spaces: 4 UTF-8 LF () TypeScript Port: 5500



Subscribe

typescript-tut

TS generic.ts U X

JS generic.js U X

TS generic.ts > Users17

```
9 // let onlyFruit=fruitsAny("apple")
10 // let onlyNum=fruitsAny(100)
11 // let onlyBool =fruitsAny(true)
12
13 let onlyFruit=fruits10("apple")
14 let onlyNum=fruits10(100)
15 let onlyBool =fruits10(true)
16
17 function Users17<T>(data:T[]):T {
18   return data[0]
19 }
20
21 let userCollection = Users17(['anil','sam'])
22
```

JS generic.js > ...

```
1 "use strict";
2 function fruits10(name) {
3   return name;
4 }
5 function fruitsAny(name) {
6   return name;
7 }
8 // let onlyFruit=fruitsAny("apple")
9 // let onlyNum=fruitsAny(100)
10 // let onlyBool =fruitsAny(true)
11 let onlyFruit = fruits10("apple");
12 let onlyNum = fruits10(100);
13 let onlyBool = fruits10(true);
14 function Users17(data) {
15   return data[0];
16 }
17 let userCollection = Users17(['anil', 'sam']);
18
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[1:00:05 AM] File change detected. Starting incremental compilation...

[1:00:05 AM] Found 0 errors. Watching for file changes.

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Ln 17, Col 28 Spaces: 4 UTF-8 LF () TypeScript Port: 5500

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keyOf in TypeScript

- What is keyof?
- Example
- How to use object keys with keyof
- Interview Questions



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keyOf

- The `keyof` keyword in TypeScript is used to get the keys of a type as a union of string.
- It is primarily used for type safety when working with objects.





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keyOf

- The `keyof` keyword in TypeScript is used to get the keys of a type as a union of string.
- It is primarily used for type safety when working with objects.





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EXPLORER

OPEN EDITORS

GROUP 1

TS keyof.ts U

GROUP 2

JS keyof.js o... U

TYPESCRIPT-TUT

TS generic.ts

TS get_set.ts

TS inheritance.ts

inputs.html

TS inputs.ts

TS interface.ts

TS intersection.ts

TS keyof.ts U

TS literal.ts

TS main.ts

TS never.ts

JS null.js

TS null.ts

JS number.js

TS number.ts

OUTLINE

TIMELINE

TS keyof.ts U X

TS keyof.ts > PersonData

1 type PersonT={

2 name:string,

3 age:number,

4 isEmp:boolean

5 }

6 let PersonData:PersonT={

7 name:'Anil sidhu',

8 age:30,

9 isEmp:true

10 }

JS keyof.js U X

output > JS keyof.js > ...

1 "use strict";

2 let PersonData = {

3 name: 'Anil sidhu',

4 age: 30,

5 isEmp: true

6 };

7

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[11:42:05 AM] File change detected. Starting incremental compilation...

[11:42:05 AM] Found 0 errors. Watching for file changes.

Ln 10, Col 2 Spaces: 2 UTF-8 LF () TypeScript Live

Code.Steps

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typescript-tut

EXPLORER

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GROUP 1

TS keyof.ts 1, U

GROUP 2

JS keyof.js o... U

TYPESCRIPT-TUT

TS generic.ts

TS get_set.ts

TS inheritance.ts

inputs.html

TS inputs.ts

TS interface.ts

TS intersection.ts

TS keyof.ts 1, U

TS literal.ts

TS main.ts

TS never.ts

JS null.js

TS null.ts

JS number.js

TS number.ts

OUTLINE

TIMELINE

TS keyof.ts 1, U

TS keyof.ts > ...

6 let PersonData:PersonT={

8 age:30,

9 isEmp:true

10 }

11 type PersonX=keyof PersonT;

12 let PersonDataX:PersonX;

13

14 PersonDataX="name";

15 PersonDataX="age";

16 PersonDataX="isEmp";

17 PersonDataX="city";

18

19

JS keyof.js U

output > JS keyof.js > ...

1 "use strict";

2 let PersonData = {

3 name: 'Anil sidhu',

4 age: 30,

5 isEmp: true

6 };

7 let PersonDataX;

8 PersonDataX = "name";

9 PersonDataX = "age";

10 PersonDataX = "isEmp";

11

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[11:44:01 AM] File change detected. Starting incremental compilation...

[11:44:01 AM] Found 0 errors. Watching for file changes.

first-program* Launchpad 1 0

Ln 17, Col 18 Spaces: 2 UTF-8 LF TypeScript Live



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EXPLORER

OPEN EDITORS

GROUP 1

TS keyof.ts U

GROUP 2

JS keyof.js o... U

TYPESCRIPT-TUT

TS generic.ts

TS get_set.ts

TS inheritance.ts

inputs.html

TS inputs.ts

TS interface.ts

TS intersection.ts

TS keyof.ts U

TS literal.ts

TS main.ts

TS never.ts

JS null.js

TS null.ts

JS number.js

TS number.ts

OUTLINE

TIMELINE

TS keyof.ts U X

TS keyof.ts > [0] UserX

6 let PersonData:PersonT={

10 }

11 type PersonX=keyof PersonT;

12 // let PersonDataX:PersonX;

13 let PersonDataX:keyof PersonT;;

14 PersonDataX="name";

15 PersonDataX="age";

16 PersonDataX="isEmp";

17

18 let UserX:keyof type PersonData="name"

19

20

21

JS keyof.js U X

output > JS keyof.js > ...

1 "use strict";

2 let PersonData = {

3 name: 'Anil sidhu',

4 age: 30,

5 isEmp: true

6 };

7 // let PersonDataX:Perso

8 let PersonDataX;

9 ;

10 PersonDataX = "name";

11 PersonDataX = "age";

12 PersonDataX = "isEmp";

13 let UserX = "name";

14

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

[11:45:29 AM] File change detected. Starting incremental compilation...

[11:45:29 AM] Found 0 errors. Watching for file changes.

Ln 18, Col 23 (6 selected)

Spaces: 2

UTF-8

LF

() TypeScript

Live



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Index signature in TypeScript

- What is Index Signature.
- How to use it with Flexible Object Shapes.
- How to use readonly key.
- Interview Questions.



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Index signature in TypeScript

- An **index signature** in TypeScript allows you to define objects with dynamic keys while specifying the type of their values.



Subscribe

typescript-tut

TS index_signature.ts U X

JS index_signature.js U X

TS index_signature.ts > [0] userData7

output > JS index_signature.js > ...

```
1 type userData7Type={
2   mobile:number,
3   id:number,
4   marks:number,
5   age:number
6 }
7
8 var userData7:userData7Type={
9   mobile:9999,
10  id:10,
11  marks:90,
12  age:40
13 }
```

```
1 "use strict";
2 var userData7 = {
3   mobile: 9999,
4   id: 10,
5   marks: 90,
6   age: 40
7 };
8
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[5:03:59 PM] File change detected. Starting incremental compilation...

[5:03:59 PM] Found 0 errors. Watching for file changes.

node + ... ^ X



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Ln 11, Col 14 Spaces: 4 UTF-8 LF TypeScript Go Live

Subscribe

typescript-tut

TS index_signature.ts 1, U

JS index_signature.js U

TS index_signature.ts > [U] userData7

output > JS index_signature.js > ...

1 type userData7Type={

3 id:number,

4 marks:number,

5 age:number,

6 semester:number

7 }

8

9 var userData7:userData7Type={

10 mobile:9999,

11

12 marks:90,

13 age:40,

14 semester:3

15 }

1 "use strict";

2 var userData7 = {

3 mobile: 9999,

4 id: 10,

5 marks: 90,

6 age: 40,

7 semester: 3

8 };

9

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[5:04:33 PM] File change detected. Starting incremental compilation...

[5:04:34 PM] Found 0 errors. Watching for file changes.

node + ... ^ x



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Ln 11, Col 2 Spaces: 4 UTF-8 LF () TypeScript Go Live

Subscribe

typescript-tut

TS index_signature.ts 2, U

JS index_signature.js U

TS index_signature.ts > ...

output > JS index_signature.js > ...

1 type userData7Type={

2 [key:string]:number

3 }

4

5 var userData7:userData7Type={

6 mobile:9999,

7 }

8

9 }

1 "use strict";

2 var userData7 = {

3 mobile: 9999,

4 };

5

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

node + ... ^ x

7

index_signature.ts:9:1 - error TS1128: Declaration or statement expected.

9 }

[5:05:25 PM] Found 2 errors. Watching for file changes.

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Ln 7, Col 4 Spaces: 4 UTF-8 LF TypeScript Go Live



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typescript-tut

TS index_signature.ts U X

JS index_signature.js U X

TS index_signature.ts > [0] userData7

output > JS index_signature.js > ...

```
1 type userData7Type={
2   [key:string]:number|string
3 }
4
5 var userData7:userData7Type={
6   mobile:9999,
7   id:10,
8   marks:90,
9   age:40,
10  semester:3,
11  name:'anil'
12 }
```

```
1 "use strict";
2 var userData7 = {
3   mobile: 9999,
4   id: 10,
5   marks: 90,
6   age: 40,
7   semester: 3,
8   name: 'anil'
9 };
10
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS GITLENS SPELL CHECKER

11 name:'anil'

index_signature.ts:2:5
[key:string]:number

The expected type comes from this index signature.

[5:05:48 PM] Found 1 error. Watching for file changes.

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Ln 8, Col 14 Spaces: 4 UTF-8 LF TypeScript Go Live



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typescript-tut

TS index_signature.ts 1, U X

JS index_signature.js U X

TS index_signature.ts > [⌘] userData7Type

output > JS index_signature.js > ...

```
10 // semester:3,
11 // name:'anil'
12 // }
13
14 type userData7Type = {
15   name:string,
16   id:number,
17   mobile:number,
18   [key: string]: number | string
19 }
20
21 var userData7: userData7Type = {
22   name: 'anil',
23   mobile: 9999,
24   // id: 10,
```

```
1 "use strict";
2 // type userData7Type={
3 //   [key:string]:number|string
4 // }
5 var userData7 = {
6   name: 'anil',
7   mobile: 9999,
8   // id: 10,
9   marks: 90,
10  age: 40,
11  semester: 3,
12 };
13
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

node + ... ^ X



21 var userData7: userData7Type = {

index_signature.ts:16:5

16 id:number,

'id' is declared here.

[5:07:05 PM] Found 1 error. Watching for file changes.

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Ln 24, Col 15 Spaces: 4 UTF-8 LF () TypeScript Go Live

typescript-tut

TS index_signature.ts 1, U X

JS index_signature.js U X

TS index_signature.ts > [0] userData7Type

output > JS index_signature.js > ...

```
14 type userData7Type = {
16   id:number,
17   mobile:number,
18   readonly [key: string]: number | string
19 }
20
21 var userData7: userData7Type = {
22   name: 'anil',
23   mobile: 9999,
24   id: 10,
25   marks: 90,
26   age: 40,
27   semester: 3,
28 }
29 userData7.city = 'noida'
```

```
1 "use strict";
2 // type userData7Type={
3 //   [key:string]:number|string
4 // }
5 var userData7 = {
6   name: 'anil',
7   mobile: 9999,
8   id: 10,
9   marks: 90,
10  age: 40,
11  semester: 3,
12 };
13 userData7.city = 'noida';
14
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[5:08:08 PM] File change detected. Starting incremental compilation...

index_signature.ts:29:1 - error TS2542: Index signature in type 'userData7Type' only permits reading.

29 userData7.city = 'noida'

[5:08:08 PM] Found 1 error. Watching for file changes.

node + ... ^ X



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Ln 18, Col 13 (8 selected) Spaces: 4 UTF-8 LF TypeScript Go Live









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Utility Types in TypeScript



- What is Utility Types ?
 - Partial
 - Required
 - Readonly
 - Pick
- Example
- Interview Questions

< >

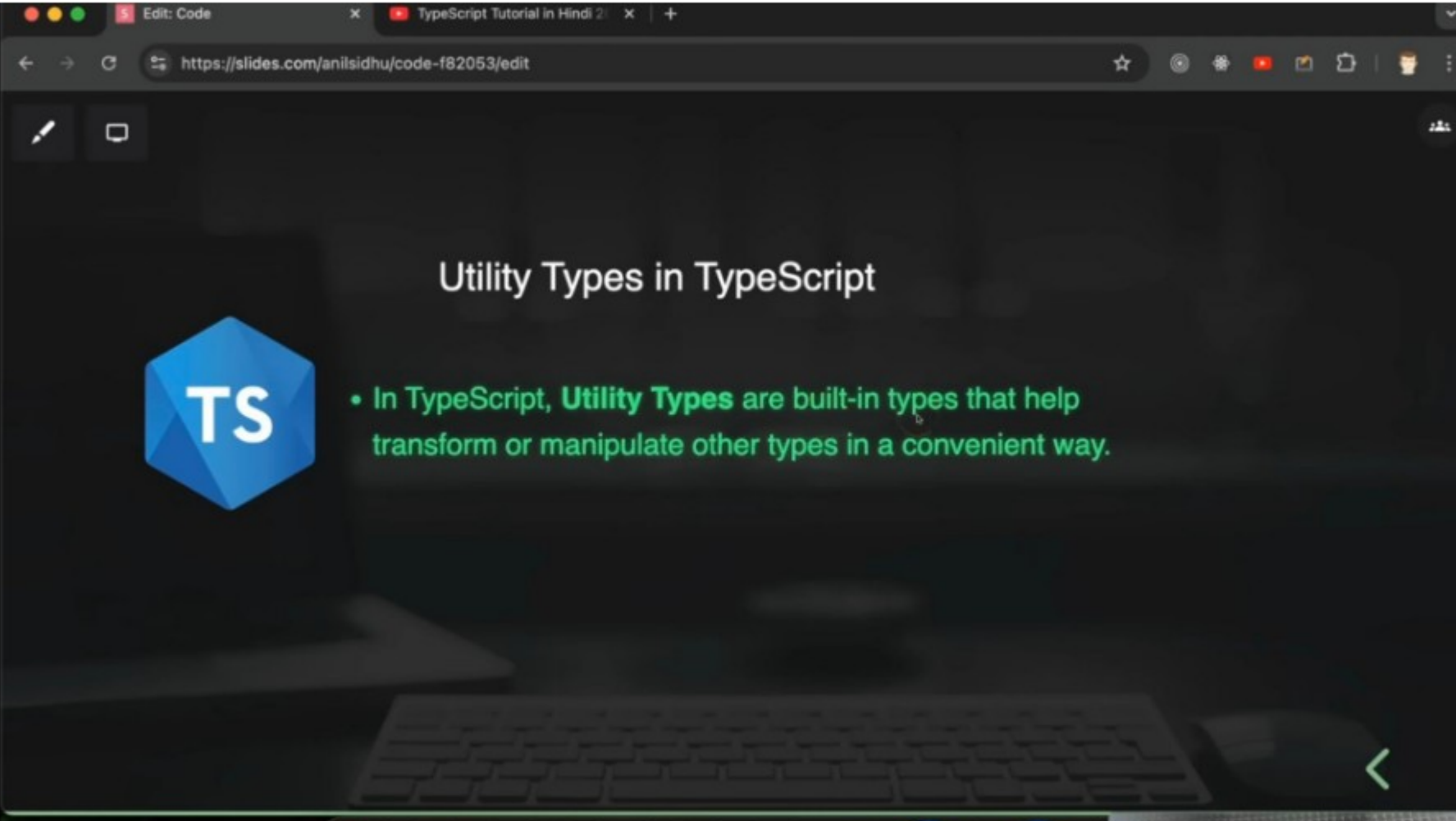
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Utility Types in TypeScript



- In TypeScript, **Utility Types** are built-in types that help transform or manipulate other types in a convenient way.

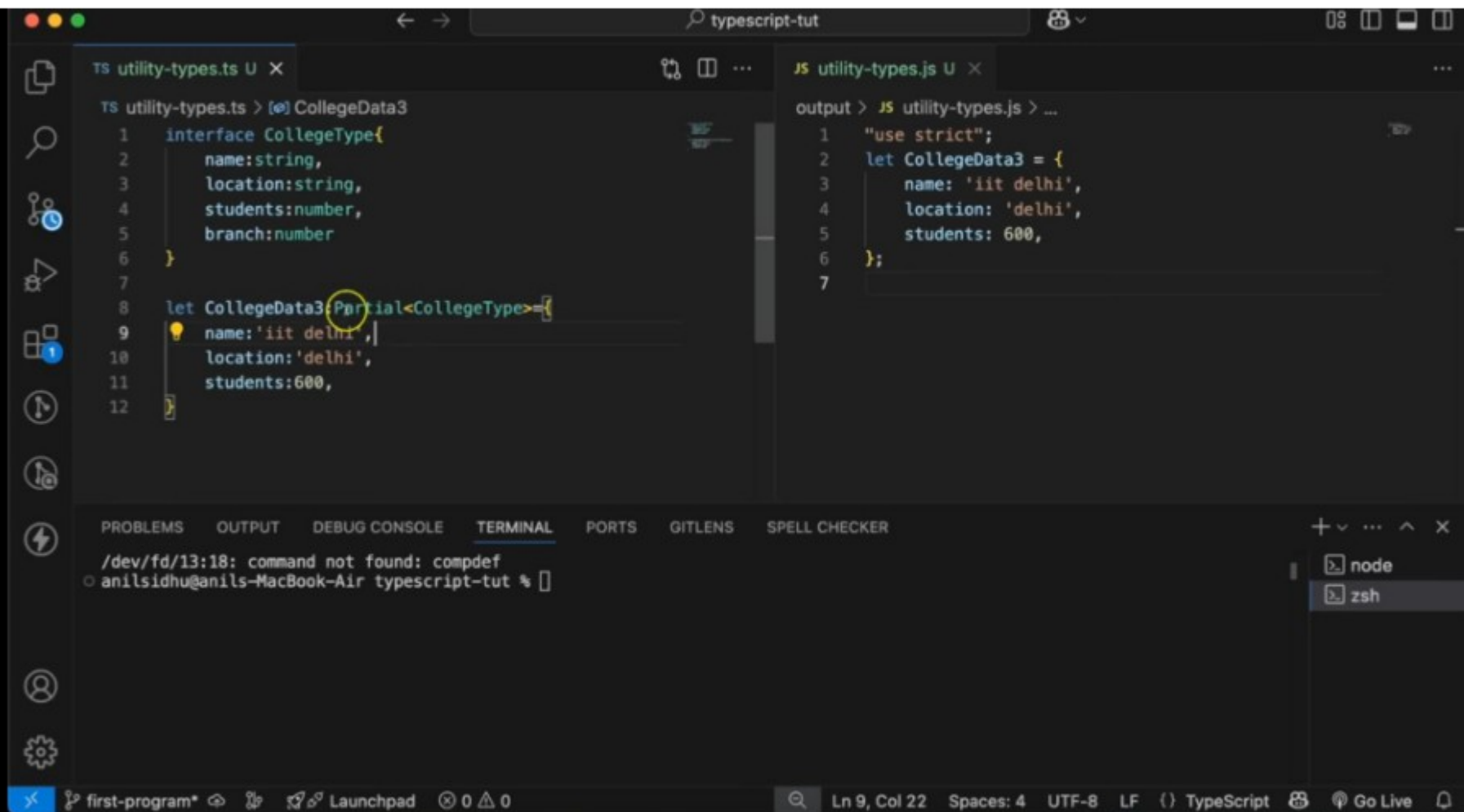


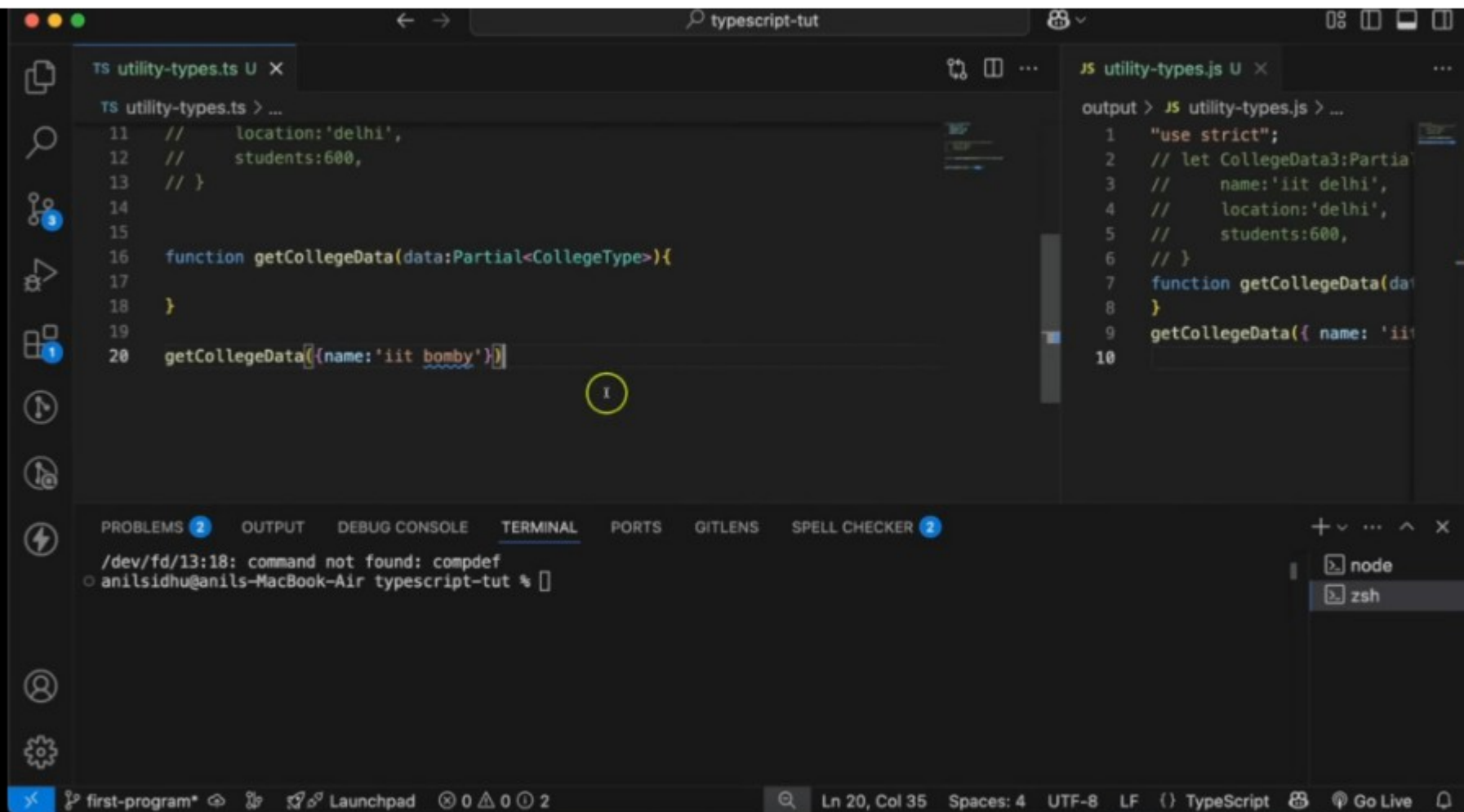














Utility Types in TypeScript



- - Omit
 - Exclude
 - Extract
 - NonNullable
 - Record
- Example
- Interview Questions



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TS

Namespace in Typescript

- What is namespace
- Make two name space
- How to run class, functions with namespace
- how to import namespace
- Interview Question

Subscribe

typescript-tut

TS namespace.ts U X

TS namespace.ts > Auth

```
1 class Auth {
2   login() {
3     console.log("user login function");
4   }
5 }
6
7 function getList() {
8   console.log("list of users");
9 }
10 }
```

JS namespace.js U X

output > JS namespace.js > ...

```
1 "use strict";
2 class Auth {
3   login() {
4     console.log("user login function");
5   }
6 }
7 function getList() {
8   console.log("list of users");
9 }
10 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

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Ln 6, Col 2 Spaces: 4 UTF-8 LF {} TypeScript

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typescript-tut

TS namespace.ts U X

```
TS namespace.ts > {} UserNameSpace
1 namespace UserNameSpace{
2   class Auth{
3     login(){
4       console.log("user login function");
5     }
6   }
7   function getList(){
8     console.log("list of users");
9   }
10 }
11
12
13
14
```

JS namespace.js U X

```
output > JS namespace.js > ...
1 "use strict";
2 var UserNameSpace;
3 (function (UserNameSpace) {
4   class Auth {
5     login() {
6       console.log("user login function");
7     }
8   }
9   function getList() {
10     console.log("list of users");
11   }
12 })(UserNameSpace || (UserNameSpace = {}));
13
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

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typescript-tut

TS namespace.ts U X

TS namespace.ts > {} UserNameSpace

```
1 namespace UserNameSpace{
8   function getList(){
11
12
13
14
15 namespace AdminNameSpace{
16   class Auth{
17     login(){
18       console.log("user login function");
19
20     }
21   }
22   function getList(){
23     console.log("list of users");
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut %

JS namespace.js U X

output > JS namespace.js > ...

```
1 "use strict";
2 var UserNameSpace;
3 (function (UserNameSpace) {
4   class Auth {
5     login() {
6       console.log("user login function");
7     }
8   }
9   function getList() {
10     console.log("list of users");
11   }
12 })(UserNameSpace || (UserNameSpace = {}));
13 var AdminNameSpace;
14 (function (AdminNameSpace) {
15   class Auth {
```

+ v ... ^ X



Ln 13, Col 2 Spaces: 4 UTF-8 LF () TypeScript

Subscribe

typescript-tut

TS namespace.ts U

```
1 namespace UserNameSpace{
2   export class Auth{
3     login(){
4       console.log("user login function");
5     }
6   }
7 }
8 export function getList(){
9   console.log("list of users");
10 }
11
12
13
14
15 namespace AdminNameSpace{
```

JS namespace.js U

```
1 "use strict";
2 var UserNameSpace;
3 (function (UserNameSpace) {
4   class Auth {
5     login() {
6       console.log("user login function");
7     }
8   }
9   UserNameSpace.Auth = Auth;
10  function getList() {
11    console.log("list of users");
12  }
13 })(UserNameSpace || (UserNameSpace = {}));
14 var AdminNameSpace;
15 (function (AdminNameSpace) {
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 1

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Ln 8, Col 11 Spaces: 4 UTF-8 LF () TypeScript



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typescript-tut

TS namespace.ts U X

```
1 namespace UserNameSpace{
2   export class Auth{
3     login(){
4       console.log("user login function");
5     }
6   }
7 }
8 export function getList(){
9   console.log("list of users");
10 }
11
12
13
14
15 namespace AdminNameSpace{
```

JS namespace.js U X

```
1 "use strict";
2 var UserNameSpace;
3 (function (UserNameSpace) {
4   class Auth {
5     login() {
6       console.log("user login function");
7     }
8   }
9   UserNameSpace.Auth = Auth;
10   function getList() {
11     console.log("list of users");
12   }
13 })(UserNameSpace || (UserNameSpace = {}));
14 var AdminNameSpace;
15 (function (AdminNameSpace) {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

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Ln 8, Col 12 Spaces: 4 UTF-8 LF () TypeScript



Subscribe

typescript-tut

TS namespace.ts U X

TS namespace.ts > ...
15 namespace AdminNameSpace{
27 }
28
29
30 var user8=new UserNameSpace.Auth()
31 user8.login()

JS namespace.js U X

output > JS namespace.js > ...
1 "use strict";
2 var UserNameSpace;
3 (function (UserNameSpace) {
4 class Auth {
5 login() {
6 console.log("user login function");
7 }
8 }
9 UserNameSpace.Auth = Auth;
10 function getList() {
11 console.log("list of users");
12 }
13 UserNameSpace.getList = getList;
14 })(UserNameSpace || (UserNameSpace = {}));
15 var AdminNameSpace;

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

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first-program* Launchpad 0 0 0 Ln 31, Col 13 Spaces: 4 UTF-8 LF () TypeScript

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Decorators in Typescript



- What are decorators?
- How to use it ?
- Example with class and property
- Interview Question



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Decorators in Typescript



- In TypeScript, **decorators** are a special kind of declaration that can be attached to classes, methods, **properties**, or parameters to modify their behavior.

```
1 @Logger
2 class Person {
3   constructor(public name: string) {}
4 }
```



typescript-tut

TS decorators.ts U X

```
1 class CustomMaths{
2   value1:number;
3   value2:number;
4   constructor(x:number,y:number){
5     this.value1=x;
6     this.value2=y;
7   }
8 }
9
10 var cm1 = new CustomMaths(10,20);
```

JS decorators.js U X

```
1 "use strict";
2 class CustomMaths {
3   value1;
4   value2;
5   constructor(x, y) {
6     this.value1 = x;
7     this.value2 = y;
8   }
9 }
10 var cm1 = new CustomMaths(10, 20);
11
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

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Ln 9, Col 1 Spaces: 2 UTF-8 LF () TypeScript



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typescript-tut

TS decorators.ts U X

TS decorators.ts > ...
1 function classLogger(constructor:Function){
2 console.log(constructor.name);
3
4 }
5 |
6 class CustomMaths{
7 value1:number;
8 value2:number;
9 constructor(x:number,y:number){
10 this.value1=x;
11 this.value2=y;
12 }
13 }
14
15 var cm1 = new CustomMaths(10,20);

JS decorators.js U X

output > JS decorators.js > ...
1 "use strict";
2 function classLogger(constructor) {
3 console.log(constructor.name);
4 }
5 class CustomMaths {
6 value1;
7 value2;
8 constructor(x, y) {
9 this.value1 = x;
10 this.value2 = y;
11 }
12 }
13 var cm1 = new CustomMaths(10, 20);
14

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

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anilsidhu@anils-MacBook-Air output %

first-program* Launchpad 0 0

Ln 5, Col 1 Spaces: 2 UTF-8 LF () TypeScript





typescript-tut

TS decorators.ts U X

```
1 function classLogger(constructor:Function){
2     console.log(constructor.name);
3 }
4
5
6 @classLogger
7 class CustomMaths{
8
9     value1:number;
10    value2:number;
11    constructor(x:number,y:number){
12        this.value1=x;
13        this.value2=y;
14    }
15 }
```

JS decorators.js U X

```
1 "use strict";
2 var __esDecorate = (this && this.__
3     function accept(f) { if (f !==
4         var kind = contextIn.kind, key =
5         var target = !descriptorIn && c
6         var descriptor = descriptorIn |
7         var __, done = false;
8         for (var i = decorators.length -
9             var context = {};
10             for (var p in contextIn) con
11             for (var p in contextIn.acce
12             context.addInitializer = fun
13             var result = (0, decorators
14             if (kind === "accessor") {
15                 if (result === void 0) {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

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anilsidhu@anils-MacBook-Air output % node decorators.js

Ln 8, Col 3 Spaces: 2 UTF-8 LF () TypeScript



Subscribe

typescript-tut

TS decorators.ts U X

TS decorators.ts > CustomMaths

```
5 function getKeyDetails(target:any,key:any){
6     console.log(key);
7 }
8
9
10
11 @classLogger
12 class CustomMaths{
13
14     @getKeyDetails
15     value1:number;
16     value2:number;
17     constructor(x:number,y:number){
18         this.value1=x;
19         this.value2=y;
20     }
21 }
```

JS decorators.js U X

output > JS decorators.js > ...

```
1 "use strict";
2 var __esDecorate = (this && this._
3     function accept(f) { if (f !==
4         var kind = contextIn.kind, key
5         var target = !descriptorIn && c
6         var descriptor = descriptorIn |
7         var _, done = false;
8         for (var i = decorators.length
9             var context = {};
10             for (var p in contextIn) co
11             for (var p in contextIn.acce
12                 context.addInitializer = fu
13             var result = (0, decorators
14             if (kind === "accessor") {
15                 if (result === void 0) {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

CustomMaths123

- anilsidhu@anils-MacBook-Air output % node decorators.js
CustomMaths
- anilsidhu@anils-MacBook-Air output % node decorators.js
{
 kind: 'field',
 name: 'value1',
 static: false,
 private: false,
 access: { has: [Function: has], get: [Function: get], set: [Function: set] },
 metadata: undefined,
 addInitializer: [Function (anonymous)]
}

first-program* Launchpad 0 0

Ln 13, Col 3 Spaces: 2 UTF-8 LF {} TypeScript



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TS

Override function with Decorator in Typescript



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typescript-tut

EXPLORER

OPEN EDITORS

GROUP 1

- TS decorator... M

GROUP 2

- JS decorator... M

TYPESCRIPT-TUT

- type
- TS any_unknown.ts
- JS app.js
- TS app.ts
- TS array.ts
- TS bigint.ts
- TS collegeData.ts
- TS decorator-with... M
- TS decorators.ts
- dom.html
- TS dom.ts
- TS enum.ts
- TS function.ts
- TS functionParams.ts
- TS generic.ts
- OUTLINE
- TIMELINE

TS decorator-with-function.ts M X

TS decorator-with-function.ts > CustomMaths1 > sum

You, 5 seconds ago | 1 author (You)

```
1 class CustomMaths1 {
2
3   sum(x:number,y:number) {
4     return x+y;
5   }
6
7   var cm2= new CustomMaths1();
8   console.log(cm2.sum(10,20));
9 }
```

You, 6 seconds ago • Uncommitted changes

JS decorator-with-function.js M X

output > JS decorator-with-function.js > ...

```
2 class CustomMaths1 {
3   sum(x, y) {
4     return x + y;
5   }
6 }
7 var cm2 = new CustomMaths
8 console.log(cm2.sum(10, 2
9
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air output % node decorator-with-function.js

30

anilsidhu@anils-MacBook-Air output %

first-program* Launchpad 0 0 Not Committed Yet Ln 5, Col 6 Spaces: 4 UTF-8 LF TypeScript 5500

Code Steps

Subscribe

typescript-tut

TS decorator-with-function.ts M X

TS decorator-with-function.ts > updatedSum

```
1 You, 1 second ago | 1 author (You)
2 function updatedSum(target:any,key:string,descriptor:PropertyDescriptor)
3   |
4   |
5 You, 1 second ago | 1 author (You)
6 class CustomMaths1{
7   sum(x:number,y:number){
8     return x+y
9   }
10 }
11 var cm2= new CustomMaths1();
12 console.log(cm2.sum(10,20));
13
```

JS decorator-with-function.js M X

output > JS decorator-with-function.js > ...

```
3 }
4 You, 4 seconds ago | 1 author (You)
5 class CustomMaths1 {
6   sum(x, y) {
7     return x + y;
8   }
9 }
10 var cm2 = new CustomMaths1();
11 console.log(cm2.sum(10, 20));
12
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air output % node decorator-with-function.js

30

anilsidhu@anils-MacBook-Air output %

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typescript-tut

TS decorator-with-function.ts M X

TS decorator-with-function.ts > CustomMaths1

```
1 function updatedSum(target:any,key:string,descriptor:PropertyDescriptor)
2     descriptor.value=function sum(x:number,y:number){
3         let output=x+y
4         return "the output of "+x+" and "+y+" is : " + output
5     }
6 }
7
8 class CustomMaths1{
9     You, 1 second ago • Uncommitted changes
10     sum(x:number,y:number){
11         return x+y
12     }
13 }
14 var cm2= new CustomMaths1();
15 console.log(cm2.sum(10,20));
```

JS decorator-with-function.js M X

```
2 function updatedSum(target, key,
3     descriptor.value = function
4
5         return "the output of "
6     };
7 }
8
9 class CustomMaths1 {
10     sum(x, y) {
11         return x + y;
12     }
13 }
14 var cm2 = new CustomMaths1();
15 console.log(cm2.sum(10, 20));
```

output > JS decorator-with-function.js > ...

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

```
● anilsidhu@anils-MacBook-Air output % node decorator-with-function.js
30
● anilsidhu@anils-MacBook-Air output % node decorator-with-function.js
30
○ anilsidhu@anils-MacBook-Air output %
```

first-program* Launchpad 0 0 Not Committed Yet Ln 9, Col 1 Spaces: 4 UTF-8 LF TypeScript 5500



typescript-tut

TS decorator-with-function.ts M X

TS decorator-with-function.ts > CustomMaths1 > sum

```
1 function updatedSum(target:any,key:string,descriptor:PropertyDescriptor)
2     descriptor.value=function sum(x:number,y:number){
3         let output=x+y
4         return "the output of "+x+" and "+y+" is : " + output
5     }
6 }
7
8 class CustomMaths1{
9
10     @updatedSum
11     sum(x:number,y:number){
12         return x+y
13     }
14 }
15 var cm2= new CustomMaths1();
```

JS decorator-with-function.js M X

```
2 var __decorate = (this && this.
4     if (typeof Reflect === "obje
5     else for (var i = decorators
6     return c > 3 && r && Object.
7 };
8 function updatedSum(target, key,
9     descriptor.value = function
10         let output = x + y;
11         return "the output of "
12     };
13 }
14 class CustomMaths1 {
15     sum(x, y) {
16         return x + y;
17     }
18 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

```
anilsidhu@anils-MacBook-Air output % node decorator-with-function.js
30
anilsidhu@anils-MacBook-Air output % node decorator-with-function.js
30
anilsidhu@anils-MacBook-Air output % node decorator-with-function.js
the output of 10 and 20 is : 30
anilsidhu@anils-MacBook-Air output %
```

first-program* Launchpad 0 0 Not Committed Yet Ln 10, Col 16 Spaces: 4 UTF-8 LF TypeScript 5500



typescript-tut

EXPLORER

OPEN EDITORS

GROUP 1

- TS decorator... M
- tsconfig.j... M

GROUP 2

- JS decorator... M

TYPESCRIPT-TUT

- package.json
- product.ts
- readonly.ts
- script.txt
- static.ts
- string.js
- string.ts
- symbol.ts
- test.ts
- tsconfig.json M
- tuple.ts
- type.ts
- typeGuards.ts
- undefine.js

OUTLINE

TIMELINE

tsconfig.json > {} compilerOptions

```
2 "compilerOptions": {
9 // "disableSourceOfProjectReferenceRedirect": true, /
10 // "disableSolutionSearching": true, /
11 // "disableReferencedProjectLoad": true, /
12
13 /* Language and Environment */
14 "target": "ES2024", /
15 // "lib": [], /
16 // "jsx": "preserve", /
17 // "libReplacement": true, /
18 "experimentalDecorators": true, /* E
19 // "emitDecoratorMetadata": true, /
20 // "jsxFactory": "", /
21 // "jsxFragmentFactory": "", /
22 // "jsxImportSource": "", /
23 // "reactNamespace": "", /
24 // "noLib": true, /
```

JS decorator-with-function.js

```
2 var __decorate = (this &&
4 if (typeof Reflect ==
5 else for (var i = dec
6 return c > 3 && r &&
7 );
8 function updatedSum(target
9 descriptor.value = fu
10 let output = x +
11 return "the output
12 };
13
14 You, 23 seconds ago | 1 author (You)
15 class CustomMaths1 {
16 sum(x, y) {
17 return x + y;
18 }
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 1

```
anilsidhu@anils-MacBook-Air output % node decorator-with-function.js
30
anilsidhu@anils-MacBook-Air output % node decorator-with-function.js
30
anilsidhu@anils-MacBook-Air output % node decorator-with-function.js
the output of 10 and 20 is : 30
anilsidhu@anils-MacBook-Air output %
```

first-program* Launchpad 0 0 1 Ln 18, Col 2 (38 selected) Spaces: 2 UTF-8 LF {} JSON with Comments 5500



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Typed Promise in TypeScript

- What is Promise.
- How to define type for promise.
- Define Custom type in promise.
- Interview Question







typescript-tut

EXPLORER

OPEN EDITORS

TS promise.ts U

TYPESCRIPT-TUT

output

useNamespace.js

utility-types.d.ts

utility-types.js

type

type.ts

any_unknown.ts

app.js

app.ts

array.ts

bigint.ts

collegeData.ts

decorator-with... M

decorators.ts

dom.html

dom.ts

enum.ts

OUTLINE

TIMELINE

TS promise.ts U X

TS promise.ts > test1

1 function test1(){

2 setTimeout(() => {

3 console.log("test1");

4 }, 2000);

5 }

6

7

8 function test2() {

9 console.log("test2");

10 }

11

12

13 test1();

14 test2();

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

/dev/fd/13:18: command not found: compdef

anilsidhu@anils-MacBook-Air typescript-tut % cd output

anilsidhu@anils-MacBook-Air output % node promise.js

test1

test2

anilsidhu@anils-MacBook-Air output % node promise.js

test2

test1

anilsidhu@anils-MacBook-Air output %

node

zsh output

first-program* Launchpad 0 0

Ln 4, Col 13 Spaces: 4 UTF-8 LF () TypeScript Go Live

EXPLORED

OPEN EDITORS

TS promise.ts U

TYPESCRIPT-TUT

output

useNameSpace.js

utility-types.d.ts

utility-types.js

type

type.ts

any_unknown.ts

app.js

app.ts

array.ts

bigint.ts

collegeData.ts

decorator-with... M

decorators.ts

dom.html

dom.ts

enum.ts

function.ts

OUTLINE

TIMELINE

TS promise.ts U X

TS promise.ts > ...

```
1 function complexLogic() {
2   return new Promise((resolved)=>{
3     setTimeout(() => {
4       resolved("result is here")
5     }, 2000);
6   })
7 }
8
9
10 complexLogic().then((data)=>{
11   console.log(data);
12
13   test2()
14 })
15 function test2() {
16   console.log("test2");
17 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER

node

zsh output

anilsidhu@anils-MacBook-Air output % node promise.js
test1
test2
anilsidhu@anils-MacBook-Air output % node promise.js
test2
test1
anilsidhu@anils-MacBook-Air output % node promise.js
result is here
test2
anilsidhu@anils-MacBook-Air output %

first-program*

Launchpad

0 0 0

Ln 8, Col 1

Spaces: 4

UTF-8

LF

() TypeScript

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EXPLORED

OPEN EDITORS

TS promise.ts U

TYPESCRIPT-TUT

output

JS useNamespace.js

TS utility-types.d.ts

JS utility-types.js

type

TS type.ts

TS any_unknown.ts

JS app.js

TS app.ts

TS array.ts

TS bigint.ts

TS collegeData.ts

TS decorator-with... M

TS decorators.ts

dom.html

TS dom.ts

TS enum.ts

TS function.ts

OUTLINE

TIMELINE

TS promise.ts U X

TS promise.ts > then() callback

1 function complexLogic():Promise<string> {

2 return new Promise((resolved)=>{

3 setTimeout(() => {

4 resolved("result is here")

5 }, 2000);

6 })

7 }

8

9

10 complexLogic().then((data:string)=>{

11 console.log(data);

12 })

13

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER

node

zsh output

anilsidhu@anils-MacBook-Air output % node promise.js

test1

test2

anilsidhu@anils-MacBook-Air output % node promise.js

test2

test1

anilsidhu@anils-MacBook-Air output % node promise.js

result is here

test2

anilsidhu@anils-MacBook-Air output %

first-program*

Launchpad

0 0

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Spaces: 4

UTF-8

LF

() TypeScript

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API call in TypeScript

- Write code for API call.
- Define type for response.
- Apply data type to API response.
- Interview Question

Navigation: <

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Google Search Results for "dummy api call"

Results for **Block F, Carterpur Village, Sector 23A, Gurugram** · Choose area

JSONPlaceholder
<https://jsonplaceholder.typicode.com>

JSONPlaceholder - Free Fake REST API

JSONPlaceholder is a free online REST API that you can use whenever you need some fake data. It can be in a README on GitHub, for a demo on CodeSandbox, in ...

[My JSON Server](#) · [Guide](#) · [Typicode's blog](#)

DummyJSON
<https://dummyjson.com>

DummyJSON - Free Fake REST API for Placeholder JSON Data

DummyJSON provides a free fake REST API with placeholder JSON data for development, testing, and prototyping. Access realistic data quickly for your ...

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Beeceptor
<https://beeceptor.com/docs/sample-api-for-testing>

Sample API For Testing

YouTube Subscribe button

Try it

Run this code here, in a console or from any site:

```
fetch('https://jsonplaceholder.typicode.com/todos/1')  
  .then(response => response.json())  
  .then(json => console.log(json))
```

Run script

```
{}
```

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Try it

Run this code here, in a console or from any site:

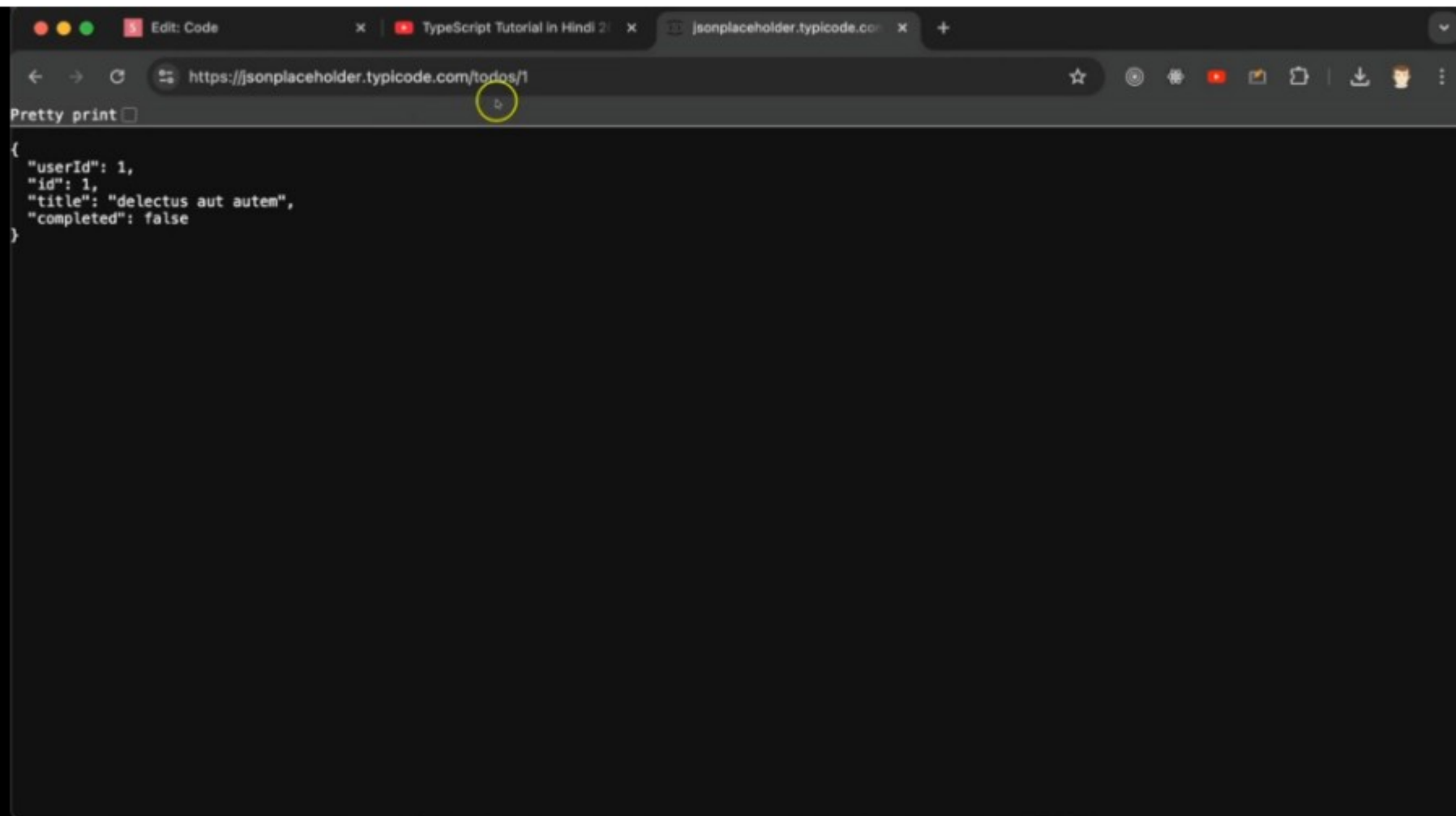
```
fetch('https://jsonplaceholder.typicode.com/todos/1')  
  .then(response => response.json())  
  .then(json => console.log(json))
```

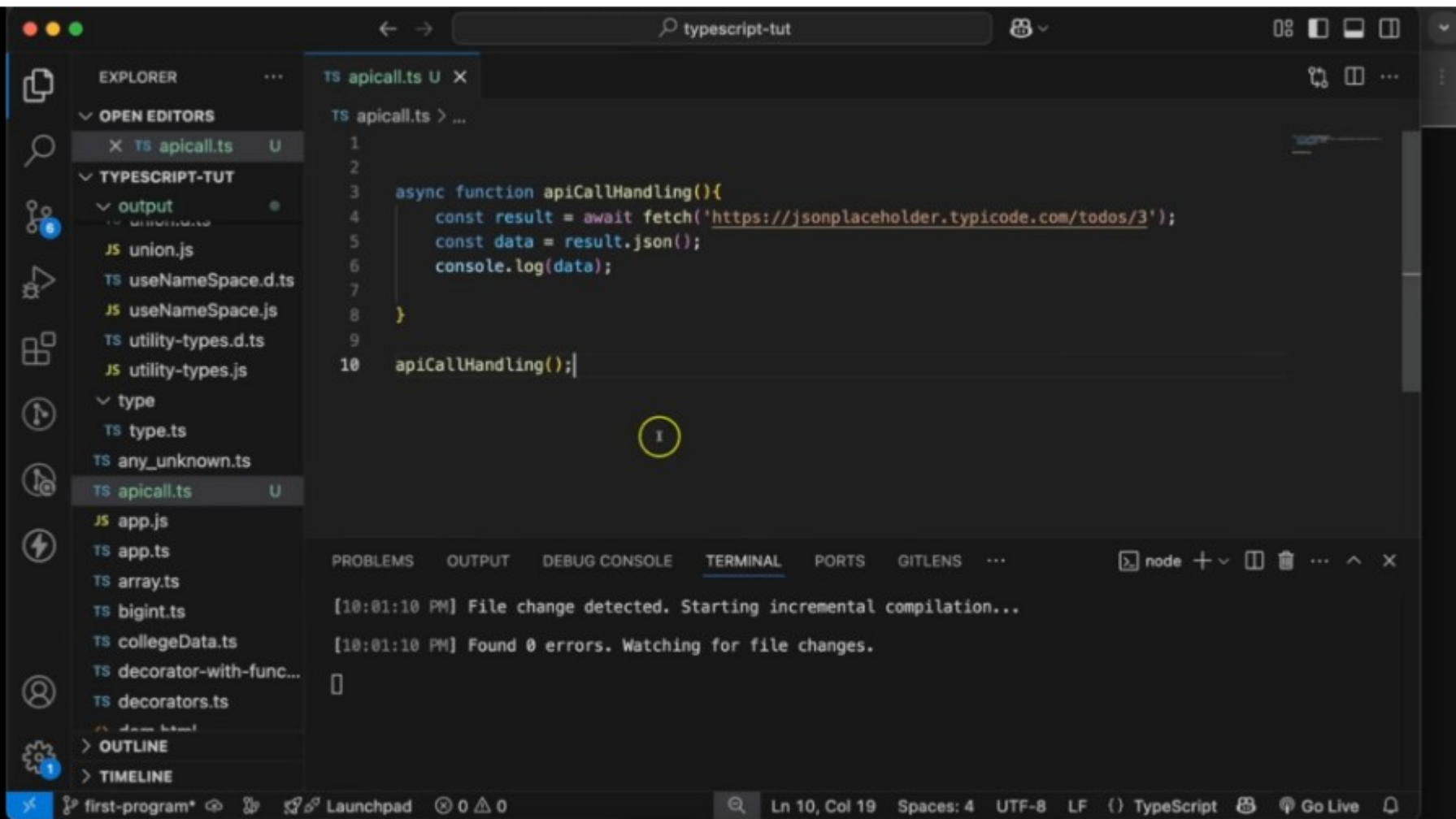
Run script

```
{}
```

AI-powered ad

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typescript-tut

TS apicall.ts U X

TS apicall.ts > apiCallHandling

```
1
2
3 async function apiCallHandling(){
4     const result = await fetch('https://jsonplaceh
5     const data = await result.json();
6     console.log(data);
7
8
9
10 apiCallHandling();
```

TS apicall.ts U

JS apicall.js U X

output > JS apicall.js > ...

```
1 "use strict";
2 async function apiCallHandling() {
3     const result = await fetch('https://jsonplaceh
4     const data = await result.json();
5     console.log(data);
6 }
7 apiCallHandling();
8
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

/dev/fd/13:18: command not found: compdef

● anilsidhu@anils-MacBook-Air typescript-tut % cd output

● anilsidhu@anils-MacBook-Air output % node apicall.js

Promise { <pending> }

○ anilsidhu@anils-MacBook-Air output %

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zsh output

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typescript-tut

TS apicall.ts U

apicall.ts U JS apicall.js U

TS apicall.ts > apiCallHandling

```
2 type APIType={
3   userId: string
4   id: number,
5   title: string
6   completed: boolean
7 }
8 async function apiCallHandling():Promise<APIType>{
9   const result = await fetch('https://jsonplaceholder.typicode.com/todos/4');
10  const data = await result.json();
11  console.log(data);
12  return data
13 }
14
15
16 apiCallHandling();
```

output > JS apicall.js > ...

```
1 "use strict";
2 async function apiCall
3   const result = awa
4   const data = await
5   console.log(data);
6 }
7 apiCallHandling();
8
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

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● anilsidhu@anils-MacBook-Air output % node apicall.js

Promise { <pending> }

● anilsidhu@anils-MacBook-Air output % node apicall.js

{ userId: 1, id: 3, title: 'fugiat veniam minus', completed: false }

● anilsidhu@anils-MacBook-Air output % node apicall.js

{ userId: 1, id: 4, title: 'et porro tempora', completed: true }

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TS apicall.ts U X

TS apicall.ts > ...

```
7 }
8 async function apiCallHandling():Promise<APIType>{
9   const result = await fetch('https://jsonplaceholder.typicode.com/todos/4');
10  const data = await result.json();
11  // console.log(data);
12  return data
13 }
14 }
15
16 apiCallHandling().then((data)=>{
17   console.log(data);
18 })
19 })
20
```

apicall.ts U

JS apicall.js U X ...

output > JS apicall.js > ...

```
1 "use strict";
2 async function apiCall
3   const result = awa
4   const data = await
5   // console.log(dat
6   return data;
7 }
8 apiCallHandling().then
9   console.log(data);
10 });
11
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

Promise { <pending> }

● anilsidhu@anils-MacBook-Air output % node apicall.js
{ userId: 1, id: 3, title: 'fugiat veniam minus', completed: false }

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Promise { <pending> }

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{ userId: 1, id: 4, title: 'et porro tempora', completed: true }

○ anilsidhu@anils-MacBook-Air output %

node

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Key Differences

Feature	<code>string</code>	<code>String</code>
Type	Primitive	Object wrapper
Memory/Performance	Lightweight and faster	Heavier due to object overhead
Usage	Recommended for typing	Avoid using in type annotations
Example	<code>let x: string = 'abc';</code>	<code>let x: String = new String('abc');</code>



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